

From Varieties to Schemes: An Original Editorial Companion

Prerequisite bridge, mastery route, solved integrative problems and Stacks capstone

Independent editorial companion; source references credited individually

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About this editorial companion

This is the **English translation** of the original editorial companion. Its independent post-translation source review is complete. Build and publication remain separate gates; this text does not by itself claim a completed public English release or an English-edition DOI.

This book is an **original editorial companion** for moving from varieties and coordinate rings to sheaves, schemes, and cohomology. Its bridge explanations, new solutions, integrative problems, and capstone were independently written by **OpenAI Codex gpt-5.6-sol, Ultra**. This is not an additional lecture course by Holger Brenner or a retranslation of the source course.

The referenced lectures, problem statements, theorems, and public solutions are by **Holger Brenner**, chiefly from *Bündel, Garben und Kohomologie (Osnabrück 2019-2020)*. Links to the [complete BGK reader](#) are retained for the separate HTML reader with all its source credits in the integrated edition. Their assigned targets have passed independent source review; build and publication status is recorded separately. These links identify specific sections, not answers that must be sought elsewhere. Source solutions remain source solutions; new editorial answers are not assigned the source author's name or endorsement.

Organisation and study route

Begin with the **prerequisite bridge**. It distinguishes closed points from all prime points, functions from sections, stalks from fibres, and local rings from residue fields. Then follow the mastery bank in the order of BGK Units 2–15, followed by 23–27. These remain the source course numbers; the jump from 15 to 23 neither rennumbers the course nor claims that Units 16–22 do not exist.

Each unit on this route has exactly three mastery items. There are **44 new editorial solutions + 13 public source solutions = 57 items** across **19 route units**. Seventeen units require new solution files; Units 8 and 11 already have three public solutions each counted here, so no new bank was created for either unit.

BGK unit	New editorial solutions	Source solutions counted	Total
2	2 solutions	2.4	3
3	2 solutions	3.1	3
4	3 solutions	None	3
5	2 solutions	5.5	3
6	3 solutions	None	3
7	2 solutions	7.14	3
8	0	8.3, 8.4, 8.11	3
9	3 solutions	None	3
10	3 solutions	None	3
11	0	11.9, 11.13, 11.14	3
12	1 solution	12.5, 12.10	3
13	3 solutions	None	3
14	3 solutions	None	3
15	3 solutions	None	3

BGK unit	New editorial solutions	Source solutions counted	Total
23	3 solutions	None	3
24	3 solutions	None	3
25	2 solutions	25.1	3
26	3 solutions	None	3
27	3 solutions	None	3
Total	44	13	57

Attempt each problem before opening its solution. When checking your answer, write down the domain and codomain of every map, the base ring, and the hypotheses that allow a denominator to be inverted. The pitfall and check notes in each item help identify steps that still need proof; they are not additional problems in the count of 57 items.

Twelve integrative problems and one capstone

After the mastery bank, work through the following twelve integrative problems, each with a complete solution. They are independently written synthesis problems, not additions to Brenner’s problem list.

1. From varieties and coordinate rings to affine schemes.
2. Affine gluing and compatibility conditions.
3. The projective line from two affine charts.
4. The affine line with doubled origin.
5. Morphisms of locally ringed spaces.
6. Krull dimension and geometric examples.
7. Quasicoherent sheaves and module localisation.
8. A Čech calculation with explicit signs and quotient.
9. A computable example of projective cohomology.
10. Euler characteristic and an exact sequence.
11. A classical projective curve and its scheme.
12. Tracing sources and reconstructing hypotheses.

Finish with **one integrated capstone** on permanent Stacks tags and proving that a neighbourhood is affine. Its seven stages are parts of **one** exercise, not seven capstones. The complete answers and oral-proof rubric are for self-study; the rubric does not require a human assessor’s presence or approval. **The Stacks Project Authors** are credited as the authors of the downstream reference; Stacks is not a translated source course.

Thus the 57 mastery items, 12 integrative problems, one capstone, and prerequisite bridge are distinct companion layers. No new course units or official PDF pages are added: the two source courses still total **60 units and 602 official PDF pages**, as frozen in the source edition. This companion does not change source problem, theorem, or unit numbers.

Terminology, sources, and usage rights

Use the [BGK terminology guide](#) alongside local definitions. Sheaf, presheaf, stalk, fibre, section, restriction, ring, field, and ringed space denote different types of objects; matching words do not replace hypotheses. Each new item points to the source problem or theorem it uses, together with the relevant revision identity or frozen witness. Source notes correcting formulas must still be read alongside the statements.

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From varieties to schemes: what changes?

This bridge connects the starting point of a reader of classical algebraic geometry with the language of sheaves and schemes. It is an independently written bridge explanation, not a translation of an additional lecture by Holger Brenner. The algebraic prerequisites are prime and maximal ideals, quotient rings, and localisation. All rings here are commutative with identity; ring homomorphisms preserve the identity.

1. Coordinate rings are not discarded

For the classical comparison in this section only, take an algebraically closed field k and a nonempty affine algebraic set $V \subseteq k^n$. Its coordinate ring is

$$A = k[V] = k[x_1, \dots, x_n]/I(V),$$

where $I(V)$ is the ideal of all polynomials vanishing on V . Then A is reduced: if f^m vanishes on V , every value $f(a)$ in the field k is also zero, so $f \in I(V)$. If the term *variety* in the source requires irreducibility, impose that condition too; in that case A is an integral domain. Reduced and integral domain are not synonyms.

The prerequisite theorem connecting algebra with classical points is the Nullstellensatz: for an ideal $J \subseteq k[x_1, \dots, x_n]$, $I(V(J)) = \sqrt{J}$, and maximal ideals have the form $(x_1 - a_1, \dots, x_n - a_n)$ with $a_i \in k$. Consequently,

$$a \in V \iff \mathfrak{m}_a = \{f \in A : f(a) = 0\} \in \text{MaxSpec}(A).$$

The scheme construction keeps the same A , but its space is

$$X = \text{Spec}(A) = \{\mathfrak{p} \subset A : \mathfrak{p} \text{ is a prime ideal}\}.$$

On this space, $V(J) = \{\mathfrak{p} : J \subseteq \mathfrak{p}\}$ is closed, while the sets $D(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}$ form an open basis. On a spectrum, $V(J)$ denotes a set of prime ideals; do not immediately read it as a set of tuples in k^n .

2. Closed points and generic points

The closure of a point $\mathfrak{p}$ is $V(\mathfrak{p})$: a closed set $V(J)$ contains $\mathfrak{p}$ exactly when $J \subseteq \mathfrak{p}$. Thus closed points are precisely maximal ideals. The classical comparison above identifies V with the subspace of closed points of X , not with all of X .

For example, for $A = k[t]$ with k algebraically closed, the points are $(t - a)$, $a \in k$, and one extra point $\eta = (0)$. The closure of η is the entire spectrum. This point records the whole line as one irreducible closed subspace; it is not a new number to be added to k . More generally, $\mathfrak{p}$ is the generic point of $V(\mathfrak{p})$.

If k is not algebraically closed, closed points need not be valued in k . For example, $(t^2 + 1)$ is a maximal ideal of $\mathbb{R}[t]$ with residue field $\mathbb{R}[t]/(t^2 + 1) \cong \mathbb{C}$. Always specify the base field before identifying closed points with k -rational points. The topological foundation is found in [Brenner, Proposition 8.5](#).

3. From functions to sheaf sections

A presheaf $\mathcal{F}$ assigns an object $\mathcal{F}(U)$ to every open set and restriction maps to smaller open sets. Identity restrictions must be identities, and successive restriction must equal direct restriction. A sheaf adds two conditions: sections that are locally equal are equal; local sections agreeing on every overlap have exactly one gluing. This is the content needed from [Brenner, Definition 4.1](#).

For $X = \text{Spec}(A)$, the structure sheaf $\mathcal{O}_X$ is determined on the open basis by

$$\Gamma(D(f), \mathcal{O}_X) = A_f, \quad \Gamma(X, \mathcal{O}_X) = A.$$

The denominator f may be inverted precisely where it does not belong to the prime ideal of the point. On a general open set, a section is represented by compatible local fractions; do not require one denominator to work on the entire open set. The basis formula and its restrictions hold for any commutative ring, including rings with zero divisors; see [Brenner, Lemma 9.12](#) and [Stacks, tag 01HV](#).

The term *function* remains useful, but *section* emphasises that the object belongs to a sheaf on an open set. On a general scheme, a section is not merely a list of values in one fixed field.

4. Stalks, local rings, residue fields, and fibres

The stalk $\mathcal{F}_{\mathfrak{p}}$ records germs of sections around $\mathfrak{p}$: two representatives agree if their restrictions agree on some smaller neighbourhood. For the structure sheaf,

$$\mathcal{O}_{X,\mathfrak{p}} = A_{\mathfrak{p}}, \quad \mathfrak{m}_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}, \quad \kappa(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \cong \text{Frac}(A/\mathfrak{p}).$$

The local ring records all germs, whereas the residue field records only values at the point. Evaluation is the composite $\mathcal{O}_X(U) \rightarrow \mathcal{O}_{X,\mathfrak{p}} \rightarrow \kappa(\mathfrak{p})$ for $\mathfrak{p} \in U$. The affine line illustrates the difference:

Point of $\text{Spec}(k[t])$	Local ring	Residue field
$\eta = (0)$	$k(t)$	$k(t)$
$(t - a), a \in k$	$k[t]_{(t-a)}$	k

At a , the germ $t - a$ is nonzero in the local ring, but its value in the residue field is zero. For a sheaf of modules $\mathcal{F}$, the fibre at the point is $\mathcal{F}_{\mathfrak{p}} \otimes_{\mathcal{O}_{X,\mathfrak{p}}} \kappa(\mathfrak{p})$, not the stalk itself. For example, the stalk of $\mathcal{O}_X$ at a is $k[t]_{(t-a)}$, while its fibre is k . This sheaf fibre must also be distinguished from the fibre of a scheme morphism. See [Brenner, Definition 3.22](#), [Definition 7.13](#), and [Lemma 9.10](#).

5. Why nilpotents must not be silently removed

Take any field k and $B = k[\varepsilon]/(\varepsilon^2)$. Every prime ideal contains ε , so $\text{Spec}(B)$ has only the point (ε) . Its topological space is the same as the one-point space $\text{Spec}(k)$, but their global section rings differ: they are B and k , respectively. The section ε is nonzero, although its value in the only residue field is zero. Thus even all point values need not determine a section.

A ring is *reduced* if it has no nonzero nilpotents; a scheme is reduced if all its local rings are reduced. The scheme $\text{Spec}(B)$ above is nonreduced because B itself is local and $\varepsilon \neq 0$ in it. Replacing a ring A by $A_{\text{red}} = A/\sqrt{(0)}$ preserves the topological space of its spectrum: all prime ideals contain $\sqrt{(0)}$, and the prime-ideal correspondence for the quotient preserves closed sets. Its structure sheaf can nevertheless change, as the example B shows. This is the information lost if geometry is remembered only as the zero set of equations.

6. Gluing objects and pulling back sections

A ringed space is a pair $(X, \mathcal{O}_X)$; it is *locally ringed* if every stalk of its structure sheaf is a local ring. A scheme is a space locally isomorphic to a spectrum with its structure sheaf. Here *isomorphic* concerns both space and sheaf, not merely a homeomorphism.

To construct a scheme from pieces X_i , specify open subsets $U_{ij} \subseteq X_i$ and isomorphisms $\varphi_{ij} : U_{ij} \rightarrow U_{ji}$. Besides identities and inverses, on every triple overlap the domains must agree and $\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$ must hold. Gluing points alone is insufficient: the structure sheaves must also be glued by the same isomorphisms. Every point then still has an affine neighbourhood. This local condition does not ensure that the entire glued space is affine. The source definition is [Brenner, Definition 10.1](#); the capstone that follows examines a global example.

A scheme morphism $f : X \rightarrow Y$ consists of a continuous map and a sheaf map

$$f^\# : \mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X, \quad (f_*\mathcal{O}_X)(V) = \mathcal{O}_X(f^{-1}(V)),$$

inducing at every x a local homomorphism $\mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$. Local means that the inverse image of the target maximal ideal is the source maximal ideal. This is not merely an extra condition on the point map; it connects that map with pullback of sections. Compare [Brenner, Definition 7.15](#).

In the affine case, $\alpha : A \rightarrow B$ gives a map in the opposite direction, $f : \text{Spec}(B) \rightarrow \text{Spec}(A)$, with $f(\mathfrak{q}) = \alpha^{-1}(\mathfrak{q})$. Indeed, $f^{-1}(D(a)) = D(\alpha(a))$, and the section map is $A_a \rightarrow B_{\alpha(a)}$. On stalks, the map $A_{\alpha^{-1}(\mathfrak{q})} \rightarrow B_{\mathfrak{q}}$ is local because a numerator belongs to the source prime ideal exactly when its image belongs to $\mathfrak{q}$. Existence and

uniqueness of this morphism are [Brenner, Corollary 10.10](#). For geometry over k , use k -algebra homomorphisms, not ring homomorphisms that forget the base structure.

A short orientation check with answers

Is (0) on the affine line closed? No: its closure is the whole line. Is t zero as a germ at (t) ? No; only its evaluation is zero. Must two schemes with one-point topological spaces be isomorphic? No: $\text{Spec}(k)$ and $\text{Spec}(k[\varepsilon]/(\varepsilon^2))$ have different section rings. Is it correct to replace the direction $A \rightarrow B$ with $\text{Spec}(A) \rightarrow \text{Spec}(B)$? No; the spectrum direction is reversed. These four answers test four main transitions; they do not add counted exercises to the mastery bank.

Provenance, rights, and source route

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BGK Unit 2 mastery bank

The problem statements below come from Holger Brenner's course. The worked solutions and checking notes were prepared independently by **OpenAI Codex gpt-5.6-sol, Ultra.**; they are not translations of Brenner's public solutions. The author and source-contributor credits remain applicable, including **Bocardodarapti** as the contributor to the frozen worksheet revision. This editorial material is licensed under **CC BY-SA 4.0** and implies neither endorsement by the author or source institutions nor human authorship or review.

This unit counts three mastery items: two new editorial solutions to Exercises 2.1 and 2.2, and the already translated [public source solution to Exercise 2.4](#). Solution 2.4 continues to count as a source solution and is neither copied nor labelled as new editorial work here. Neither selected exercise has a public solution in the frozen source map.

The prerequisites are the definitions of a real vector bundle and local trivialisation, and [Definition 2.1 on continuous sections](#). The base space is not assumed to be Hausdorff.

New item 1 - Nowhere-zero sections and triviality of line bundles

Source: [Exercise 2.1](#). Statement identifier: *Reelles Geradenbündel/Trivial/Nullstellenfreier Schnitt/Aufgabe*. Fixed witness: [revision 1048817](#), source page 111598. The exercise number is determined by frozen worksheet revision 602852; the revision of the transcluded statement is recorded separately above.

Source statement

Show that a real line bundle $L \rightarrow X$ over a topological space X is trivial if and only if it has a continuous section that is nowhere zero.

Complete editorial solution

Write the bundle projection as $p : L \rightarrow X$. Each fibre $L_x = p^{-1}(x)$ is a one-dimensional real vector space. The nowhere-zero condition means $s(x) \neq 0_x$ in each fibre, not avoidance of a single zero point common to the whole total space.

First direction. If the bundle is trivial, there is a bundle isomorphism

$$\tau : L \longrightarrow X \times \mathbb{R}$$

that preserves the base point and is linear on each fibre. Set $s(x) := \tau^{-1}(x, 1)$. This map is continuous as the composite of $x \mapsto (x, 1)$ and τ^{-1} . Since τ is over X , we have $p(s(x)) = x$, so s is a section. The linear isomorphism $\tau_x : L_x \rightarrow \mathbb{R}$ sends 0_x to 0, whereas $\tau_x(s(x)) = 1$. Thus the section is nowhere zero.

Conversely. Let s be a nowhere-zero continuous section. Define

$$\Phi : X \times \mathbb{R} \longrightarrow L, \quad \Phi(x, t) = t s(x).$$

Since $s(x)$ is a basis of the one-dimensional space L_x , the map $t \mapsto t s(x)$ is a linear isomorphism $\mathbb{R} \rightarrow L_x$. Thus Φ is bijective and linear on each fibre. We must still prove that Φ and its inverse are continuous; a continuous bijection alone is not enough to give a bundle isomorphism.

Take any local trivialisation

$$\tau_U : p^{-1}(U) \longrightarrow U \times \mathbb{R}$$

with $U \subseteq X$ open. In these coordinates, the section s has the form

$$\tau_U(s(x)) = (x, a(x)), \quad x \in U,$$

for a continuous function $a : U \rightarrow \mathbb{R}$. Its continuity follows from that of $\tau_U \circ s|_U$ and the second-coordinate projection. Since s is nowhere zero, $a(x) \neq 0$ throughout U . In these coordinates, Φ and its inverse are given by

$$(x, t) \mapsto (x, t a(x)), \quad (x, b) \mapsto \left(x, \frac{b}{a(x)}\right).$$

Both formulas are continuous: multiplication in $\mathbb{R}$ is continuous, and $x \mapsto 1/a(x)$ is continuous because a is nowhere zero. The sets $U \times \mathbb{R}$ form an open cover of the domain of Φ , while the sets $p^{-1}(U)$ form an open cover of the domain of its inverse. Continuity on an open cover gives global continuity in both directions.

Hence Φ is a bundle isomorphism $X \times \mathbb{R} \cong L$, and L is trivial. Both directions have been proved.

Pitfalls and checks

Choosing a nonzero vector separately in every fibre does not yet produce a **continuous** section. It is continuity that ensures the coordinate function a and the inverse trivialisation are continuous. Check the inverse formula in one chart using

$$\frac{t a(x)}{a(x)} = t, \quad \frac{b}{a(x)} a(x) = b.$$

Rank one is used precisely when a single nonzero vector is declared to be a basis. In higher rank, a single nowhere-zero section does not by itself trivialise the whole bundle.

New item 2 - The image of a section is a closed subspace

Source: [Exercise 2.2](#). Statement identifier: *Reelles Vektorbündel/Schnitt/Abgeschlossene Teilmenge/Aufgabe*. Fixed witness: [revision 1048838](#), source page 111631. The exercise number comes from frozen worksheet revision 602852.

Source statement

Let $s : X \rightarrow V$ be a continuous section of a real vector bundle $p : V \rightarrow X$ over a topological space X . Show that the image $s(X) \subseteq V$ is a closed subset homeomorphic to X .

Complete editorial solution

Homeomorphism with the base. If $s(x) = s(y)$, apply p and use $p \circ s = \text{Id}_X$ to obtain $x = y$. Thus the map

$$\bar{s} : X \longrightarrow s(X), \quad x \longmapsto s(x),$$

is bijective. It is continuous for the subspace topology on $s(X)$: if O is open in V , then $\bar{s}^{-1}(O \cap s(X)) = s^{-1}(O)$ is open in X . Its inverse is the continuous restriction

$$p|_{s(X)} : s(X) \longrightarrow X.$$

Indeed, $p(s(x)) = x$, and if $v = s(x) \in s(X)$, then $s(p(v)) = s(x) = v$. Hence $\bar{s}$ is a homeomorphism.

Closedness in the total space. Take a local trivialisation

$$\tau_U : p^{-1}(U) \longrightarrow U \times \mathbb{R}^r.$$

There is a continuous map $f : U \rightarrow \mathbb{R}^r$ with $\tau_U(s(x)) = (x, f(x))$. Moreover,

$$s(X) \cap p^{-1}(U) = s(U),$$

because $s(x) \in p^{-1}(U)$ exactly when $x = p(s(x)) \in U$. Therefore $\tau_U(s(U))$ is the graph

$$\text{Graph}(f) = \{(x, v) \in U \times \mathbb{R}^r \mid v = f(x)\}.$$

The map

$$h : U \times \mathbb{R}^r \longrightarrow \mathbb{R}^r, \quad h(x, v) = v - f(x)$$

is continuous. Since $\{0\}$ is closed in $\mathbb{R}^r$, this graph is the closed subset

$$\text{Graph}(f) = h^{-1}(\{0\})$$

of $U \times \mathbb{R}^r$. Via the homeomorphism τ_U , it follows that $s(X) \cap p^{-1}(U)$ is closed in $p^{-1}(U)$.

The sets $p^{-1}(U)$ from all the trivialisations form an open cover of V . On each such set,

$$(V \setminus s(X)) \cap p^{-1}(U)$$

is open in $p^{-1}(U)$, hence also open in V . Their union is $V \setminus s(X)$. Thus the complement of $s(X)$ is open, and $s(X)$ is closed in V .

Pitfalls and checks

Do not infer closedness from $p \circ s = \text{Id}_X$ alone. For general continuous maps, this identity gives a topological embedding, but does not by itself give a closed image. Here the proof uses the local model $U \times \mathbb{R}^r$ and the closedness of $\{0\}$ in a real vector space. The base space X need not be Hausdorff.

As a check, if s is the zero section, then $f = 0$ and $h(x, v) = v$. The local statement becomes the closedness of $U \times \{0\}$, exactly as expected.

BGK Unit 3 mastery bank

The problem statements come from Holger Brenner's course. The following worked solutions and checking notes were prepared independently by **OpenAI Codex gpt-5.6-sol, Ultra.**; they are not translations of public source solutions. The source credits remain applicable, including **Bocardodarapti** as the contributor to the frozen worksheet revision. This editorial material is licensed under **CC BY-SA 4.0** and implies neither endorsement by the author or source institutions nor human authorship or review.

The three mastery items for Unit 3 comprise the two new editorial solutions here to Exercises 3.3 and 3.16, together with the already translated [public source solution to Exercise 3.1](#). Solution 3.1 continues to count as a source solution and is not repeated here. The frozen source map states that Exercises 3.3 and 3.16 have no public solutions.

The prerequisites used are the gluing data for the Möbius strip from Unit 2, the definition of the tensor product of bundles, and the definitions of a presheaf and a stalk in terms of germ classes. Neither the sheaf property nor cohomology is required.

New item 1 - The tensor square of the Möbius strip

Source: [Exercise 3.3](#). Statement identifier: *Möbiusband/Tensorprodukt/Trivial/Aufgabe*. Fixed witness: [revision 846097](#), source page 111727. The exercise number is determined by frozen worksheet revision 619301.

Source statement

Show that the tensor product of the Möbius strip with itself is a trivial line bundle.

Complete editorial solution

Write $L \rightarrow S^1$ for the real line bundle of the Möbius strip, with

$$S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}.$$

Use the cover and gluing data from [Example 2.11](#):

$$U = S^1 \setminus \{(0, 1)\}, \quad V = S^1 \setminus \{(0, -1)\}.$$

On $U \cap V$ we have $x \neq 0$. If t_U and t_V are the fibre coordinates in the two trivialisations, the transition convention is

$$t_V = \varepsilon(x, y)t_U, \quad \varepsilon(x, y) = \begin{cases} 1, & x > 0, \\ -1, & x < 0. \end{cases}$$

The function ε is continuous on the intersection, since its two components are open and the formula is constant on each component.

By [Definition 3.2](#), the transition map of $L \otimes L$ is obtained by tensoring the two transition maps. On a fibre, the map on pure tensors is

$$a \otimes b \mapsto (\varepsilon a) \otimes (\varepsilon b) = \varepsilon^2(a \otimes b) = a \otimes b.$$

Pure tensors span the tensor product, so this map is the identity on all of $\mathbb{R} \otimes_{\mathbb{R}} \mathbb{R}$. Under the linear identification $a \otimes b \mapsto ab$, the fibre is $\mathbb{R}$, and the tensor-bundle transition is always 1, on both components of the intersection.

Here is the resulting global trivialisation. Let $e_U(p)$ and $e_V(p)$ be the local basis vectors of L_p with coordinate 1 in their respective charts. Since coordinates change according to the formula above, on the intersection we have

$$e_U(p) = \varepsilon(p)e_V(p).$$

Hence

$$e_U(p) \otimes e_U(p) = \varepsilon(p)^2 e_V(p) \otimes e_V(p) = e_V(p) \otimes e_V(p).$$

Thus the following two rules define a single well-defined map:

$$\begin{aligned}\Phi : L \otimes L &\longrightarrow S^1 \times \mathbb{R}, \\ a e_U(p) \otimes e_U(p) &\longmapsto (p, a) \quad (p \in U), \\ a e_V(p) \otimes e_V(p) &\longmapsto (p, a) \quad (p \in V).\end{aligned}$$

In each chart, Φ is a trivialisation, so it is continuous and a linear bijection on every fibre. Its inverse is given by the two local formulas

$$(p, a) \longmapsto a e_U(p) \otimes e_U(p), \quad (p, a) \longmapsto a e_V(p) \otimes e_V(p),$$

which are continuous and agree on the intersection. Local continuity on an open cover gives global continuity in both directions. Thus Φ is an isomorphism of line bundles, and

$$L \otimes L \cong S^1 \times \mathbb{R}.$$

Pitfalls and checks

The tensor product is not the direct sum. The rank of $L \otimes L$ is $1 \cdot 1 = 1$, whereas the rank of $L \oplus L$ is $1 + 1 = 2$. The calculation $(-1)(-1) = 1$ here is a calculation of **tensor-product transitions**, not a new gluing instruction for a picture of the strip.

As a check, the local sections $e_U \otimes e_U$ and $e_V \otimes e_V$ agree on the intersection and are nonzero in every fibre. Together they give a nowhere-zero global continuous section. New item 1 for Unit 2 provides a second check that this line bundle is trivial.

New item 2 - Stalks of the product of two presheaves

Source: [Exercise 3.16](#). Statement identifier: *Prägarbe/Produkt/Halm/Aufgabe*. Fixed witness: [revision 1083990](#), source page 111822. The exercise number comes from frozen worksheet revision 619301.

Source statement

Let $\mathcal{F}$ and $\mathcal{G}$ be presheaves on a topological space X , and let $\mathcal{F} \times \mathcal{G}$ be their product presheaf. Show that for every point $P \in X$,

$$(\mathcal{F} \times \mathcal{G})_P = \mathcal{F}_P \times \mathcal{G}_P.$$

Complete editorial solution

We prove the source equality by constructing a canonical bijection, meaning a bijection independent of the choice of neighbourhoods or germ representatives. There is no hypothesis that $\mathcal{F}$ or $\mathcal{G}$ is a sheaf; it is enough that both are presheaves of sets.

On an open set U , the value of the product presheaf is $\mathcal{F}(U) \times \mathcal{G}(U)$, and restriction is componentwise. For $V \subseteq U$, the formula is

$$(s, t)|_V = (s|_V, t|_V).$$

The identity and composition properties of restrictions follow from those of the two presheaves. Thus the product object is indeed a presheaf.

By [Definition 3.21](#) and [Definition 3.22](#), a germ $s_P \in \mathcal{F}_P$ is represented by a section $s \in \mathcal{F}(U)$ on an open neighbourhood U of P . Representatives $s \in \mathcal{F}(U)$ and $s' \in \mathcal{F}(U')$ give the same germ exactly when there is an open neighbourhood W with $P \in W \subseteq U \cap U'$ and $s|_W = s'|_W$.

The canonical map. Define

$$\Theta : (\mathcal{F} \times \mathcal{G})_P \longrightarrow \mathcal{F}_P \times \mathcal{G}_P, \quad (s, t)_P \longmapsto (s_P, t_P).$$

If the pairs (s, t) and (s', t') have the same germ, they agree after restriction to some common neighbourhood. Their components also agree there, so $s_P = s'_P$ and $t_P = t'_P$. Hence Θ is well-defined.

Surjectivity. Take any $(a, b) \in \mathcal{F}_P \times \mathcal{G}_P$. Choose a representative $s \in \mathcal{F}(U)$ of a and $t \in \mathcal{G}(V)$ of b , with $P \in U$ and $P \in V$. They may not yet have the same domain. Since $U \cap V$ is still an open neighbourhood of P , we have a pair

$$(s|_{U \cap V}, t|_{U \cap V}) \in (\mathcal{F} \times \mathcal{G})(U \cap V).$$

The germ of this pair is sent by Θ to (a, b) : restriction to a smaller neighbourhood does not change the germ. Thus Θ is surjective.

Injectivity. Suppose $(s, t) \in (\mathcal{F} \times \mathcal{G})(U)$ and $(s', t') \in (\mathcal{F} \times \mathcal{G})(U')$ have the same image under Θ . This means

$$s_P = s'_P, \quad t_P = t'_P.$$

The first equality gives an open neighbourhood $P \in W_{\mathcal{F}} \subseteq U \cap U'$ with $s|_{W_{\mathcal{F}}} = s'|_{W_{\mathcal{F}}}$. The second equality gives an open neighbourhood $P \in W_{\mathcal{G}} \subseteq U \cap U'$ with $t|_{W_{\mathcal{G}}} = t'|_{W_{\mathcal{G}}}$. Set

$$W := W_{\mathcal{F}} \cap W_{\mathcal{G}}.$$

The set W is still an open neighbourhood of P . Restricting once more and using composition of restrictions, both equalities hold simultaneously on W . Hence

$$(s, t)|_W = (s|_W, t|_W) = (s'|_W, t'|_W) = (s', t')|_W.$$

Thus $(s, t)_P = (s', t')_P$, and Θ is injective. Together with surjectivity, this proves the required canonical bijection.

Explicitly, its inverse sends the pair of germs (s_P, t_P) to the germ $(s|_{U \cap V}, t|_{U \cap V})_P$. The injectivity just proved ensures that the result is independent of the chosen representatives. Thus all choices in the construction of the inverse have been checked.

Pitfalls and checks

Do not pair two representatives before making their domains agree. A section on U and a section on V yield a section of the product only after both have been restricted to $U \cap V$.

The decisive step is taking a **finite intersection** of neighbourhoods. The intersection of two open neighbourhoods is still open and contains P . The same proof does not automatically apply to infinite products, because an infinite intersection of open neighbourhoods need not be open. To check the inverse, starting with $(s, t)_P$ on one neighbourhood U gives back $(s|_U, t|_U)_P = (s, t)_P$; starting with (s_P, t_P) gives back the two original germs.

BGK 4 mastery bank: gluing and stalks

The following problem statements come from Holger Brenner's course. The complete solutions, explanations, and brief checks are new editorial material prepared by **OpenAI Codex gpt-5.6-sol, Ultra.**, not translated source solutions. In the Unit 4 freeze, there are no public source solutions for these three selected exercises. Credits to the author and source contributors remain applicable; this text does not claim human authorship or review. The problem statements are unchanged. This material is licensed under **CC BY-SA 4.0**.

The main prerequisite is [Definition 4.1 on sheaves](#): sections that agree locally are equal, and a family of sections agreeing on intersections has a unique gluing. For a sheaf of sets, $\mathcal{F}(\emptyset)$ has exactly one element; for a sheaf of groups, this value is the trivial group. Both include the empty cover in the sheaf axiom.

New solution 1 - The product of two sheaves, Exercise 4.1

Source: [Exercise 4.1 in the reader](#), identifier Garbe/Produkt/Aufgabe, page 111820, [fixed revision 1082920](#). The exercise number follows Worksheet 4, revision 1003857; the identity of the transcluded exercise page is recorded separately in the Unit 4 manifest.

Brenner's problem statement. Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on a topological space X . Prove that the assignment

$$\mathcal{H}(U) := \mathcal{F}(U) \times \mathcal{G}(U)$$

with the natural product maps as restrictions defines a sheaf on X .

Independent editorial solution. For open sets $V \subseteq U$, define

$$\rho_{U,V}^{\mathcal{H}}(s, t) := (\rho_{U,V}^{\mathcal{F}}(s), \rho_{U,V}^{\mathcal{G}}(t)).$$

Restriction from U to itself is the identity. If $W \subseteq V \subseteq U$, then for every pair (s, t) ,

$$\rho_{V,W}^{\mathcal{H}} \rho_{U,V}^{\mathcal{H}}(s, t) = (s|_W, t|_W) = \rho_{U,W}^{\mathcal{H}}(s, t).$$

Thus $\mathcal{H}$ is, to begin with, a presheaf.

Take an open cover $U = \bigcup_{i \in I} U_i$. If two pairs $(s, t), (s', t') \in \mathcal{H}(U)$ have equal restrictions to every U_i , equality of pairs means

$$s|_{U_i} = s'|_{U_i}, \quad t|_{U_i} = t'|_{U_i}$$

for every i . The local equality axiom for $\mathcal{F}$ gives $s = s'$, and the same axiom for $\mathcal{G}$ gives $t = t'$. Hence the pairs are equal.

Now take a compatible family $(s_i, t_i) \in \mathcal{H}(U_i)$. Compatibility on $U_i \cap U_j$ means precisely the two systems of equalities

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}, \quad t_i|_{U_i \cap U_j} = t_j|_{U_i \cap U_j}.$$

The sheaf property of $\mathcal{F}$ gives a unique $s \in \mathcal{F}(U)$ restricting to all the s_i . Likewise, there is a unique $t \in \mathcal{G}(U)$ restricting to all the t_i . The pair $(s, t) \in \mathcal{H}(U)$ glues the original family and is unique by the local equality property just proved.

Finally, $\mathcal{H}(\emptyset)$ is a product of two singleton sets and is therefore a singleton. Thus the empty cover also satisfies the axiom. Both sheaf conditions hold for every cover, and $\mathcal{H}$ is a sheaf.

Pitfall and check. Here the product is taken on each open set, with componentwise restrictions. Specifying the sets $\mathcal{F}(U) \times \mathcal{G}(U)$ alone does not specify a presheaf. As a check, the two projections $(s, t) \mapsto s$ and $(s, t) \mapsto t$ commute with restrictions precisely because of the definition above.

New solution 2 - Sections on two disjoint pieces, Exercise 4.2

Source: [Exercise 4.2 in the reader](#), identifier Garbe/UnzusammenhängenderRaum/Produkt/Aufgabe, page 111900, [fixed revision 1082921](#). The exercise number follows Worksheet 4, revision 1003857.

Brenner's problem statement. Let $\mathcal{G}$ be a sheaf on a disconnected topological space decomposed as

$$X = U \sqcup V,$$

where U, V are open, nonempty, and disjoint. Show that

$$\mathcal{G}(X) = \mathcal{G}(U) \times \mathcal{G}(V).$$

Independent editorial solution. The equality in the statement is a canonical identification through restriction. The map to be proved bijective is

$$R : \mathcal{G}(X) \longrightarrow \mathcal{G}(U) \times \mathcal{G}(V), \quad s \longmapsto (s|_U, s|_V).$$

If $R(s) = R(t)$, then $s|_U = t|_U$ and $s|_V = t|_V$. Since U, V cover X , the local equality axiom gives $s = t$. Thus R is injective.

For surjectivity, choose any pair $(a, b) \in \mathcal{G}(U) \times \mathcal{G}(V)$. The only intersection on which a comparison is needed is $U \cap V = \emptyset$. The two restrictions

$$a|_{\emptyset}, b|_{\emptyset} \in \mathcal{G}(\emptyset)$$

are automatically equal, since $\mathcal{G}(\emptyset)$ is a singleton. Thus (a, b) is a compatible family on the cover $\{U, V\}$. The gluing axiom gives a section $s \in \mathcal{G}(X)$ with $s|_U = a$ and $s|_V = b$. Therefore $R(s) = (a, b)$, so R is surjective.

The inverse of R sends (a, b) to its unique gluing. No additional choice enters this construction; that is why the identification is canonical. If $\mathcal{G}$ is a sheaf of groups or rings, restrictions are homomorphisms, so R is also an isomorphism for that algebraic structure, not merely a bijection of sets.

Pitfall and check. For a cover whose members are not disjoint, not every pair of sections can be glued. The permissible pairs must satisfy $a|_{U \cap V} = b|_{U \cap V}$. In this exercise the condition is automatic because the intersection is empty, not because gluing ignores compatibility. Nor does the statement literally identify sections with pairs before the map R has been specified.

New solution 3 - The skyscraper sheaf, Exercise 4.9

Source: [Exercise 4.9 in the reader](#), identifier Wolkenkratzergarbe/Gruppe/Garbeneigenschaft/Aufgabe, page 1398 88, [fixed revision 1081969](#). The exercise number follows Worksheet 4, revision 1003857. The German typo in the source's final instruction does not change the mathematical statement.

Brenner's problem statement. Let X be a topological space, $P \in X$, and G a commutative group. For each open set $U \subseteq X$, set

$$\mathcal{G}(U) = \begin{cases} G, & P \in U, \\ 0, & P \notin U. \end{cases}$$

With the natural restrictions, prove that $\mathcal{G}$ is a sheaf of commutative groups, determine $\mathcal{G}_P$, and, if P is closed, determine $\mathcal{G}_Q$ for every $Q \neq P$.

Independent editorial solution. For $V \subseteq U$, the restriction $\rho_{U,V}$ is the identity $G \rightarrow G$ if $P \in V$; the unique homomorphism $G \rightarrow 0$ if $P \in U$ but $P \notin V$; and the identity of the trivial group $0 \rightarrow 0$ if $P \notin U$. The case $P \notin U$ but $P \in V$ cannot occur. These three cases immediately give the identity restriction and composition laws: once a restriction is zero, every subsequent restriction remains zero. Thus $\mathcal{G}$ is a presheaf of commutative groups.

Take an open cover $U = \bigcup_{i \in I} U_i$. If $P \notin U$, all the groups $\mathcal{G}(U)$, $\mathcal{G}(U_i)$, and those on intersections are trivial. There is only one possible compatible family, and its gluing is unique. This also includes $U = \emptyset$.

If $P \in U$, there is at least one index i_0 with $P \in U_{i_0}$. Take a compatible family $g_i \in \mathcal{G}(U_i)$. For every i with $P \in U_i$, the point P also belongs to $U_i \cap U_{i_0}$. Both restrictions to this intersection are identities on G , so compatibility gives

$$g_i = g_{i_0} \quad \text{in } G.$$

For indices with $P \notin U_i$, the section g_i must be zero. Thus $g := g_{i_0} \in G = \mathcal{G}(U)$ restricts to every g_i : via the identity on pieces containing P , and via $G \rightarrow 0$ on the others. This gluing is unique, because restriction to U_{i_0} is the identity. The same uniqueness argument shows that two sections over U agreeing locally are equal. Hence $\mathcal{G}$ is a sheaf of commutative groups.

For the stalk at P , every open neighbourhood W of P has $\mathcal{G}(W) = G$, and all restrictions between such neighbourhoods are identities. Concretely, a germ represented by $g \in G$ on a neighbourhood is determined only by g : two representatives give the same germ exactly when those elements of G agree. Hence

$$\mathcal{G}_P \cong G.$$

Finally, suppose P is closed and $Q \neq P$. The set $X \setminus \{P\}$ is open and contains Q . Every germ at Q can be represented by a section on an open neighbourhood W of Q . After restriction to the smaller neighbourhood

$$W \cap (X \setminus \{P\}),$$

the section lies in the zero group. Thus every germ at Q is zero, and

$$\mathcal{G}_Q = 0 \quad (Q \neq P, P \text{ closed}).$$

Pitfall and check. The hypothesis that P is closed is needed in the last step, not in constructing the sheaf. If every neighbourhood of Q instead contains P , all the groups used to form the stalk at Q are G with identity restrictions, so that stalk is also isomorphic to G . Do not assume that every point of a topological space is closed. As another check, when $G = 0$, the entire sheaf and all its stalks are indeed zero.

BGK 5 mastery bank: sheafification and quotient stalks

The following problem statements come from Holger Brenner's course. The complete solutions, explanations, and brief checks are new editorial material prepared by **OpenAI Codex gpt-5.6-sol, Ultra.**, not translated source solutions. Credits to the author and source contributors remain applicable; this text does not claim human authorship or review. No endorsement by the author or source institutions is implied. This material is licensed under **CC BY-SA 4.0**.

Unit 5 already has a [public source solution to Exercise 5.5](#). That solution continues to count as a source solution and is not recreated here. The two new solutions below, to Exercises 5.2 and 5.11, complete the three mastery items for Unit 5. Both problem statements are preserved.

The notation s_P denotes the germ of a section s at P , while $\mathcal{F}_P$ is the stalk of the presheaf or sheaf $\mathcal{F}$. Equality of two germs means that their representatives agree after restriction to a sufficiently small open neighbourhood; they need not already agree on their original neighbourhoods.

New solution 1 - The universal property of sheafification, Exercise 5.2

Source: [Exercise 5.2 in the reader](#), identifier Prägarbe/Vergarbung/UniverselleEigenschaft/Aufgabe, page 11 1906, [fixed revision 1083991](#). The exercise number follows Worksheet 5, revision 619386; the revision of the transcluded exercise page is recorded separately in the Unit 5 manifest.

Brenner's problem statement. Let $\mathcal{F}$ be a presheaf on a topological space X , and let $\widetilde{\mathcal{F}}$ be its sheafification. For every presheaf morphism $\psi : \mathcal{F} \rightarrow \mathcal{G}$ to a sheaf $\mathcal{G}$, prove that there is exactly one morphism

$$\widetilde{\psi} : \widetilde{\mathcal{F}} \rightarrow \mathcal{G}$$

factoring ψ through the canonical morphism $\eta : \mathcal{F} \rightarrow \widetilde{\mathcal{F}}$. That is, $\widetilde{\psi} \circ \eta = \psi$.

Independent editorial solution. We construct the component $\widetilde{\psi}_U$ for each open set U , then prove that all components commute with restrictions. Use [Definition 5.1](#): a section $\sigma \in \widetilde{\mathcal{F}}(U)$ is a family of germs $(\sigma_P)_{P \in U}$ locally arising from sections of $\mathcal{F}$. Thus there is an open cover $U = \bigcup_i U_i$ and $s_i \in \mathcal{F}(U_i)$ with

$$\sigma|_{U_i} = \eta_{U_i}(s_i), \quad (s_i)_Q = \sigma_Q \quad (Q \in U_i).$$

We want to glue the sections

$$t_i := \psi_{U_i}(s_i) \in \mathcal{G}(U_i).$$

To check compatibility, take $Q \in U_i \cap U_j$. The germs $(s_i)_Q$ and $(s_j)_Q$ agree. Since ψ commutes with restrictions, it induces a stalk map ψ_Q , so

$$(t_i)_Q = \psi_Q((s_i)_Q) = \psi_Q((s_j)_Q) = (t_j)_Q.$$

By [Lemma 4.4, the stalkwise test for equality of sections](#), applied to the sheaf $\mathcal{G}$ over $U_i \cap U_j$, we obtain

$$t_i|_{U_i \cap U_j} = t_j|_{U_i \cap U_j}.$$

Since $\mathcal{G}$ is a sheaf, the family (t_i) has a unique gluing $t \in \mathcal{G}(U)$. Define $\tilde{\psi}_U(\sigma) := t$.

This construction is independent of the cover or local representatives. Indeed, suppose (V_j, r_j) is another choice of representatives for σ . At each point $Q \in U_i \cap V_j$, we have $(s_i)_Q = (r_j)_Q = \sigma_Q$. The same stalk argument gives

$$\psi_{U_i}(s_i)|_{U_i \cap V_j} = \psi_{V_j}(r_j)|_{U_i \cap V_j}.$$

The two glued sections have equal restrictions on the refinement cover $\{U_i \cap V_j\}_{i,j}$ of U . Local uniqueness in $\mathcal{G}$ shows that the results are equal. Thus $\tilde{\psi}_U$ is well-defined.

Now take an open set $W \subseteq U$. The restriction $\sigma|_W$ is represented by $s_i|_{U_i \cap W}$ on the cover $\{U_i \cap W\}_i$. Since ψ is a presheaf morphism,

$$\psi_{U_i \cap W}(s_i|_{U_i \cap W}) = \psi_{U_i}(s_i)|_{U_i \cap W}.$$

Thus $\tilde{\psi}_W(\sigma|_W)$ and $\tilde{\psi}_U(\sigma)|_W$ have equal local restrictions. Uniqueness of gluing gives

$$\tilde{\psi}_W(\sigma|_W) = \tilde{\psi}_U(\sigma)|_W.$$

Hence this family of components is a sheaf morphism.

To check the factorisation, take $s \in \mathcal{F}(U)$. The section $\eta_U(s)$ has representative s on all of U , so the construction with the one-member cover gives

$$\tilde{\psi}_U(\eta_U(s)) = \psi_U(s).$$

Thus $\tilde{\psi} \circ \eta = \psi$.

For uniqueness, suppose $\theta : \tilde{\mathcal{F}} \rightarrow \mathcal{G}$ is another morphism with $\theta \circ \eta = \psi$. For a section σ and the local representatives s_i above, naturality of θ gives

$$\theta_U(\sigma)|_{U_i} = \theta_{U_i}(\eta_{U_i}(s_i)) = \psi_{U_i}(s_i) = \tilde{\psi}_U(\sigma)|_{U_i}.$$

The sections are equal by the sheaf property of $\mathcal{G}$. This holds for all U and σ , so $\theta = \tilde{\psi}$. For $U = \emptyset$, both sheaves take singleton values, so the component and all its identities are also unique. The complete proof does not assume that $\mathcal{F}$ is already a sheaf.

Pitfall and check. Do not define $\tilde{\psi}_U$ only on sections coming from $\eta_U(\mathcal{F}(U))$: η_U need not be surjective. Representatives are available locally, and it is the sheaf property of the *target* $\mathcal{G}$ that permits gluing. If $\mathcal{F}$ is already a sheaf, η is an isomorphism by [Lemma 5.2\(4\)](#); the formula above then gives $\tilde{\psi} = \psi \circ \eta^{-1}$, as expected.

New solution 2 - Stalks of a quotient sheaf, Exercise 5.11

Source: [Exercise 5.11 in the reader](#), identifier GarbenvonGruppen/Untergarbe/Quotientengarbe/Halm/Aufgabe, page 112024, [fixed revision 1082924](#). The exercise number follows Worksheet 5, revision 619386.

Brenner's problem statement. Let $\mathcal{G}$ be a sheaf of commutative groups, $\mathcal{F} \subseteq \mathcal{G}$ a subsheaf of groups, and $\mathcal{G}/\mathcal{F}$ its quotient sheaf. Prove that, for every point $P \in X$,

$$(\mathcal{G}/\mathcal{F})_P = \mathcal{G}_P/\mathcal{F}_P.$$

Independent editorial solution. This equality means a canonical group isomorphism. We must not replace the quotient sheaf by the quotient of sections on each open set. Following [Definition 5.8](#), first form the presheaf of groups

$$\mathcal{Q}(U) := \mathcal{G}(U)/\mathcal{F}(U), \quad \rho_{U,V}^{\mathcal{Q}}([g]) := [g|_V].$$

These restrictions are well-defined: if $g - h \in \mathcal{F}(U)$, then $g|_V - h|_V \in \mathcal{F}(V)$ because $\mathcal{F}$ is a subsheaf. The quotient sheaf is $\widetilde{\mathcal{Q}}$. By [Lemma 5.2\(2\)](#), which applies to every presheaf, the canonical map induces an isomorphism

$$\mathcal{Q}_P \xrightarrow{\sim} \widetilde{\mathcal{Q}}_P = (\mathcal{G}/\mathcal{F})_P.$$

It therefore suffices to determine $\mathcal{Q}_P$.

The quotient homomorphisms on each open set form a presheaf morphism $q : \mathcal{G} \rightarrow \mathcal{Q}$. On stalks, it gives

$$q_P : \mathcal{G}_P \longrightarrow \mathcal{Q}_P, \quad g_P \longmapsto [g]_P.$$

If two sections $g \in \mathcal{G}(U)$ and $h \in \mathcal{G}(V)$ represent the same germ at P , they agree on some open neighbourhood $P \in W \subseteq U \cap V$. Their quotient classes then also agree in $\mathcal{Q}(W)$, so the formula for q_P is independent of the representative. Addition of germs is computed after shrinking to a common neighbourhood; since every q_U is a homomorphism, so is q_P .

The map q_P is surjective. Every element $\xi \in \mathcal{Q}_P$ has a representative $a \in \mathcal{Q}(U)$ on some open neighbourhood U of P . By the definition of a quotient group, there is $g \in \mathcal{G}(U)$ with $a = [g]$. Thus $\xi = [g]_P = q_P(g_P)$.

Next, the inclusion $\mathcal{F} \subseteq \mathcal{G}$ induces a stalk inclusion $\mathcal{F}_P \subseteq \mathcal{G}_P$, by [Lemma 4.5](#). We show that this subgroup is exactly the kernel of q_P .

If $g_P \in \ker q_P$, then $[g]_P = 0$ in $\mathcal{Q}_P$. By the definition of equality of germs, there is an open neighbourhood W of P , contained in the domain of the representative g , such that

$$[g|_W] = 0 \quad \text{in } \mathcal{G}(W)/\mathcal{F}(W).$$

This means precisely that $g|_W \in \mathcal{F}(W)$, so the original germ g_P belongs to $\mathcal{F}_P$. Conversely, every element of $\mathcal{F}_P$ has a representative $f \in \mathcal{F}(W)$. The image of f in $\mathcal{Q}(W)$ is zero, so $q_P(f_P) = 0$. Thus

$$\ker q_P = \mathcal{F}_P.$$

The first isomorphism theorem for commutative groups now gives the explicit isomorphism

$$\mathcal{G}_P/\mathcal{F}_P \xrightarrow{\sim} \mathcal{Q}_P, \quad g_P + \mathcal{F}_P \longmapsto [g]_P.$$

Composing it with the sheafification isomorphism yields

$$\mathcal{G}_P/\mathcal{F}_P \xrightarrow{\sim} (\mathcal{G}/\mathcal{F})_P,$$

giving the required canonical identification. The construction uses a representative on a sufficiently small neighbourhood; it makes no claim that a global representative is always available.

Pitfall and check. The germ equality $[g]_P = 0$ means that there is a neighbourhood W with $g|_W \in \mathcal{F}(W)$; it does not immediately mean that $g \in \mathcal{F}(U)$ on its original domain. Sheafification can change global sections but not stalks. As boundary checks, if $\mathcal{F} = 0$, the formula gives the stalk $\mathcal{G}_P$; if $\mathcal{F} = \mathcal{G}$, both sides are the zero group.

BGK 6 mastery exercises

The following three problem statements come from Holger Brenner's course on Wikiversity. The complete solutions and learning checks are new editorial material prepared independently by **OpenAI Codex gpt-5.6-sol, Ultra**. These are not translated public solutions by Brenner: the frozen source map records the absence of public solutions for all three exercises. Their mathematical statements and hypotheses are unchanged.

This text is licensed under **CC BY-SA 4.0**, with source credits preserved. No claim of human authorship, endorsement, or review is made for these editorial solutions. The learning sequence moves from sections on open sets to stalks and then to sheaf morphisms.

New item 1 - Brenner Exercise 6.4: a split sequence

Source: [Exercise 6.4 in the BGK reader](#). Frozen entity: TopologischeGruppen/SpaltendeSequenz/Garbenversion/Aufgabe, pageid 112025, [revision 1050308](#). The exercise number follows [Worksheet 6, revision 900086](#).

Source problem statement

Let F and H be commutative topological groups, $G = F \times H$ with the product topology, and

$$0 \longrightarrow F \longrightarrow G \longrightarrow H \longrightarrow 0$$

the associated product short exact sequence. Prove that, for every topological space X , there is a short exact sequence of sheaves

$$0 \longrightarrow C^0(-, F) \longrightarrow C^0(-, G) \longrightarrow C^0(-, H) \longrightarrow 0,$$

and that the rightmost map remains surjective after global evaluation on X .

Complete independent solution

Use additive notation for all three groups. The inclusion and projection in question are $j(a) = (a, 0_H)$ and $p(a, b) = b$. For each open $U \subseteq X$, define

$$\begin{aligned} j_U : C^0(U, F) &\longrightarrow C^0(U, G), & j_U(a)(x) &= (a(x), 0_H), \\ p_U : C^0(U, G) &\longrightarrow C^0(U, H), & p_U(b)(x) &= p(b(x)). \end{aligned}$$

Both maps are group homomorphisms. They also produce continuous functions, since the inclusion j and projection p are continuous. Composition with a fixed function commutes with restriction to an open set, so the families j_U, p_U are presheaf morphisms.

The presheaf $C^0(-, F)$ is a sheaf: local functions agreeing on intersections determine exactly one function on their union, and that function is continuous because continuity can be tested on an open cover. The same argument applies to G and H . Thus these are indeed sheaf morphisms.

Now check exactness for every U . If $j_U(a) = 0$, then $(a(x), 0_H) = (0_F, 0_H)$ for every $x \in U$, so $a = 0$; hence j_U is injective. The composite $p_U j_U$ is zero. Conversely, if $p_U(b) = 0$, each $b(x)$ has the form $(a(x), 0_H)$. The function $a = \text{pr}_F \circ b$ is continuous and $j_U(a) = b$. Thus

$$\ker p_U = \text{im } j_U.$$

For surjectivity, take any $h \in C^0(U, H)$ and set

$$\sigma_U(h)(x) := (0_F, h(x)).$$

This function is continuous, and $p_U \sigma_U(h) = h$. The family σ_U itself is compatible with restrictions. Thus σ is a sheaf morphism with $p\sigma = \text{Id}$: the sheaf sequence even splits. Every section in the kernel comes from a section on the left, and every section on the right has a preimage on the same open set; this proves sheaf exactness, not just a formal statement about a complex.

In particular, for $U = X$, the formula $h \mapsto (x \mapsto (0_F, h(x)))$ gives a right inverse on global sections. Hence

$$C^0(X, G) \longrightarrow C^0(X, H)$$

is surjective, as required. The entire argument also applies to $U = \emptyset$, when the group of maps has just one element.

Pitfall and check

In general, surjectivity of a sheaf morphism guarantees only local preimages, not global ones. Here the decisive extra step is the existence of a **single global right inverse** $H \rightarrow F \times H$, $h \mapsto (0_F, h)$. Check directly that $\sigma_U(h)|_V = \sigma_V(h|_V)$ for $V \subseteq U$; this is why the splittings on all open sets constitute a sheaf splitting.

New item 2 - Brenner Exercise 6.9: pushforward from a point

Source: [Exercise 6.9 in the BGK reader](#). Frozen entity: TopologischerRaum/Punkt/Vorschub/Wolkenkratzergerbe/Aufgabe, pageid 112028, [revision 1084500](#). The exercise number follows Worksheet 6, revision 900086.

Source problem statement

Let X be a topological space, $P \in X$, and $i : \{P\} \rightarrow X$ the inclusion. For a sheaf of commutative groups $\mathcal{F}$ on $\{P\}$, describe $i_*\mathcal{F}$ on the open sets of X . Determine its stalks when P is a closed point.

Complete independent solution

Write $A := \mathcal{F}(\{P\})$. Since $\mathcal{F}$ is a sheaf of groups, $\mathcal{F}(\emptyset) = 0$: the gluing condition for the empty cover gives exactly one section on the empty set. These two open sets give all the sheaf data on a one-point space.

By [Definition 6.9](#), for open $U \subseteq X$,

$$(i_*\mathcal{F})(U) = \mathcal{F}(i^{-1}(U)) = \begin{cases} A, & P \in U, \\ 0, & P \notin U. \end{cases}$$

If $V \subseteq U$, there are three possibilities. If $P \in V$, both section groups are A and the restriction is the identity $A \rightarrow A$. If $P \in U$ but $P \notin V$, the restriction is the unique homomorphism $A \rightarrow 0$. If $P \notin U$, the restriction is $0 \rightarrow 0$. The possibility $P \notin U$ but $P \in V$ cannot occur. Thus the entire restriction structure is determined. This pushforward is a sheaf by [Lemma 6.10](#), which applies to every continuous map and every sheaf on its domain.

At P , every open neighbourhood contains P . All the groups forming the stalk are A , and all restriction homomorphisms are identities. The map $A \rightarrow (i_*\mathcal{F})_P$ sends a to the germ of the section a on X . It is surjective because every germ representative comes from a copy of A , and injective because restrictions never identify two distinct elements. Thus

$$(i_*\mathcal{F})_P \cong A.$$

Now suppose $\{P\}$ is closed and take $Q \neq P$. The set $X \setminus \{P\}$ is open and contains Q . Every open neighbourhood U of Q can be shrunk to $U \cap (X \setminus \{P\})$. On that smaller neighbourhood, the pushforward section group is zero. Consequently every germ representative at Q becomes zero after restriction, so

$$(i_*\mathcal{F})_Q = 0 \quad (Q \neq P).$$

Thus, for a closed point P , this sheaf has stalk A only at P and zero stalk at every other point. This is the skyscraper sheaf with value A at P .

Pitfall and check

Do not drop the hypothesis that P is closed when concluding that the stalks away from P are zero. If Q lies in the closure of $\{P\}$, every open neighbourhood of Q contains P , so the same stalk computation instead gives A . The zero-stalk proof uses a **neighbourhood of Q not containing P** , not merely $Q \neq P$.

New item 3 - Brenner Exercise 6.13: the pullback–pushforward morphism bijection

Source: [Exercise 6.13 in the BGK reader](#). Frozen entity: TopologischeRäume/StetigeAbbildung/RückzugundVorschub/Morphismen/Aufgabe, pageid 116425, [revision 1081982](#). The exercise number follows Worksheet 6, revision 900086.

Source problem statement

Let $\varphi : X \rightarrow Y$ be continuous, $\mathcal{F}$ a sheaf on X , and $\mathcal{G}$ a sheaf on Y . Prove that there is a natural bijection

$$\mathrm{Hom}_X(\varphi^{-1}\mathcal{G}, \mathcal{F}) \cong \mathrm{Hom}_Y(\mathcal{G}, \varphi_*\mathcal{F}).$$

Complete independent solution

We construct both directions, then check that they are inverse to each other. The argument first applies to sheaves of sets. If the sheaves carry commutative group structures, all maps constructed from the original homomorphisms remain homomorphisms, so the same proof applies in that category.

By [Definition 6.12](#), the pullback presheaf is

$$\mathcal{P}(U) := \operatorname{colim}_{\substack{V \subseteq Y \text{ open} \\ U \subseteq \varphi^{-1}(V)}} \mathcal{G}(V).$$

Write a representative of a colimit element as $[V, s]$, with $s \in \mathcal{G}(V)$. Restriction to $U' \subseteq U$ retains the representative $[V, s]$. Two representatives are equal when their restrictions agree on a smaller open set still containing $\varphi(U)$. By [Definition 6.13](#), $\varphi^{-1}\mathcal{G} = \mathcal{P}^a$, the sheafification of $\mathcal{P}$. Write $\lambda : \mathcal{P} \rightarrow \mathcal{P}^a$ for the canonical map.

We use the universal property of sheafification: every presheaf morphism $\mathcal{P} \rightarrow \mathcal{F}$, with $\mathcal{F}$ already a sheaf, extends uniquely to a morphism $\mathcal{P}^a \rightarrow \mathcal{F}$. The reason is that sections of $\mathcal{P}^a$ are locally represented by sections of $\mathcal{P}$; the images of these representatives agree locally on intersections, so they glue to exactly one section of $\mathcal{F}$. Equality of germs ensures independence of the choice of representatives or cover.

From right to left. Given $\theta : \mathcal{G} \rightarrow \varphi_*\mathcal{F}$, define

$$\alpha_U : \mathcal{P}(U) \longrightarrow \mathcal{F}(U), \quad [V, s] \mapsto \theta_V(s)|_U.$$

The formula has the correct types because $\theta_V(s) \in \mathcal{F}(\varphi^{-1}(V))$ and $U \subseteq \varphi^{-1}(V)$. If two representatives become equal after restriction to $W \subseteq V \cap V'$ with $U \subseteq \varphi^{-1}(W)$, compatibility of θ with restrictions shows that the two images in $\mathcal{F}(U)$ agree. Thus the formula is well-defined. Restriction from U to U' also commutes with the formula, so α is a presheaf morphism. The universal property gives exactly one

$$\psi_\theta : \varphi^{-1}\mathcal{G} \longrightarrow \mathcal{F}, \quad \psi_\theta \lambda = \alpha.$$

From left to right. Given $\psi : \varphi^{-1}\mathcal{G} \rightarrow \mathcal{F}$, every $s \in \mathcal{G}(V)$ determines an element $[V, s]$ of $\mathcal{P}(\varphi^{-1}(V))$. Set

$$(\theta_\psi)_V(s) := \psi_{\varphi^{-1}(V)}(\lambda_{\varphi^{-1}(V)}([V, s])) \in \mathcal{F}(\varphi^{-1}(V)).$$

If $W \subseteq V$, the representative $[V, s]$ restricted to $\varphi^{-1}(W)$ equals $[W, s|_W]$ in the colimit. Since λ and ψ respect restrictions, this formula gives a sheaf morphism $\theta_\psi : \mathcal{G} \rightarrow \varphi_*\mathcal{F}$.

The two constructions are inverse. Start with θ . For $s \in \mathcal{G}(V)$, the first construction satisfies

$$(\theta_{\psi_\theta})_V(s) = \alpha_{\varphi^{-1}(V)}([V, s]) = \theta_V(s).$$

Thus it returns exactly θ . Conversely, start with ψ and construct θ_ψ . For a representative $[V, s] \in \mathcal{P}(U)$, compatibility with restrictions gives

$$\begin{aligned} \alpha_U([V, s]) &= (\theta_\psi)_V(s)|_U \\ &= \psi_U(\lambda_U([V, s])). \end{aligned}$$

Hence the new morphism ψ_{θ_ψ} and ψ agree after composition with λ . Uniqueness in the universal property of sheafification gives $\psi_{\theta_\psi} = \psi$.

Finally, this bijection is natural. If $u : \mathcal{F} \rightarrow \mathcal{F}'$ and $v : \mathcal{G}' \rightarrow \mathcal{G}$ are sheaf morphisms, the formulas on representatives give, respectively,

$$\theta_{u \circ \psi} = (\varphi_* u) \circ \theta_\psi, \quad \theta_{\psi \circ \varphi^{-1} v} = \theta_\psi \circ v.$$

The first equality simply applies u to the image of a section; the second replaces $[V, s']$ by $[V, v_V(s')]$. Thus the bijection commutes with postcomposition in $\mathcal{F}$ and precomposition in $\mathcal{G}$. This is the entire naturality claim and completes the proof of the required bijection.

Pitfall and check

Do not replace $\varphi^{-1}\mathcal{G}(U)$ by the single value $\mathcal{G}(\varphi(U))$: the image $\varphi(U)$ need not be open, and the colimit presheaf must still be sheafified. In every formula, check where a section lives before restricting it. The condition $U \subseteq \varphi^{-1}(V)$ is exactly what makes $\theta_V(s)|_U$ legitimate.

BGK 7 mastery exercises

The following two problem statements come from Holger Brenner's course on Wikiversity. The solutions and learning checks are new editorial material prepared independently by **OpenAI Codex gpt-5.6-sol, Ultra**. Neither is a public source solution: the frozen map records the absence of public solutions for Exercises 7.5 and 7.16. The [public solution to Exercise 7.14](#) remains a separate source solution and is not counted as new writing here.

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New item 1 - Brenner Exercise 7.5: the sheaf of units

Source: [Exercise 7.5 in the BGK reader](#). Frozen entity: Beringter Raum/Einheiten/Garbe/Aufgabe, pageid 116370, [revision 1081774](#). The exercise number follows [Worksheet 7, revision 618943](#).

Source problem statement

Let $(X, \mathcal{O}_X)$ be a ringed space. Prove that the assignment

$$U \mapsto \mathcal{O}_X(U)^\times = (\Gamma(U, \mathcal{O}_X))^\times$$

on open sets $U \subseteq X$, together with the natural restrictions, is a sheaf of commutative groups. This sheaf is denoted by $\mathcal{O}_X^\times$ and called the sheaf of units.

Complete independent solution

For each U , the units of the commutative ring $\mathcal{O}_X(U)$ form a commutative group under **multiplication**, with identity 1_U and multiplicative inverses. Ring restriction homomorphisms preserve multiplication and the identity. Thus, if $s, t \in \mathcal{O}_X(U)$ satisfy $st = 1_U$, then for $V \subseteq U$,

$$(s|_V)(t|_V) = 1_V.$$

That is, restriction sends units to units and inverses to inverses. The identity and composition properties of restrictions are inherited from $\mathcal{O}_X$. The assignment therefore gives, to begin with, a presheaf of commutative groups.

To prove the sheaf property, take an open cover $U = \bigcup_i U_i$. If two units $s, t \in \mathcal{O}_X(U)^\times$ have equal restrictions on every U_i , the uniqueness property of the sheaf $\mathcal{O}_X$ gives $s = t$. This proves uniqueness of gluing.

For existence, take units $s_i \in \mathcal{O}_X(U_i)^\times$ satisfying

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j} \quad \text{for all } i, j.$$

Since $\mathcal{O}_X$ is a sheaf, there is exactly one $s \in \mathcal{O}_X(U)$ with $s|_{U_i} = s_i$. However, we must still prove that s is a unit, not merely a ring section.

Write $t_i = s_i^{-1}$. On $U_i \cap U_j$, the restrictions of t_i and t_j are inverses of the same element. Inverses in a group are unique, so

$$t_i|_{U_i \cap U_j} = t_j|_{U_i \cap U_j}.$$

Thus the t_i glue to $t \in \mathcal{O}_X(U)$. Now st and 1_U have equal restrictions to every U_i :

$$(st)|_{U_i} = s_i t_i = 1_{U_i} = 1_U|_{U_i}.$$

Sheaf uniqueness gives $st = 1_U$. Since the ring is commutative, also $ts = 1_U$, so $s \in \mathcal{O}_X(U)^\times$ with inverse t . This proves existence of gluing within the presheaf of units itself.

On the empty set, the section ring of the sheaf has just one element; its unit group is also the one-element group. Thus the empty-cover axiom introduces no exception. All axioms for a sheaf of commutative groups are satisfied.

Pitfall and check

Gluing the s_i merely as sections of $\mathcal{O}_X$ is not enough: one must glue **their inverses** to prove that the resulting section is a unit. Moreover, $\mathcal{O}_X^\times$ is not in general a subsheaf of the additive group $\mathcal{O}_X$. For example, 1 and -1 are units in $\mathbb{R}$, but their sum 0 is not a unit. The correct group operation in this exercise is multiplication.

New item 2 - Brenner Exercise 7.16: the residue field of continuous functions

Source: [Exercise 7.16 in the BGK reader](#). Frozen entity: TopologischerRaum/StetigeFunktionen/Restekörper/Aufgabe, pageid 112082, [revision 848530](#). The exercise number follows Worksheet 7, revision 618943.

Source problem statement

Let X be a topological space with its sheaf of real-valued continuous functions $\mathcal{C} = C^0(-, \mathbb{R})$. Prove that the residue field at each point $P \in X$ is $\mathbb{R}$, through the canonical evaluation isomorphism.

Complete independent solution

Fix $P \in X$ and write $A := \mathcal{C}_P$ for the stalk. Elements of A are germs $[U, f]_P$, where $P \in U$ is open and $f : U \rightarrow \mathbb{R}$ is continuous. Two representatives give the same germ if their functions agree on a smaller open neighbourhood of P . Since that neighbourhood contains P , the two function values at P agree. Thus evaluation

$$\text{ev}_P : A \longrightarrow \mathbb{R}, \quad [U, f]_P \longmapsto f(P)$$

is well-defined. Addition and multiplication of germs are computed after restricting representatives to a common neighbourhood, so evaluation is a ring homomorphism preserving 1. It is surjective: each $r \in \mathbb{R}$ is the value of the germ of the constant function $x \mapsto r$.

Its kernel is the ideal

$$\mathfrak{m}_P := \{[U, f]_P \in A \mid f(P) = 0\}.$$

To verify that this is the stalk's unique maximal ideal, we characterise all its units. If $f(P) \neq 0$, the set

$$W := f^{-1}(\mathbb{R} \setminus \{0\}) \subseteq U$$

is an open neighbourhood of P . The function $g : W \rightarrow \mathbb{R}$, $g(x) = 1/f(x)$, is continuous, since multiplicative inversion is continuous on $\mathbb{R} \setminus \{0\}$. Since $fg = 1$ on W , the germ $[U, f]_P$ is a unit in A .

Conversely, if $a \in A$ is a unit with inverse b , applying evaluation to $ab = 1$ gives

$$\text{ev}_P(a) \text{ev}_P(b) = 1.$$

Hence $\text{ev}_P(a) \neq 0$. Therefore

$$A^\times = A \setminus \mathfrak{m}_P.$$

The ideal $\mathfrak{m}_P$ is proper because it does not contain the germ of the constant function 1. The surjective evaluation homomorphism gives $A/\mathfrak{m}_P \cong \mathbb{R}$, so $\mathfrak{m}_P$ is maximal. If $\mathfrak{n}$ is another maximal ideal, it cannot contain a unit; since every element outside $\mathfrak{m}_P$ is a unit, $\mathfrak{n} \subseteq \mathfrak{m}_P$. Maximality of $\mathfrak{n}$ and properness of $\mathfrak{m}_P$ force $\mathfrak{n} = \mathfrak{m}_P$. This also proves that A is a local ring.

By [Definition 7.13](#), the residue field at P is $\kappa(P) = A/\mathfrak{m}_P$. The required isomorphism is

$$\begin{aligned} \kappa(P) &\longrightarrow \mathbb{R}, \\ [U, f]_P + \mathfrak{m}_P &\longmapsto f(P). \end{aligned}$$

This formula is independent of both the germ representative and the quotient-class representative. Its inverse sends r to the class of the germ of the constant function r . The first composite plainly returns r . For the other composite, $f - f(P)$ vanishes at P , so its germ lies in $\mathfrak{m}_P$; hence the class of the germ of f equals that of the constant $f(P)$. Both composites are indeed identities.

No Hausdorff or manifold assumption on X is required. The only properties used are continuity of real functions and the definition of a stalk. If X is empty, the statement about every point holds vacuously.

Pitfall and check

The stalk $\mathcal{C}_P$ is not its residue field. For $X = \mathbb{R}$ and $P = 0$, the germ of $x \mapsto x$ is nonzero: the function is not identically zero on any neighbourhood of 0. Yet its image in $\kappa(0)$ is 0, since its value at 0 is zero. This example distinguishes information about a function **near a point**, retained by the stalk, from its value **at the point**, retained by the residue field.

BGK 9 mastery exercises

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1. Localisation as a quotient ring

Source: [Exercise 9.4](#), [worksheet revision 612139](#). Exact identifier: KommutativeRingtheorie/Nenneraufnahme/EinElement/Restklassendarstellung/Aufgabe; source page 19608, [fixed revision 1098291](#).

Brenner's exercise. For a commutative ring R and $f \in R$, prove the R -algebra isomorphism

$$R_f \cong R[T]/(Tf - 1).$$

Independent solution. Write $A = R[T]/(Tf - 1)$ and let t be the class of T . In A we have $tf = 1$, so the image of f is a unit with inverse t . Define

$$\alpha : R_f \longrightarrow A, \quad a/f^n \longmapsto at^n.$$

This map is well-defined even if R has zero divisors. Indeed, if $a/f^n = b/f^m$, there is $k \geq 0$ with $f^k(f^m a - f^n b) = 0$ in R . After mapping to A , multiply by t^{k+m+n} ; this gives $at^n = bt^m$. The formulas for addition and multiplication of fractions directly show that α is an R -algebra homomorphism.

Conversely, evaluation $R[T] \rightarrow R_f$ at $T = f^{-1}$ annihilates $Tf - 1$. It therefore induces

$$\beta : A \longrightarrow R_f, \quad t \longmapsto 1/f.$$

The composite $\beta\alpha$ sends every a/f^n back to a/f^n . The composite $\alpha\beta$ fixes the image of every element of R and fixes t . Since these elements generate A as a ring, $\alpha\beta = \text{id}_A$. Thus the homomorphisms are inverse to each other.

Check. If $f = 0$, the ideal $(Tf - 1)$ is all of $R[T]$ and both sides are the zero ring. Do not assume that $R \rightarrow R_f$ is injective; the proof above does not require that assumption.

2. When does a localisation vanish?

Source: [Exercise 9.6](#), [worksheet revision 612139](#). Exact identifier: Nenneraufnahme/f/Nilpotent/Aufgabe; source page 94310, [fixed revision 1045587](#).

Brenner's exercise. For a commutative ring R and $f \in R$, prove that f is nilpotent exactly when R_f is the zero ring.

Independent solution. If $f^n = 0$ for some $n \geq 1$, the image of f in R_f is invertible. Multiply $f^n = 0$ by the inverse f^{-n} in R_f to obtain $1 = 0$. Every element u then satisfies $u = u \cdot 1 = u \cdot 0 = 0$, so R_f is the zero ring.

Conversely, if R_f is the zero ring, the fractions $1/1$ and $0/1$ are equal. The definition of equality in a localisation gives $k \geq 0$ such that $f^k(1 - 0) = 0$ in R . If $k \geq 1$, this is exactly nilpotence. If $k = 0$, then $1 = 0$ in R ; in that case R itself is zero and $f = 0$ is also nilpotent. Both directions have been proved.

Check. In $R = K[\varepsilon]/(\varepsilon^2)$, localisation at ε is indeed zero although R is not the zero ring. In contrast, $R = \mathbb{Z}/6\mathbb{Z}$ and $f = 2$ give a nonzero localisation: the map $R \rightarrow \mathbb{F}_3$ sends 2 to a unit and extends to R_f . Thus “zero divisor” cannot replace “nilpotent”.

3. Intermediate rings over a principal ideal domain

Source: [Exercise 9.8](#), [worksheet revision 612139](#). Exact identifier: Hauptidealbereich/ZwischenringinQuotientenkörper/IstNenneraufnahme/Aufgabe; source page 20756, [fixed revision 1061311](#).

Brenner's exercise. Let R be a principal ideal domain, Q its field of fractions, and $R \subseteq S \subseteq Q$ an intermediate ring. Prove that S is a localisation of R .

Independent solution. Take the multiplicative set

$$M = \{b \in R \setminus \{0\} : 1/b \in S\}.$$

This set contains 1, and if $b, c \in M$, then $1/(bc) = (1/b)(1/c) \in S$. Since all elements of M are nonzero, we can regard $M^{-1}R$ as a subring of Q . The definition of M directly gives $M^{-1}R \subseteq S$.

For the converse, take $q \in S$ and write $q = a/b$ with $b \neq 0$. Since R is a principal ideal domain, dividing numerator and denominator by a generator of the ideal (a, b) allows us to choose a, b with $(a, b) = R$. There are then $u, v \in R$ with

$$ua + vb = 1.$$

Divide this equality by b in Q . We obtain

$$1/b = u(a/b) + v = uq + v \in S.$$

Thus $b \in M$ and $q = a/b \in M^{-1}R$. Since q was arbitrary, $S = M^{-1}R$ as subrings of Q , not merely as abstractly isomorphic rings.

Check. The key step is the Bézout identity $ua + vb = 1$ for a reduced fraction. Unique factorisation alone does not guarantee this identity; do not replace the principal ideal domain hypothesis without an additional proof.

BGK 10 mastery exercises

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1. Quasi-affine but not affine

Source: [Exercise 10.1](#), [worksheet revision 612138](#). Exact identifier: Quasiaffines Schema/Nicht affin/Aufgabe; source page 112256, [fixed revision 847500](#).

Brenner's exercise. Give an example of a quasi-affine scheme that is not affine.

Independent solution. Choose any field K , write $A = K[x, y]$, and take

$$U = \operatorname{Spec}(A) \setminus \{(x, y)\} = D(x) \cup D(y).$$

The last equality holds because the only prime ideal containing x and y is the maximal ideal (x, y) . As an open subset of an affine scheme, U is quasi-affine. We will show that U is not affine by computing its global sections.

The structure sheaf on the cover $D(x), D(y)$ gives

$$\Gamma(U, \mathcal{O}_U) = A_x \cap A_y \subseteq K(x, y).$$

Indeed, a section is a pair of elements of A_x, A_y agreeing in A_{xy} ; all these maps are injective because A is an integral domain. If $a/x^m = b/y^n$ lies in the intersection, then $y^n a = x^m b$. In the unique factorisation domain $K[x, y]$, x is prime and does not divide y , so x^m divides a . Hence $a/x^m \in A$. The reverse inclusion is clear, so $\Gamma(U, \mathcal{O}_U) = A$ and the restriction map from A is the identity under this identification.

Suppose U were affine. The inclusion $j : U \rightarrow \operatorname{Spec}(A)$ would be a morphism between two affine schemes inducing an isomorphism on global sections. It would have to be an isomorphism: apply [Theorem 10.9](#) to the inverse of the global homomorphism to construct an inverse morphism; uniqueness in the theorem ensures that both composites are identities. But j is not surjective, since (x, y) is not in its image. This is a contradiction.

Check. The space U is even quasi-compact, being a union of two affine open sets. Thus the failure of affineness in this example is not caused by a failure of quasi-compactness.

2. Quasi-affine but not quasi-compact

Source: [Exercise 10.2](#), [worksheet revision 612138](#). Exact identifier: QuasiaffinesSchema/Nichtquasikompakt/Aufgabe; source page 112258, [fixed revision 847501](#).

Brenner's exercise. Give an example of a quasi-affine scheme that is not quasi-compact.

Independent solution. For a field K , take the polynomial ring in infinitely many variables

$$A = K[x_1, x_2, x_3, \dots], \quad U = \bigcup_{i \geq 1} D(x_i) \subseteq \operatorname{Spek}(A).$$

Each polynomial still involves only finitely many variables. The set U is open in an affine scheme, hence quasi-affine by [Definition 10.4](#).

The open cover $\{D(x_i)\}_{i \geq 1}$ has no finite subcover. To prove this, take any finite index set $F \subset \mathbb{N}_{\geq 1}$ and choose $j \notin F$. The ideal

$$\mathfrak{p}_F = (x_i : i \in F)$$

is prime because the quotient ring $A/\mathfrak{p}_F$ is the polynomial ring over K in the remaining variables, which is an integral domain. The element x_j does not belong to $\mathfrak{p}_F$, so $\mathfrak{p}_F \in D(x_j) \subseteq U$. In contrast, every x_i with $i \in F$ belongs to $\mathfrak{p}_F$, so $\mathfrak{p}_F \notin \bigcup_{i \in F} D(x_i)$. Thus no finite choice covers U . This is the failure of quasi-compactness.

Check. A single open cover without a finite subcover suffices to prove that a space is not quasi-compact. Merely saying “the cover is infinite” is not enough, since an infinite cover may still have a finite subcover.

3. Morphisms over a base and algebra homomorphisms

Source: [Exercise 10.6](#), [worksheet revision 612138](#). Exact identifier: Algebramorphismus/Basischema/Morphismus/Aufgabe; source page 112315, [fixed revision 1082198](#).

Brenner's exercise. For a commutative ring R and commutative R -algebras A, B , prove that an R -algebra homomorphism $A \rightarrow B$ is the same data as a scheme morphism $\operatorname{Spek}(B) \rightarrow \operatorname{Spek}(A)$ over $\operatorname{Spek}(R)$.

Independent solution. Write the algebra structure maps as $\alpha : R \rightarrow A$ and $\beta : R \rightarrow B$, and the scheme structure morphisms as p_A and p_B . By [Theorem 10.9](#), applied to the locally ringed space $\operatorname{Spek}(B)$ and affine target $\operatorname{Spek}(A)$, every ring homomorphism $\varphi : A \rightarrow B$ determines exactly one morphism ψ with global homomorphism $\psi^\# = \varphi$. Conversely, global sections of any ψ give that homomorphism; uniqueness in the theorem shows that the two operations are inverse to each other.

It remains to prove that the conditions involving the base correspond. By definition, φ is an R -algebra homomorphism if and only if

$$\varphi \circ \alpha = \beta.$$

The composite $p_A \circ \psi$ has global homomorphism $\varphi \circ \alpha$, whereas p_B has global homomorphism β . Both morphisms have affine target $\operatorname{Spek}(R)$. Again, uniqueness in Theorem 10.9 gives the equivalence

$$\varphi \circ \alpha = \beta \iff p_A \circ \psi = p_B.$$

The condition on the right says exactly that ψ is a morphism over $\operatorname{Spek}(R)$. Thus the general correspondence restricts to the required bijection, with ring arrows pointing in the opposite direction to scheme arrows.

Check. Commutativity of the maps on points alone is not enough. For example, complex conjugation gives a ring automorphism of $\mathbb{C}$ and a scheme automorphism of $\operatorname{Spek}(\mathbb{C})$; its topological map is the identity on a one-point space, but it is not a morphism over $\operatorname{Spek}(\mathbb{C})$ with the identity base structure, since it does not fix every scalar.

BGK 12 mastery: reading the projective spectrum

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The three mastery items for Unit 12 comprise the new solution below and [source solution 12.5](#) and [source solution 12.10](#). Both source solutions retain their source-solution status and are not rewritten here.

New item 1: the coordinate cross yields two projective points

Source: [BGK Exercise 12.8](#), identifier Achsenkreuz/ProjektivesSpektrum/Aufgabe, [revision 1082163](#). This statement uses the standard grading and an arbitrary field K , without assuming that K is algebraically closed.

Source exercise

Determine the projective spectrum of the coordinate cross

$$\mathrm{Spek}(K[X, Y]/(XY))$$

with the standard grading.

Independent solution

Write

$$A = K[X, Y]/(XY), \quad x = \overline{X}, \quad y = \overline{Y}.$$

The degrees of x and y are one, so the irrelevant ideal is $A_+ = (x, y)$. We seek all homogeneous prime ideals not containing (x, y) , together with their scheme structure.

Determining the points. For every prime ideal $\mathfrak{p}$ of A , the equality $xy = 0$ gives $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$. A prime ideal that is a point of $\mathrm{Proj}(A)$ cannot contain both. Suppose $x \in \mathfrak{p}$ but $y \notin \mathfrak{p}$. In

$$A/(x) \cong K[y],$$

the ideal $\mathfrak{p}/(x)$ is a homogeneous prime ideal not containing y . The only such ideal is (0) : every nonzero homogeneous polynomial in one variable has the form cy^d ; if a proper homogeneous ideal contains such an element, then $d > 0$, and primality forces y into the ideal. Hence $\mathfrak{p} = (x)$. Interchanging x and y , the other case gives $\mathfrak{p} = (y)$. Thus the set of points is exactly

$$\mathrm{Proj}(A) = \{(x), (y)\}.$$

This argument determines all homogeneous prime points, not just points already expressed in K -coordinates.

Determining the scheme structure. The standard opens $D_+(x)$ and $D_+(y)$ cover the projective spectrum. Since x becomes a unit in A_x , the equation $xy = 0$ forces $y = 0$. Therefore

$$A_x \cong K[x, x^{-1}], \quad (A_x)_0 = K.$$

[Lemma 12.9](#), for the homogeneous element x of degree one, gives

$$D_+(x) \cong \mathrm{Spek}(K).$$

Likewise, $D_+(y) \cong \mathrm{Spek}(K)$. Their intersection is empty, since a prime ideal cannot omit both x and y when $xy = 0$. Hence

$$\mathrm{Proj}(K[X, Y]/(XY)) \cong \mathrm{Spek}(K) \amalg \mathrm{Spek}(K).$$

The point (y) belongs to $D_+(x)$ and has coordinates $[1 : 0]$; the point (x) belongs to $D_+(y)$ and has coordinates $[0 : 1]$. Each point is both open and closed, its local ring is K , and there is no hidden nilpotent structure. As an additional check, the global section ring is $K \times K$, since sections on the two disjoint components can be chosen independently.

Checks and common mistakes

The ideal (x, y) represents the origin of the affine coordinate cross, but is not a point of the projective spectrum because it contains the irrelevant ideal. Each affine axis, on the other hand, contributes one projective point, not a projective line. The answer does not depend on K being algebraically closed: the coordinate ring of each affine open is already exactly K .

BGK 13 mastery: stalks, matrices, and invertible sheaves

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The Indonesian term *ruang bergelanggang*, rendered here as *ringed space*, follows the edition glossary and denotes the object called *ruang berdering* in the Indonesian source translation. This change in terminology does not add a requirement that the stalk rings be local.

New item 1: the module structure on a stalk

Source: [BGK Exercise 13.5](#), identifier Beringter Raum/Modul/Halm/Aufgabe, [revision 1082395](#).

Source exercise

Let $\mathcal{F}$ be an $\mathcal{O}_X$ -module on a ringed space $(X, \mathcal{O}_X)$. Prove that, for every $P \in X$, the stalk $\mathcal{F}_P$ is an $\mathcal{O}_{X,P}$ -module.

Independent solution

An element $a_P \in \mathcal{O}_{X,P}$ is represented by a section $a \in \mathcal{O}_X(U)$ on an open neighbourhood U of P . Likewise, $s_P \in \mathcal{F}_P$ has a representative $s \in \mathcal{F}(V)$ for some open neighbourhood V of P . We define

$$a_P s_P := ((a|_{U \cap V})(s|_{U \cap V}))_P.$$

The multiplication on the right is defined because $\mathcal{F}(U \cap V)$ is a module over $\mathcal{O}_X(U \cap V)$.

We must check that the result is independent of the representatives. Suppose a' is another representative of a_P and s' another representative of s_P . Equality of germs means that on a neighbourhood W_a of P , the restrictions of a and a' agree, and on a neighbourhood W_s , the restrictions of s and s' agree. Intersect these neighbourhoods with all the representative domains. On this intersection, the two products agree. Compatibility of scalar multiplication with restrictions, which is part of [Definition 13.5](#), ensures that the original two products determine the same germ. Thus the operation is well-defined.

Addition on $\mathcal{F}_P$ is constructed in the same way: restrict two representatives to a common neighbourhood, then add them. The abelian group axioms hold because each axiom involves only finitely many representatives; they can all be restricted to a common neighbourhood where the axiom already holds in $\mathcal{F}(W)$.

The same method proves the module axioms. For $a_P, b_P \in \mathcal{O}_{X,P}$ and $s_P, t_P \in \mathcal{F}_P$, choose representatives of all of them on a single W . The axioms for the section module on W give

$$\begin{aligned}
(a_P + b_P)s_P &= a_P s_P + b_P s_P, \\
a_P(s_P + t_P) &= a_P s_P + a_P t_P, \\
(a_P b_P)s_P &= a_P(b_P s_P), \\
1_P s_P &= s_P.
\end{aligned}$$

Hence $\mathcal{F}_P$ has a natural $\mathcal{O}_{X,P}$ -module structure. No assumption that the ringed space is locally ringed is required.

Checks and common mistakes

Multiplication of representatives with different domains must be preceded by restriction to a common neighbourhood. Nor is the stalk $\mathcal{F}_P$ the fibre: on a locally ringed space, the fibre defined in [Definition 13.8](#) is $\mathcal{F}_P \otimes_{\mathcal{O}_{X,P}} \kappa(P)$, which still requires a change of scalars to the residue field.

New item 2: unit determinants and isomorphisms of free sheaves

Source: [BGK Exercise 13.10](#), identifier BeringterRaum/FreierModul/Festlegungssatz/Determinante/Isomorphismus/Aufgabe, [revision 1097130](#).

Source exercise

Let $(X, \mathcal{O}_X)$ be a ringed space and

$$s_i = (s_{i1}, \dots, s_{in}) \in \Gamma(X, \mathcal{O}_X)^n, \quad 1 \leq i \leq n.$$

Prove that $\det(s_{ij})$ is a unit in $\Gamma(X, \mathcal{O}_X)$ if and only if the associated homomorphism

$$\varphi : \mathcal{O}_X^n \longrightarrow \mathcal{O}_X^n, \quad e_i \longmapsto s_i,$$

is an isomorphism.

Independent solution

Write $R = \Gamma(X, \mathcal{O}_X)$ and $A = (s_{ij})$. [Theorem 13.10](#) ensures that these sections determine exactly one homomorphism of sheaves of modules. If coordinate vectors are written as columns, its matrix is $B = A^\top$, since the i th column contains the coordinates of $\varphi(e_i) = s_i$. In particular, $\det B = \det A$.

Suppose $d = \det A$ is a unit in R . The adjugate identity for matrices over a commutative ring gives

$$B \operatorname{adj}(B) = \operatorname{adj}(B) B = d I_n.$$

Hence the matrix

$$C = d^{-1} \operatorname{adj}(B) \in M_n(R)$$

satisfies $BC = CB = I_n$. The entries of C are global sections. Restricting them to each open U gives a $\Gamma(U, \mathcal{O}_X)$ -module homomorphism on $\Gamma(U, \mathcal{O}_X)^n$. These homomorphisms are compatible with restrictions and therefore determine a sheaf homomorphism $\psi : \mathcal{O}_X^n \rightarrow \mathcal{O}_X^n$. The matrix identities remain valid after restriction, so $\varphi\psi = \psi\varphi = \operatorname{id}$. Thus φ is an isomorphism.

Conversely, suppose φ is an isomorphism of sheaves of modules with inverse ψ . Evaluating both composites on X gives inverse R -module homomorphisms on R^n . The matrix of ψ_X in the standard basis is some $C \in M_n(R)$, so

$$BC = CB = I_n.$$

Taking determinants gives

$$\det A \cdot \det C = \det B \cdot \det C = 1.$$

Hence $\det A$ is a unit, with inverse $\det C$. Both directions have been proved without treating the global section ring as a field.

Checks and common mistakes

The condition is a *unit* determinant, not merely a nonzero determinant. The matrix (2) over $\mathbb{Z}$, for example, is not invertible over $\mathbb{Z}$. The transpose above merely records the row/column convention; transposition does not change the determinant. The converse uses an already existing sheaf inverse, not an assumption that any isomorphism on global sections gives an isomorphism of arbitrary sheaves.

New item 3: the dual of an invertible sheaf

Source: [BGK Exercise 13.16](#), identifier BeringterRaum/InvertierbareGarben/DualeGarbe/Invertierbar/Aufgabe, revision 1082386.

Source exercise

Let $\mathcal{L}$ be an invertible sheaf on a ringed space $(X, \mathcal{O}_X)$. Prove that the dual sheaf

$$\mathcal{L}^* = \mathcal{H}om(\mathcal{L}, \mathcal{O}_X)$$

is also invertible.

Independent solution

By [Definition 13.17](#), there is an open cover $X = \bigcup_i U_i$ with $\mathcal{L}|_{U_i} \cong \mathcal{O}_X|_{U_i}$. Choose a local basis $e_i \in \mathcal{L}(U_i)$ corresponding to the section 1 under this trivialisation. For each open $V \subseteq U_i$, every section of $\mathcal{L}(V)$ is uniquely written as $be_i|_V$, with $b \in \mathcal{O}_X(V)$.

A dual section on V is not merely a function on global sections: by [Definition 13.13](#), it is a homomorphism of sheaves of modules

$$\lambda : \mathcal{L}|_V \longrightarrow \mathcal{O}_X|_V.$$

This homomorphism determines an element $a = \lambda_V(e_i|_V) \in \mathcal{O}_X(V)$. Conversely, each $a \in \mathcal{O}_X(V)$ determines such a homomorphism: on each $W \subseteq V$, define

$$\lambda_W^a(be_i|_W) = b(a|_W).$$

Uniqueness of representation in the basis $e_i|_W$ makes this formula well-defined. It is $\mathcal{O}_X(W)$ -linear and compatible with every restriction. The two constructions are inverse. Moreover, when V is restricted to a smaller open set, evaluation on e_i and the construction of λ^a restrict in the same way. Thus we obtain an isomorphism of *sheaves* of modules

$$\mathcal{L}^*|_{U_i} \cong \mathcal{O}_X|_{U_i}.$$

The same cover $\{U_i\}$ therefore shows that $\mathcal{L}^*$ is locally free of rank one, that is, invertible.

We can also check its transition maps. On $U_i \cap U_j$, write $e_j = g_{ij}e_i$, with g_{ij} a unit. If e_i^* is the dual basis sending e_i to 1, then

$$e_j^* = g_{ij}^{-1}e_i^*,$$

because $(g_{ij}^{-1}e_i^*)(g_{ij}e_i) = 1$. Thus the dual transitions are also multiplication by units, as invertibility requires.

Checks and common mistakes

The proof chooses a basis only on each U_i , not a global basis. Invertibility of $\mathcal{L}^*$ does not say that the sheaf is globally trivial. Also distinguish the sheaf $\mathcal{H}om(\mathcal{L}, \mathcal{O}_X)$ from a homomorphism module formed only from the two modules of global sections: the local computation above uses homomorphisms on all smaller open sets.

BGK 14 mastery: localisation and global sections

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New item 1: zero at a point, zero on a neighbourhood

Source: [BGK Exercise 14.3](#), identifier `KommutativerRing/Modul/Endlicherzeugter/0inPunkt/Umgebung/Aufgabe, revision 1039028`.

Source exercise

Let R be a commutative ring and M a finitely generated R -module. If $\mathfrak{p} \in \text{Spek}(R)$ satisfies $M_{\mathfrak{p}} = 0$, prove that there is $f \notin \mathfrak{p}$ with $M_f = 0$.

Independent solution

Choose generators $m_1, \dots, m_r$ for M . If $M = 0$, the choice $f = 1$ already gives the conclusion. For each generator in the general case, the assumption $M_{\mathfrak{p}} = 0$ gives

$$\frac{m_i}{1} = 0 \quad \text{in } M_{\mathfrak{p}}.$$

By the definition of equality in module localisation at the multiplicative set $R \setminus \mathfrak{p}$, there is $s_i \notin \mathfrak{p}$ such that $s_i m_i = 0$ in M . Take

$$f = s_1 s_2 \cdots s_r.$$

Since $\mathfrak{p}$ is prime and contains none of the s_i , we have $f \notin \mathfrak{p}$. For every i ,

$$f m_i = \left(\prod_{j \neq i} s_j \right) (s_i m_i) = 0.$$

Every element of M is a linear combination of the m_i , so $fM = 0$. In M_f , the element f becomes a unit. For any fraction $m/f^k \in M_f$,

$$\frac{m}{f^k} = \frac{f m}{f^{k+1}} = 0.$$

Hence $M_f = 0$. The principal open $D(f)$ contains $\mathfrak{p}$ because $f \notin \mathfrak{p}$. In sheaf language, further localisation at any point $\mathfrak{q} \in D(f)$ is also zero, so $\widetilde{M}|_{D(f)} = 0$. This explains the geometric meaning of the neighbourhood found.

Checks and common mistakes

Finiteness is used to form one product f annihilating all generators. The proof has no Noetherian or integral domain hypothesis. Without finiteness, a common denominator can fail to exist: for $M = \mathbb{Q}/\mathbb{Z}$ and $\mathfrak{p} = (0) \subset \mathbb{Z}$, we have $M_{(0)} = 0$ because every class is annihilated by a nonzero integer. However, for $f \neq 0$, choose a prime number ℓ not dividing f . The class $1/\ell + \mathbb{Z}$ remains nonzero after localisation at f , since none of the f^k/ℓ are integers. Thus $M_f \neq 0$ for every such choice.

New item 2: locally surjective, but not the unit ideal globally

Source: [BGK Exercise 14.10](#), identifier [PunktierteEbene/Festlegungssatz/KeinEinheitsideal/Surjektiv/Aufgabe](#), revision 1081689.

Source exercise

Let K be a field and $U = \mathbb{A}_K^2 \setminus \{(0,0)\}$ the punctured affine plane, with its structure sheaf $\mathcal{O}_U$. Give global sections $s_1, s_2 \in \Gamma(U, \mathcal{O}_U)$ such that (s_1, s_2) is not the unit ideal, but the homomorphism of sheaves of modules

$$\mathcal{O}_U^2 \longrightarrow \mathcal{O}_U, \quad e_i \mapsto s_i,$$

is surjective. Here $\mathcal{O}_U$ is the restriction of the affine plane's structure sheaf, also denoted by $\mathcal{O}_X$ in the source exercise.

Independent solution

Take $R = K[X, Y]$ and regard the origin as the maximal ideal (X, Y) . Since a prime ideal containing X and Y must equal (X, Y) ,

$$U = D(X) \cup D(Y).$$

We will use the restrictions of the coordinate functions $s_1 = X$ and $s_2 = Y$.

Computing the global section ring. By the description of structure-sheaf sections on principal opens, which is also the case $M = R$ of [Lemma 14.5](#),

$$\Gamma(D(X), \mathcal{O}_U) = R_X, \quad \Gamma(D(Y), \mathcal{O}_U) = R_Y.$$

The intersection is $D(XY)$, with section ring R_{XY} . All these rings are subrings of the field of fractions $K(X, Y)$. The sheaf gluing axiom gives

$$\Gamma(U, \mathcal{O}_U) = R_X \cap R_Y \quad \text{inside } K(X, Y).$$

We prove that this intersection is exactly R . If

$$\frac{a}{X^m} = \frac{b}{Y^n}, \quad a, b \in R,$$

then $aY^n = bX^m$. The element X is prime in R , since $R/(X) = K[Y]$ is an integral domain, and X does not divide Y . Hence X^m divides a , by repeatedly applying primality of X . Thus $a/X^m \in R$. The inclusion $R \subseteq R_X \cap R_Y$ is clear, so

$$\Gamma(U, \mathcal{O}_U) = K[X, Y].$$

The global ideal is not the unit ideal. In this ring, the ideal generated by s_1 and s_2 is (X, Y) , which is proper because $R/(X, Y) \cong K \neq 0$. Concretely, there are no polynomials a, b with $aX + bY = 1$: substituting $X = Y = 0$ would give $0 = 1$. This substitution merely tests an identity in the polynomial ring; we are not putting the origin back into U .

The sheaf homomorphism is surjective. The map to check is

$$\varphi(a, b) = Xa + Yb.$$

On any open $V \subseteq D(X)$, the function X has an inverse, so every $t \in \mathcal{O}_U(V)$ has a preimage $(X^{-1}t, 0)$. This formula is compatible with restrictions and gives a right inverse to $\varphi|_{D(X)}$. On $D(Y)$, a right inverse is $t \mapsto (0, Y^{-1}t)$. Since $D(X)$ and $D(Y)$ cover U , every target section can be lifted locally; equivalently, the map on each stalk is surjective. Thus φ is a surjection of sheaves of modules.

In contrast, the map on global sections is

$$R^2 \longrightarrow R, \quad (a, b) \longmapsto Xa + Yb,$$

whose image is (X, Y) and does not contain 1. This is the required example.

Checks and common mistakes

The two local right inverses need not agree on the intersection; indeed, there is no global right inverse lifting 1. Sheaf surjectivity means the existence of *local* lifts, not surjectivity on sections over every open set. This computation works over any field K and uses all prime points of the scheme, not just its rational points.

New item 3: change of scalars and the tensor–Hom adjunction

Source: [BGK Exercise 14.15](#), identifier Ringwechsel/VorgezogenerundzurückgezogenerModul/Homomorphismus/Aufgabe, [revision 1039630](#).

Source exercise

Let $\theta : A \rightarrow B$ be a homomorphism of commutative rings, M an A -module, and N a B -module. Write N' for N viewed as an A -module through θ . Prove the natural group isomorphism

$$\mathrm{Hom}_B(M \otimes_A B, N) \cong \mathrm{Hom}_A(M, N').$$

Independent solution

The scalar structure on N' is $a \cdot n = \theta(a)n$. Define the first map by evaluating on tensors whose second factor is 1:

$$\Phi(f)(m) = f(m \otimes 1).$$

This map is additive in m . For $a \in A$,

$$\begin{aligned} \Phi(f)(am) &= f(am \otimes 1) \\ &= f(m \otimes \theta(a)) \\ &= \theta(a)f(m \otimes 1) \\ &= a \cdot \Phi(f)(m). \end{aligned}$$

Thus $\Phi(f)$ is an A -module homomorphism from M to N' .

For the reverse direction, given $g \in \mathrm{Hom}_A(M, N')$, we want to define

$$\Psi(g)(m \otimes b) = b g(m).$$

The map $(m, b) \mapsto b g(m)$ is additive in each variable. It is also A -balanced, since

$$b g(am) = b \theta(a) g(m) = (\theta(a)b) g(m).$$

The universal property of the tensor product therefore gives exactly one additive map $M \otimes_A B \rightarrow N$ with this formula. It is B -linear: for $c \in B$,

$$\Psi(g)(m \otimes cb) = cb g(m) = c \Psi(g)(m \otimes b).$$

Since pure tensors generate $M \otimes_A B$ as an additive group, this check proves linearity on all elements.

The two constructions are inverse. For g ,

$$\Phi(\Psi(g))(m) = \Psi(g)(m \otimes 1) = g(m).$$

For f , B -linearity gives

$$\Psi(\Phi(f))(m \otimes b) = bf(m \otimes 1) = f(m \otimes b).$$

Equality on pure tensors extends to the whole tensor product. The formulas also respect addition of f and of g , so they genuinely give a group isomorphism, not merely a bijection of sets.

Finally, naturality can be checked without choosing a basis. If $u : M_1 \rightarrow M$ is an A -module homomorphism and $v : N \rightarrow N_1$ a B -module homomorphism, then for every $m_1 \in M_1$,

$$\begin{aligned} \Phi(v \circ f \circ (u \otimes \text{id}_B))(m_1) &= v(f(u(m_1) \otimes 1)) \\ &= (v \circ \Phi(f) \circ u)(m_1). \end{aligned}$$

Thus the isomorphism is compatible with changing M and N through homomorphisms, exactly as the word *natural* means.

Checks and common mistakes

On the right, linearity uses the A -structure on N' through θ ; do not assume that M is already a B -module. A formula on tensors must satisfy the balancing relation before it can be declared well-defined. The proof makes no finiteness, freeness, or flatness assumptions on the modules.

BGK 15 mastery exercises

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This material uses the source notation: $\widehat{M}$ is the sheaf on $\text{Proj}(R)$ associated to a graded module M ; this notation is distinguished from $\widetilde{M}$ on the affine spectrum.

1. Different modules, the same projective sheaf

Exercise source: [Exercise 15.7](#), ID br-bgk-2019-w15-ex07. Source entity *Graduierter Ring/Moduln/Realisierung auf Proj/Aufgabe*, [revision 1081620](#).

Problem statement. Let R be a $\mathbb{Z}$ -graded ring and $Y = \text{Proj}(R)$. Show that nonisomorphic graded R -modules can give isomorphic sheaves of $\mathcal{O}_Y$ -modules.

Editorial solution. The word *can* asks for an example of this phenomenon. Take a field K and the standard graded ring

$$R = K[X_0, X_1], \quad Y = \mathbb{P}_K^1.$$

Define three graded modules

$$T = R/(X_0, X_1), \quad M = R, \quad N = R \oplus T.$$

The module T is isomorphic to K in degree 0, with all other graded components zero. Hence

$$\dim_K M_0 = 1, \quad \dim_K N_0 = 2.$$

A graded module isomorphism preserves every graded component, so M and N are not isomorphic as graded modules. They remain nonisomorphic even after forgetting the grading: N has a nonzero element $(0, \bar{1})$ annihilated by X_0 , whereas multiplication by X_0 in the integral domain $R = M$ is injective.

Now consider the standard cover

$$Y = D_+(X_0) \cup D_+(X_1).$$

Since $X_i T = 0$, the localisation T_{X_i} is the zero module. Indeed, X_i is invertible in this localised module, so every $t/1$ satisfies

$$\frac{t}{1} = X_i^{-1} \frac{X_i t}{1} = 0.$$

By [Lemma 15.3\(2\)](#), the sheaf $\widehat{T}$ on $D_+(X_i)$ is associated to $(T_{X_i})_0 = 0$. Thus $\widehat{T}$ is zero on both members of the cover, and therefore on all of Y .

The projection $\pi : N \rightarrow R$, $(r, t) \mapsto r$, induces a morphism $\hat{\pi} : \widehat{N} \rightarrow \widehat{R}$. On each $D_+(X_i)$, this morphism comes from the isomorphism

$$(N_{X_i})_0 = (R_{X_i})_0 \oplus (T_{X_i})_0 = (R_{X_i})_0.$$

Its local inverse comes from $r \mapsto (r, 0)$, so these inverses are compatible on the intersection. Hence

$$\widehat{N} \cong \widehat{R} = \mathcal{O}_Y = \widehat{M}.$$

This is the required pair. The component T is visible in the graded module but disappears in every localisation used by the projective cover.

Check and pitfall. Do not conclude that $T = 0$ merely because $\widehat{T} = 0$. In this example $\bar{1} \in T$ is clearly nonzero. What vanish are all the $(T_{X_i})_0$. This example proves the possibility requested by the exercise; it does not say that all different modules have the same sheaf.

2. Ten cubic sections on the projective plane

Exercise source: [Exercise 15.12](#), ID br-bgk-2019-w15-ex12. Source entity *Getwistete Strukturgarbe/Projektive Ebene/Grad 3/Basis/Aufgabe*, [revision 659923](#).

Problem statement. For a field K , give an explicit basis of

$$\Gamma(\mathbb{P}_K^2, \mathcal{O}_{\mathbb{P}_K^2}(3))$$

as a vector space over K , then determine its dimension.

Editorial solution. [Example 15.5](#) applies to projective space of dimension at least one over a field. With $d = 2$ and $\ell = 3$, it gives the identification

$$\Gamma(\mathbb{P}_K^2, \mathcal{O}_{\mathbb{P}_K^2}(3)) \cong K[X_0, X_1, X_2]_3.$$

The right-hand side is the space of homogeneous polynomials of total degree 3. Its monomials correspond to triples of nonnegative integers (a_0, a_1, a_2) satisfying $a_0 + a_1 + a_2 = 3$. The complete list gives the basis

$$\begin{aligned} \mathcal{B} = \{ & X_0^3, X_1^3, X_2^3, \\ & X_0^2 X_1, X_0^2 X_2, X_1^2 X_0, \\ & X_1^2 X_2, X_2^2 X_0, X_2^2 X_1, \\ & X_0 X_1 X_2 \}. \end{aligned}$$

There are three exponent patterns: $(3, 0, 0)$ with its three placements; $(2, 1, 0)$ with its six placements; and $(1, 1, 1)$. Thus the list has $3 + 6 + 1 = 10$ members.

To prove that the list is indeed a basis, not merely a set of ten sections, take a homogeneous polynomial F of degree 3. Its monomial expansion writes F as a linear combination of members of $\mathcal{B}$. If a linear combination of members of $\mathcal{B}$ is zero, every monomial coefficient must vanish, since monomial expansions in a polynomial ring are unique. Thus these elements are linearly independent and span the whole space. Therefore

$$\dim_K \Gamma(\mathbb{P}_K^2, \mathcal{O}_{\mathbb{P}_K^2}(3)) = 10.$$

Locally on $D_+(X_i)$, a polynomial F can be written as

$$F = \left(\frac{F}{X_i^3} \right) X_i^3.$$

Here F/X_i^3 is a regular function of degree zero, while X_i^3 is a local generator of the sheaf $\mathcal{O}(3)$. This expression explains how the homogeneous polynomial represents a section and why the global identification above does not claim that F is an ordinary global regular function on $\mathbb{P}_K^2$.

Check and pitfall. The counting formula $\binom{3+2}{2} = 10$ checks the number. Do not include monomials of lower degree: the degree is exactly 3, not at most 3. The argument does not divide by 2 or 3, so it remains valid in every characteristic.

3. Tensoring two twists adds their degrees

Exercise source: [Exercise 15.13](#), ID br-bgk-2019-w15-ex13. Source entity *Projektives Spektrum/Getwistete Strukturargen/Tensorierung/Aufgabe*, [revision 1097158](#).

Problem statement. Let R be a standard graded commutative ring and $Y = \text{Proj}(R)$. For $\ell, m \in \mathbb{Z}$, prove

$$\mathcal{O}_Y(\ell) \otimes_{\mathcal{O}_Y} \mathcal{O}_Y(m) \cong \mathcal{O}_Y(\ell + m).$$

Editorial solution. If Y is empty, the statement holds immediately for sheaves on the empty space. Otherwise, choose degree-one generators x_i of R as an algebra over R_0 . The sets $U_i = D_+(x_i)$ cover Y . Write $A_i = (R_{x_i})_0$, so $U_i = \text{Spek}(A_i)$.

With the lecture's shift convention, the sheaf $\mathcal{O}_Y(n)$ on U_i is associated to the module

$$(R_{x_i})_n = A_i x_i^n.$$

This equality holds for every $n \in \mathbb{Z}$, including $n < 0$: x_i is invertible in R_{x_i} , and dividing a degree- n element by x_i^n gives a degree-zero element. Thus x_i^n is a basis of this free rank-one module on a nonempty chart. This is the trivialisation in [Lemma 15.6](#).

Multiplication in the graded localised ring defines the map

$$\begin{aligned} \mu_i : (A_i x_i^\ell) \otimes_{A_i} (A_i x_i^m) &\longrightarrow A_i x_i^{\ell+m}, \\ (a x_i^\ell) \otimes (b x_i^m) &\longmapsto a b x_i^{\ell+m}. \end{aligned}$$

This map is A_i -linear and respects the tensor relation $(c a x_i^\ell) \otimes (b x_i^m) = (a x_i^\ell) \otimes (c b x_i^m)$. Its inverse is explicitly

$$\nu_i(c x_i^{\ell+m}) = (c x_i^\ell) \otimes x_i^m.$$

The composite $\mu_i \nu_i$ is the identity. Conversely,

$$\nu_i \mu_i((a x_i^\ell) \otimes (b x_i^m)) = (a b x_i^\ell) \otimes x_i^m = (a x_i^\ell) \otimes (b x_i^m),$$

so $\nu_i \mu_i$ is also the identity. Passing to associated sheaves gives the desired isomorphism on U_i .

We must still check that these local isomorphisms glue. On $U_i \cap U_j$, the element x_j/x_i is a degree-zero unit, and the local bases are related by

$$x_j^n = \left(\frac{x_j}{x_i} \right)^n x_i^n.$$

For the tensor product, the change-of-basis factor is $(x_j/x_i)^\ell (x_j/x_i)^m = (x_j/x_i)^{\ell+m}$, exactly the change-of-basis factor for the target sheaf. Thus μ_i and μ_j give the same map on the intersection. They glue to a global morphism

$$\mu : \mathcal{O}_Y(\ell) \otimes_{\mathcal{O}_Y} \mathcal{O}_Y(m) \longrightarrow \mathcal{O}_Y(\ell + m).$$

This morphism is an isomorphism on an open cover, hence a global isomorphism.

Check and pitfall. Taking $m = -\ell$ gives $\mathcal{O}_Y(\ell) \otimes_{\mathcal{O}_Y} \mathcal{O}_Y(-\ell) \cong \mathcal{O}_Y$. The argument uses invertibility of x_i only *on the chart* $D_+(x_i)$, not an assumption that x_i is a global unit. Standard grading provides the cover by degree-one elements; this hypothesis cannot be dropped from the proof.

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BGK 23 mastery exercises

The exercises below come from the course by Holger Brenner and Wikiversity contributors. The solutions and checking notes are **independent editorial material**, not public source solutions. The source freeze contains no public solution pages for these three exercises; this editorial addition does not change the negative result recorded in the source edition.

We write commutative group operations additively. A group D is called divisible if, for every $n \geq 1$, multiplication by n on D is surjective. An injective module must satisfy the extension property while preserving the specified scalar structure. The Indonesian term *flasid* means *flasque*, as in the lecture: all restriction maps of the sheaf are surjective.

1. Divisible as a group, but not injective as a module

Exercise source: [Exercise 23.11](#), ID br-bgk-2019-w23-ex11. Source entity *Modul/Divisible Gruppe/Nicht injektiv/Aufgabe*, [revision 837979](#).

Problem statement. Give an example of a commutative ring R and an R -module M that is not injective, although M is divisible as a commutative group.

Editorial solution. Take

$$R = \mathbb{Q}[X], \quad M = R.$$

The additive group of M is divisible: for every polynomial $f(X) \in \mathbb{Q}[X]$ and every integer $n \geq 1$, the polynomial $f(X)/n$ still has rational coefficients and satisfies $n(f/n) = f$. This statement uses only addition and multiplication by integers.

To test injectivity as an R -module, consider the ideal inclusion $XR \subseteq R$ and the map

$$\varphi : XR \longrightarrow M, \quad Xf(X) \mapsto f(X).$$

This map is well-defined because the representation Xf determines f uniquely: X is not a zero divisor in $\mathbb{Q}[X]$. For $a, f \in R$, we have $\varphi(aXf) = af = a\varphi(Xf)$, so φ is indeed R -linear.

If M were injective, [Definition 23.1](#) would give an R -linear extension

$$\psi : R \longrightarrow M, \quad \psi|_{XR} = \varphi.$$

Write $p(X) = \psi(1)$. By linearity, every $a \in R$ satisfies $\psi(a) = ap(X)$. In particular,

$$1 = \varphi(X) = \psi(X) = Xp(X).$$

This is impossible in $\mathbb{Q}[X]$: the right-hand side has constant term zero, whereas the left-hand side has constant term one. Hence φ has no R -linear extension, so M is not injective as an R -module.

Check and pitfall. Dividing coefficients by an integer n is a legitimate operation in M ; dividing the polynomial 1 by X is not. [Lemma 23.5](#) identifies divisibility with injectivity for commutative groups, that is, $\mathbb{Z}$ -modules. It does not identify divisibility of the additive group with injectivity over an arbitrary larger ring R .

2. An injective resolution of length one for every commutative group

Exercise source: [Exercise 23.13](#), ID br-bgk-2019-w23-ex13. Source entity *Kommutative Gruppe/Kurze injektive Auflösung/Aufgabe*, [revision 1039002](#).

Problem statement. Prove that every commutative group G has an injective resolution of the form

$$0 \longrightarrow G \longrightarrow I_0 \longrightarrow I_1 \longrightarrow 0.$$

Editorial solution. We give a construction and check its exactness. Choose a generating set S for G ; the set of all elements of G itself may be used. There is a surjection

$$p : F = \mathbb{Z}^{(S)} \longrightarrow G$$

sending the basis vector e_s to the generator s . Write $H = \ker p$. The first isomorphism theorem gives $G \cong F/H$.

Embed F in the rational vector space

$$V = \mathbb{Q}^{(S)}.$$

Parentheses in the superscript denote a **direct sum**: each vector has only finitely many nonzero coordinates. We can divide such a vector by every positive integer without changing its finite-support property. Thus the additive group of V is divisible.

Since $H \subseteq F \subseteq V$, set

$$I_0 = V/H, \quad I_1 = V/F.$$

Both groups are divisible. Explicitly, for a class $v + H \in V/H$ and $n \geq 1$, the class $(v/n) + H$ satisfies $n((v/n) + H) = v + H$; the same argument applies modulo F . By [Lemma 23.5](#), every divisible commutative group is injective as a $\mathbb{Z}$ -module. Hence I_0 and I_1 are injective in the category being used.

Define

$$\begin{aligned} j : F/H &\longrightarrow V/H, & f + H &\longmapsto f + H, \\ d : V/H &\longrightarrow V/F, & v + H &\longmapsto v + F. \end{aligned}$$

Both maps are well-defined because $H \subseteq F$. The map j is injective: if the image of $f + H$ is zero in V/H , then $f \in H$, so the original class is zero in F/H . The map d is surjective, since every class $v + F$ has preimage $v + H$.

Finally,

$$\begin{aligned}\ker d &= \{v + H \in V/H \mid v \in F\} \\ &= F/H = \operatorname{im} j.\end{aligned}$$

Under the identification $G \cong F/H$, we obtain the short exact sequence

$$0 \longrightarrow G \xrightarrow{j} I_0 \xrightarrow{d} I_1 \longrightarrow 0$$

with both terms I_0, I_1 injective, exactly as required.

Check and pitfall. For $G = \mathbb{Z}$, the construction can be chosen as $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$. This sequence need not split: injectivity of I_0 does not force its subgroup G to be a direct summand. [Lemma 23.6](#) asserts splitting when the **left-hand term** of a short exact sequence is injective, not merely its middle term. The exercise's result is also specific to commutative groups; it is not a bound on the length of injective resolutions for modules over an arbitrary commutative ring.

3. The sheaf of all functions is flasque

Exercise source: [Exercise 23.19](#), ID br-bgk-2019-w23-ex19. Source entity *Kommutative Gruppe/Abbildungen/Garbe/Welk/Aufgabe*, [revision 1081885](#).

Problem statement. Let G be a commutative group and X a topological space. Prove that the sheaf

$$\mathcal{F}(U) = \operatorname{Abb}(U, G)$$

on X is flasque. The notation $\operatorname{Abb}(U, G)$ means all set maps from U to G , with no continuity requirement.

Editorial solution. Addition on $\mathcal{F}(U)$ is pointwise, and for open $U \subseteq V$ the restriction map is $r_{V,U}(s) = s|_U$. This is a group homomorphism and plainly satisfies the identity and composition compatibility of restrictions.

First check the sheaf property. Let $U = \bigcup_{i \in I} U_i$ be an open cover, and let functions $s_i : U_i \rightarrow G$ agree on each intersection $U_i \cap U_j$. For $x \in U$, choose an i with $x \in U_i$ and set $s(x) = s_i(x)$. Agreement on intersections ensures that this value is independent of the choice of i . Thus $s : U \rightarrow G$ restricts to s_i on each U_i . A function with this property is unique, since every point of U lies in some U_i . On the empty set there is just one function $\emptyset \rightarrow G$, as the sheaf axiom requires. Thus $\mathcal{F}$ is indeed a sheaf of commutative groups.

Now take open sets $U \subseteq V$ and a section $s \in \mathcal{F}(U)$. With 0_G the identity element of G , define the function

$$\tilde{s} : V \longrightarrow G, \quad \tilde{s}(x) = \begin{cases} s(x), & x \in U, \\ 0_G, & x \in V \setminus U. \end{cases}$$

Since $\mathcal{F}(V)$ contains **all** functions, $\tilde{s}$ is a legitimate section; we do not need $V \setminus U$ to be open. Clearly $r_{V,U}(\tilde{s}) = s$. Thus every section on U extends to V , so $r_{V,U}$ is surjective.

The argument applies to every $U \subseteq V$, including $U = \emptyset$. By [Definition 23.14](#), $\mathcal{F}$ is flasque.

Check and pitfall. Extension by zero above even gives a group-homomorphism right inverse to each restriction. However, the same method does not automatically work for a sheaf of **continuous functions**: assigning zero outside U can destroy continuity at the boundary of U . For example, $x \mapsto 1/x$ on $(0, 1) \subseteq \mathbb{R}$ has no real-valued continuous extension to all of $\mathbb{R}$. The absence of a continuity requirement in $\operatorname{Abb}(U, G)$ is an essential part of the proof.

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BGK 24 mastery exercises: Hom and Ext

The following three exercises come from Holger Brenner's course. The solutions here are new editorial material, not public source solutions or translations of the author's work. The source worksheet and its record of the absence of public solutions are preserved. This sequence connects left exactness of Hom, the property of projective modules, and a nonzero Ext¹ computation.

Source Exercise 24.1: left exactness of Hom

Source: [Exercise 24.1 in the BGK reader](#), entity Modul/Homomorphismenmodul/Kovariant/Linksexakt/Aufgabe, revision 1081675, page 114118. The exercise is preserved without changing its hypotheses.

Problem statement

Let R be a commutative ring, A an R -module, and

$$0 \longrightarrow L \xrightarrow{i} M \xrightarrow{p} N \longrightarrow 0$$

a short exact sequence of R -modules. Prove that the sequence

$$0 \longrightarrow \operatorname{Hom}_R(A, L) \xrightarrow{i_*} \operatorname{Hom}_R(A, M) \xrightarrow{p_*} \operatorname{Hom}_R(A, N)$$

is exact. Here $i_*(f) = i \circ f$ and $p_*(g) = p \circ g$.

Complete editorial solution

First, i_* is injective. If $i \circ f = 0$, then for every $a \in A$ we have $i(f(a)) = 0$. Since i is injective, $f(a) = 0$ for every a , so $f = 0$.

Next, $p \circ i = 0$ gives

$$p_*(i_*(f)) = p \circ i \circ f = 0.$$

Thus $\operatorname{im}(i_*) \subseteq \ker(p_*)$. For the reverse inclusion, take $g : A \rightarrow M$ with $p \circ g = 0$. Every $g(a)$ lies in $\ker(p) = \operatorname{im}(i)$. Since i is injective, there is exactly one element $f(a) \in L$ such that $i(f(a)) = g(a)$. This defines a map $f : A \rightarrow L$.

The map is linear. For $a, a' \in A$ and $r \in R$,

$$\begin{aligned} i(f(a + a')) &= g(a + a') = g(a) + g(a') = i(f(a) + f(a')), \\ i(f(ra)) &= g(ra) = rg(a) = i(rf(a)). \end{aligned}$$

Injectivity of i gives $f(a + a') = f(a) + f(a')$ and $f(ra) = rf(a)$. Hence $f \in \operatorname{Hom}_R(A, L)$ and $g = i_*(f)$. Thus $\ker(p_*) = \operatorname{im}(i_*)$, as required.

Check and pitfall

The requested sequence has no 0 on the right. Surjectivity of p does not guarantee that every map $A \rightarrow N$ lifts to M . The proof only lifts maps whose images already lie in $\ker(p)$ to the module L ; that differs from lifting maps to N .

Source Exercise 24.3: a projective module in the first argument

Source: [Exercise 24.3 in the BGK reader](#), entity Projektiver Modul/Extmoduln/Aufgabe, revision 1039771, page 114115. The definition used is [Definition 24.11](#).

Problem statement

Let R be a commutative ring, P a projective R -module, and M an R -module. Prove that

$$\operatorname{Ext}_R^n(P, M) = 0 \quad (n \geq 1).$$

Complete editorial solution

Take an injective resolution of the second argument,

$$0 \longrightarrow M \longrightarrow I^0 \xrightarrow{d^0} I^1 \xrightarrow{d^1} I^2 \longrightarrow \dots$$

By definition, $\operatorname{Ext}_R^n(P, M) = H^n(\operatorname{Hom}_R(P, I^\bullet))$. We show directly that every positive-degree cocycle is a coboundary.

Fix $n \geq 1$ and take a homomorphism $f : P \rightarrow I^n$ that is a cocycle, meaning $d^n \circ f = 0$. With $Z^n = \ker(d^n)$, the map f factors through a homomorphism $\bar{f} : P \rightarrow Z^n$. Exactness of the resolution gives a surjection

$$\bar{d}^{n-1} : I^{n-1} \longrightarrow Z^n, \quad u \longmapsto d^{n-1}(u).$$

Projectivity of P means that homomorphisms from P can be lifted through every surjection. Thus there is $g : P \rightarrow I^{n-1}$ with $\bar{d}^{n-1} \circ g = \bar{f}$. Including Z^n into I^n , we obtain

$$d^{n-1} \circ g = f.$$

Thus f is indeed a coboundary. Since every cocycle has this form, the quotient of cocycles by coboundaries is zero in every degree $n \geq 1$. This is the required statement.

Check and pitfall

It is P , the first argument, that must be projective; the injective resolution is still taken of M , the second argument. We do not assert that P is injective. Nor does the conclusion hold in degree zero: for example, $\operatorname{Ext}_R^0(R, M) = \operatorname{Hom}_R(R, M) \cong M$ can be nonzero.

Source Exercise 24.4: Ext classes detected modulo k

Source: [Exercise 24.4 in the BGK reader](#), entity Extmodul/1/ZmodkundZ/Nicht0/Aufgabe, [revision 1107271](#), page 114122. The computation below proves the source's nonvanishing conclusion and also determines the group; the hypothesis $k \geq 2$ is unchanged.

Problem statement

Using the short exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot k} \mathbb{Z} \longrightarrow \mathbb{Z}/(k) \longrightarrow 0,$$

prove that $\operatorname{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/(k), \mathbb{Z})$ is nonzero for $k \geq 2$.

Complete editorial solution

Write $A = \mathbb{Z}/(k)$ and take an injective resolution

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\iota} I^0 \xrightarrow{d^0} I^1 \xrightarrow{d^1} I^2 \longrightarrow \dots$$

We will construct an isomorphism

$$\mathbb{Z}/(k) \xrightarrow{\sim} H^1(\operatorname{Hom}_{\mathbb{Z}}(A, I^\bullet)) = \operatorname{Ext}_{\mathbb{Z}}^1(A, \mathbb{Z}).$$

For $n \in \mathbb{Z}$, define a homomorphism from the subgroup $k\mathbb{Z} \subseteq \mathbb{Z}$ to I^0 by $k \mapsto \iota(n)$. Injectivity of I^0 extends it to a homomorphism $\mathbb{Z} \rightarrow I^0$. If the image of 1 is y , then $ky = \iota(n)$. Since $d^0 \iota = 0$, the element $d^0 y$ is killed by k . Hence there is a homomorphism

$$f_n : A \longrightarrow I^1, \quad \bar{1} \longmapsto d^0 y.$$

The equality $d^1 d^0 = 0$ shows that f_n is a cocycle. Set $\delta(n) = [f_n]$. If y' is another choice, $k(y - y') = 0$, so $b : A \rightarrow I^0$, $b(\bar{1}) = y - y'$, is well-defined. The difference $f_n - f'_n = d^0 \circ b$ is a coboundary. Thus $\delta(n)$ is independent of the choice of y . Choosing $y + y'$ for $n + n'$ also shows that δ is additive.

Now compute its kernel. If $n = km$, we may take $y = \iota(m)$; then $f_n = 0$ and $k\mathbb{Z} \subseteq \ker \delta$. Conversely, suppose $\delta(n) = 0$. There is $b : A \rightarrow I^0$ with $f_n = d^0 \circ b$. For $b_0 = b(\bar{1})$ we have $kb_0 = 0$ and $d^0(y - b_0) = 0$. Exactness of the resolution gives $y - b_0 = \iota(m)$ for some $m \in \mathbb{Z}$. Multiply by k :

$$\iota(n) = ky = kb_0 + k\iota(m) = \iota(km).$$

Since ι is injective, $n = km$. Thus $\ker \delta = k\mathbb{Z}$.

Finally, δ is surjective. A cocycle $f : A \rightarrow I^1$ is determined by $z = f(\bar{1})$ with $kz = 0$ and $d^1 z = 0$. Exactness of the resolution gives $z = d^0 y$ for some $y \in I^0$. Now $d^0(ky) = kz = 0$, so $ky = \iota(n)$ for some integer n . The construction above yields $f_n = f$. Thus every cohomology class lies in the image of δ .

The first isomorphism theorem gives

$$\text{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/(k), \mathbb{Z}) \cong \mathbb{Z}/(k).$$

The class $\bar{1}$ is nonzero when $k \geq 2$, so this Ext group is nonzero. The inclusion $k\mathbb{Z} \subseteq \mathbb{Z}$ used in the construction is precisely the image of the map $\cdot k$ in the source's short exact sequence.

Check and pitfall

Do not apply a long exact sequence in the first argument of Ext without explaining why: the lecture's definition uses a resolution of the second argument. The cocycle computation above works directly with that definition. Also check $k = 1$: the result is the zero group, so the bound $k \geq 2$ is genuinely needed for the source's conclusion.

Origin and licence of the material

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BGK 25 mastery exercises: local representatives and first cohomology

The following two solutions are new editorial material for Holger Brenner's exercises, not public source solutions. Together with the [public solution to Exercise 25.1](#), they form three mastery items for Unit 25. That public solution remains in its original place and retains its source attribution "essentially Tarek Emmrich"; it is neither copied nor counted as new editorial work.

Source Exercise 25.2: gluing representatives modulo continuous functions

Source: [Exercise 25.2 in the BGK reader](#), entity Intervall/Intervallüberdeckung/2/FunktionenmodulostetigeFunktionen/Global surjektiv/Aufgabe, [revision 1096998](#), page 114224. No adaptation has been made to the exercise's hypotheses.

Problem statement

Let $I \subseteq \mathbb{R}$ be a real interval and $I = U \cup V$, where U, V are intervals relatively open in I . Consider the short exact sequence of sheaves

$$0 \longrightarrow C^0(-, \mathbb{R}) \longrightarrow \text{Abb}(-, \mathbb{R}) \longrightarrow \mathcal{Q} \longrightarrow 0, \quad \mathcal{Q} = \text{Abb}(-, \mathbb{R})/C^0(-, \mathbb{R}).$$

A section $q \in \Gamma(I, \mathcal{Q})$ is represented on U by $s : U \rightarrow \mathbb{R}$ and on V by $t : V \rightarrow \mathbb{R}$. Prove that q has a representative given by a map $r : I \rightarrow \mathbb{R}$. The notation Abb means all maps; s, t, r are not required to be continuous.

Complete editorial solution

Write $W = U \cap V$. Since s and t represent restrictions of the same section q , the image of $s|_W - t|_W$ in $\mathcal{Q}(W)$ is zero. Exactness at the middle sheaf shows that

$$f := s|_W - t|_W \in C^0(W, \mathbb{R}).$$

This step can also be read directly locally: every point of W has a neighbourhood on which $s - t$ is continuous; continuity is a local property, so $s - t$ is continuous on all of W . We are not assuming that evaluation on global sections preserves all sheaf surjections.

The result of [Exercise 25.1 and its source solution](#) gives continuous functions $g : U \rightarrow \mathbb{R}$ and $h : V \rightarrow \mathbb{R}$ with

$$f = g|_W - h|_W.$$

Here h denotes the function on V . In the public source solution, the second piecewise formula defines $-h$, not h ; with that sign convention the displayed decomposition is exactly $f = g|_W - h|_W$. If one interval is contained in the other, the decomposition can be taken directly as $g = f, h = 0$ when $U \subseteq V$, or $g = 0, h = -f$ when $V \subseteq U$. If W is empty, use $g = h = 0$. Thus the boundary cases require no additional hypothesis on the cover.

Define maps on the two members of the cover by

$$r_U = s - g, \quad r_V = t - h.$$

On the intersection,

$$r_U|_W - r_V|_W = (s|_W - t|_W) - (g|_W - h|_W) = f - f = 0.$$

Thus the formula

$$r(x) = \begin{cases} s(x) - g(x), & x \in U, \\ t(x) - h(x), & x \in V \end{cases}$$

is well-defined and gives a map $I \rightarrow \mathbb{R}$. The difference $r|_U - s = -g$ is continuous, so $r|_U$ and s have the same image in $\mathcal{Q}(U)$. Likewise, $r|_V - t = -h$ is continuous. Thus the global image of r and the section q agree on the cover U, V ; the uniqueness axiom of the sheaf $\mathcal{Q}$ says they agree on I .

Check and pitfall

What is glued is $s - g$ and $t - h$, not s and t themselves. The minus signs matter: the condition $s - t = g - h$ gives exactly $s - g = t - h$. Moreover, $\mathcal{Q}$ is the quotient in the category of sheaves; in general, $\mathcal{Q}(I)$ must not be identified from the outset with $\text{Abb}(I, \mathbb{R})/C^0(I, \mathbb{R})$. The existence of a global representative in the situation of this exercise is precisely what must be proved.

Source Exercise 25.10: the sheaf of units and the function field

Source: [Exercise 25.10 in the BGK reader](#), entity Schema/Integer/Einheitengarbe/Funktionenkörpergruppe/Erste Kohomologie/Fakt/Beweis/Aufgabe, [revision 1082082](#), page 114519. This supplies an editorial proof of the result stated as [Lemma 25.10](#).

Problem statement

Let $(X, \mathcal{O}_X)$ be an integral scheme with function field K . Let $\mathcal{O}_X^\times$ be the sheaf of units and $\mathcal{U}$ the constant sheaf with value $K^\times$. Prove the identification

$$H^1(X, \mathcal{O}_X^\times) \cong \frac{\Gamma(X, \mathcal{U}/\mathcal{O}_X^\times)}{\text{im}(K^\times \rightarrow \Gamma(X, \mathcal{U}/\mathcal{O}_X^\times))}.$$

The symbol $=$ in the source statement denotes this natural identification. Unit groups are written multiplicatively; their identity element is 1.

Complete editorial solution

Since X is integral, its underlying topological space is irreducible and all rings of nonempty affine charts are integral domains. There is a generic point η with $\mathcal{O}_{X,\eta} = K$. Every nonempty open set contains η and is irreducible, hence also connected. Consequently a locally constant function with values in $K^\times$ on a nonempty open set is constant. Thus

$$\Gamma(V, \mathcal{U}) = K^\times \quad \text{for } V \neq \emptyset, \quad \Gamma(\emptyset, \mathcal{U}) = \{1\}.$$

Every restriction between two nonempty open sets is the identity on $K^\times$; restriction to the empty set is also surjective. Thus $\mathcal{U}$ is flasque. By [Lemma 25.3](#), flasque sheaves are acyclic, and in particular $H^1(X, \mathcal{U}) = 0$.

Evaluation at the generic point gives an embedding $\mathcal{O}_X^\times \hookrightarrow \mathcal{U}$. To see injectivity, on an integral affine chart $\text{Spec}(R)$, sections on principal opens lie in localisations $R_f \subseteq K$. Two sections equal as elements of K agree on every such chart, and hence as sheaf sections. A unit section has an inverse also mapping into K , so its image lies in $K^\times$. This map is compatible with restrictions.

Write $\mathcal{D} = \mathcal{U}/\mathcal{O}_X^\times$, the quotient sheaf of abelian groups. We obtain a short exact sequence of sheaves

$$1 \rightarrow \mathcal{O}_X^\times \rightarrow \mathcal{U} \rightarrow \mathcal{D} \rightarrow 1.$$

The groups are commutative because they come from units in commutative rings. Thus [Corollary 25.2](#) applies. The beginning of the long exact cohomology sequence is

$$\Gamma(X, \mathcal{O}_X^\times) \rightarrow K^\times \xrightarrow{\alpha} \Gamma(X, \mathcal{D}) \xrightarrow{\delta} H^1(X, \mathcal{O}_X^\times) \rightarrow H^1(X, \mathcal{U}) = 0.$$

Exactness at $H^1(X, \mathcal{O}_X^\times)$ makes δ surjective. Exactness at $\Gamma(X, \mathcal{D})$ says $\ker \delta = \text{im } \alpha$. The first isomorphism theorem for abelian groups now gives the required identification.

The local meaning of the result can be explained without changing the computation. A section $d \in \Gamma(X, \mathcal{D})$ has local representatives $q_i \in K^\times$ on a cover by nonempty open sets (V_i) , with

$$q_i/q_j \in \mathcal{O}_X^\times(V_i \cap V_j).$$

The class $\delta(d)$ is zero exactly when a single $q \in K^\times$ represents d on the entire space. Locally this means

$$q_i/q \in \mathcal{O}_X^\times(V_i) \quad \text{for every } i.$$

Thus quotienting by the image of $K^\times$ disregards changes of all local representatives by the same nonzero rational function. This does not require that rational function to be a global regular unit.

Check and pitfall

A constant sheaf is not flasque on an arbitrary topological space. The proof here uses irreducibility of X to ensure that every nonempty open set is connected. Nor should the denominator $\text{im}(K^\times)$ be replaced by all of $\Gamma(X, \mathcal{D})$, since that would erase the cohomological obstruction being computed. If $X = \text{Spek}(K)$ is a single point, then $\mathcal{O}_X^\times = \mathcal{U}$, $\mathcal{D}$ is trivial, and both sides are indeed zero as abelian groups.

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BGK 26 mastery exercises

The three exercises below come from Holger Brenner's course and Wikiversity contributions whose revision identities are preserved in the edition. **The following solutions are new editorial material, not public solutions by Brenner or translations of source solutions.** The source record stating that no public solutions exist remains applicable.

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1. Zeroth cohomology and the gluing axiom

Source exercise: [Exercise 26.2 in the reader](#). Exact identifier: Cech-Kohomologie/0/GlobaleAuswertung/Aufgabe; [source revision 1082005](#). Its placement is frozen in Worksheet 26, revision 619292.

Statement. Let $X = \bigcup_{i \in I} U_i$ be an open cover of a topological space X , and $\mathcal{G}$ a sheaf of commutative groups on X . Prove that

$$\check{H}^0(\mathcal{U}, \mathcal{G}) \cong \Gamma(X, \mathcal{G}).$$

The equality in the source statement is read as the canonical identification by restriction of global sections, not literal equality of two set constructions.

Editorial solution. Order the index set as in [Definition 26.3](#). The first two terms of the complex are

$$\check{C}^0 = \prod_i \Gamma(U_i, \mathcal{G}), \quad \check{C}^1 = \prod_{i < j} \Gamma(U_i \cap U_j, \mathcal{G}).$$

The first differential has components

$$(\delta_0(s))_{ij} = s_j|_{U_i \cap U_j} - s_i|_{U_i \cap U_j}.$$

Since the complex used starts in degree 0, no nonzero coboundaries enter degree 0. Hence

$$\check{H}^0(\mathcal{U}, \mathcal{G}) = \ker \delta_0.$$

Define the homomorphism

$$\rho : \Gamma(X, \mathcal{G}) \longrightarrow \check{C}^0, \quad t \longmapsto (t|_{U_i})_i.$$

Two restrictions of the same section certainly agree on every intersection. Thus $\rho(t) \in \ker \delta_0$.

Conversely, take $(s_i)_i \in \ker \delta_0$. The equation $\delta_0(s) = 0$ says exactly that

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$$

for all i, j . The sheaf gluing axiom gives a section $t \in \Gamma(X, \mathcal{G})$ with $t|_{U_i} = s_i$ for every i . The uniqueness axiom ensures that this section is unique. Thus ρ is surjective onto the kernel and also injective: if all restrictions of t are zero, uniqueness of gluing forces $t = 0$.

Restriction preserves addition. Gluing also preserves it, because the section gluing the family $(s_i + s'_i)$ is the sum of the two glued sections, again by uniqueness. Therefore ρ is the required group isomorphism. This proof requires neither a finite cover, connectedness of the space, nor acyclicity.

Check and pitfall. For a single open set $U_1 = X$, we have $\check{C}^1 = 0$ and immediately obtain $\check{H}^0 = \Gamma(X, \mathcal{G})$ (and, for this singleton cover, $\check{H}^1 = 0$). For a general cover, do not conclude that $\check{H}^1 = 0$ from the degree-0 argument: the sheaf axiom gives exactness in degree 0, not automatically in every degree.

2. A constant sheaf on an irreducible space

Source exercise: [Exercise 26.6 in the reader](#). Exact identifier: IrreduziblerRaum/KonstanteGarbe/Cech-Kohomologie/Aufgabe; [source revision 1081578](#). Its placement is frozen in Worksheet 26, revision 619292.

Statement. Let X be an irreducible topological space and $\mathcal{G} = \underline{G}$ the constant sheaf associated to a commutative group G . Determine the Čech complex and its cohomology for a finite open cover $\mathcal{U} = (U_i)_{i \in I}$.

Editorial solution. We use the usual convention that an irreducible space is nonempty. If another convention allows the empty space, the case $X = \emptyset$ is separate: all section groups and cohomology groups are zero. Empty members of the cover may be removed without changing the complex, since every intersection involving them contributes the zero group.

Label the remaining cover indices $\{0, 1, \dots, m-1\}$, with $m \geq 1$. In an irreducible space, any two nonempty open sets intersect. By induction, every finite intersection U_J is also nonempty. Moreover, U_J is irreducible: two nonempty relatively open subsets of it are two nonempty open subsets of X , so they intersect. In particular, U_J is connected.

Sections of the constant sheaf $\underline{G}$ on an open set can be viewed as locally constant functions to G equipped with the discrete topology. On a connected space such a function is constant: if two different values occurred, the preimage of one value and its complement would separate the space into two nonempty open sets. Thus

$$\Gamma(U_J, \underline{G}) = G,$$

and all restrictions between nonempty intersections are identities on G .

The complex is therefore

$$0 \longrightarrow G^m \xrightarrow{\delta_0} G^{\binom{m}{2}} \xrightarrow{\delta_1} \dots \xrightarrow{\delta_{m-2}} G \longrightarrow 0,$$

with G^m in degree 0 and the last G in degree $m-1$. Uniformly,

$$\check{C}^q = G^{\binom{m}{q+1}}, \quad (\delta_q s)_{i_0 \dots i_{q+1}} = \sum_{r=0}^{q+1} (-1)^r s_{i_0 \dots \widehat{i_r} \dots i_{q+1}}.$$

Terms with $q \geq m$ are zero. For $m=1$, the complex consists only of G in degree 0.

Zeroth cohomology is the diagonal

$$\ker \delta_0 = \{(g, \dots, g) : g \in G\} \cong G.$$

To show that all positive cohomology vanishes, we give an explicit homotopy. For $q \geq 1$, define $h_q : \check{C}^q \rightarrow \check{C}^{q-1}$ by

$$(h_q s)_{i_0 \dots i_{q-1}} = \begin{cases} s_{0i_0 \dots i_{q-1}}, & 0 \notin \{i_0, \dots, i_{q-1}\}, \\ 0, & 0 \in \{i_0, \dots, i_{q-1}\}. \end{cases}$$

Putting index 0 first requires no additional restrictions: all the groups involved have been identified with G .

Check a tuple $L = (i_0 < \dots < i_q)$. If $0 \in L$, the component $(h_{q+1} \delta_q s)_L$ is zero. In $(\delta_{q-1} h_q s)_L$, only the term omitting index 0 is nonzero, and that term equals s_L .

If $0 \notin L$, then

$$(h_{q+1} \delta_q s)_L = (\delta_q s)_{0i_0 \dots i_q} = s_L + \sum_{r=0}^q (-1)^{r+1} s_{0, L \setminus \{i_r\}},$$

whereas

$$(\delta_{q-1} h_q s)_L = \sum_{r=0}^q (-1)^r s_{0, L \setminus \{i_r\}}.$$

The two alternating sums cancel. In both cases,

$$\delta_{q-1}h_q + h_{q+1}\delta_q = \text{id}_{\check{C}^q}.$$

If $\delta_q s = 0$, this equation gives $s = \delta_{q-1}(h_q s)$. Thus every positive-degree cocycle is a coboundary, and

$$\boxed{\check{H}^0(\mathcal{U}, \underline{G}) \cong G, \quad \check{H}^q(\mathcal{U}, \underline{G}) = 0 \quad (q \geq 1).}$$

Check and pitfall. For $m = 2$, the complex is $G \oplus G \rightarrow G$, $(a, b) \mapsto b - a$; the kernel is the diagonal and the image is all of G . Irreducibility is used to ensure that all intersections are nonempty and connected. Connectedness of X alone does not ensure this. Also note the last degree $m - 1$: the definition of $\check{C}^q$ uses $q + 1$ indices, not q indices.

3. Solving a cocycle on two affine open sets

Source exercise: [Exercise 26.8 in the reader](#). Exact identifier: AffinesSchema/Zweierüberdeckung/Strukturgabe/Cech-Kohomologie/Aufgabe; [source revision 1038046](#). The spelling Strukturgabe in the source identifier is preserved. Its placement is frozen in Worksheet 26, revision 619292.

Statement. Let R be a commutative ring and

$$X = \text{Spek}(R) = D(f) \cup D(g).$$

Prove that

$$\check{H}^1(\{D(f), D(g)\}, \mathcal{O}_X) = 0.$$

Editorial solution. Ordering $D(f)$ before $D(g)$, the Čech complex of the structure sheaf is

$$0 \longrightarrow R_f \oplus R_g \xrightarrow{\delta_0} R_{fg} \longrightarrow 0, \quad \delta_0(a, b) = b - a.$$

Here both terms on the right are restricted to $D(fg)$ before subtraction. There are no intersections with three indices, so every element of R_{fg} is a cocycle and

$$\check{H}^1 = R_{fg} / \text{im } \delta_0.$$

We will explicitly write every cocycle as a coboundary.

The cover condition gives

$$\emptyset = X \setminus (D(f) \cup D(g)) = V(f, g).$$

If the ideal (f, g) were proper, it would be contained in a maximal ideal, giving a point of $V(f, g)$, a contradiction. Thus $(f, g) = R$. More generally, for every integer $N \geq 1$, we have $V(f^N, g^N) = V(f, g) = \emptyset$, so there are $u, v \in R$ with

$$uf^N + vg^N = 1.$$

Take $c \in R_{fg}$ and write

$$c = \frac{r}{(fg)^N}$$

for some $r \in R$ and $N \geq 1$. Even an element with denominator of exponent zero can be written this way by multiplying numerator and denominator by fg . In R_{fg} , the equality above yields

$$c = \frac{r(uf^N + vg^N)}{f^N g^N} = \frac{ur}{g^N} + \frac{vr}{f^N}.$$

Define

$$a = -\frac{vr}{f^N} \in R_f, \quad b = \frac{ur}{g^N} \in R_g.$$

Then

$$\delta_0(a, b) = b - a = \frac{ur}{g^N} + \frac{vr}{f^N} = c.$$

Thus δ_0 is surjective and the quotient group is zero.

All equalities take place in localisations, so they require neither that R be an integral domain nor that f, g be non-zero-divisors. If R is the zero ring, all modules in the complex are also zero and the conclusion remains valid.

Check and pitfall. For $R = \mathbb{Z}$, $f = 2$, $g = 3$, the identity $(-1)2 + 1 \cdot 3 = 1$ gives

$$\frac{1}{6} = -\frac{1}{3} + \frac{1}{2} = \delta_0\left(-\frac{1}{2}, -\frac{1}{3}\right).$$

If $D(f) \cup D(g)$ does not cover the whole spectrum, the equation $uf^N + vg^N = 1$ is unavailable. For example, the cover of the punctured plane by $D(X)$ and $D(Y)$ does not satisfy this hypothesis on all of $\text{Spek}(R[X, Y])$; Exercise 27.2 instead exhibits first cohomology that can be nonzero.

BGK 27 mastery exercises

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1. Three monomial components in the Čech complex

Source exercise: [Exercise 27.2 in the reader](#). Exact identifier: Polynomring/2/Cech-Komplex/Monom/Aufgabe; [source revision 1037476](#). Its placement is frozen in Worksheet 27, revision 1070033.

Statement. Let $A = R[X, Y]$ for a commutative ring R . Determine the Čech complex of the structure sheaf for the standard cover of the punctured plane for the monomials X^2Y^3 , X^5Y^{-4} , and $X^{-3}Y^{-6}$. Determine its homology in each case.

Editorial solution. The space being covered is

$$U = \text{Spek}(A) \setminus V(X, Y) = D(X) \cup D(Y),$$

not the whole affine plane. Ordering $D(X)$ before $D(Y)$, the complex is

$$0 \longrightarrow A_X \oplus A_Y \xrightarrow{\delta_0} A_{XY} \longrightarrow 0, \quad \delta_0(a, b) = b - a.$$

The cohomological degrees of the two nonzero terms are 0 and 1. The homology requested in the source exercise is computed with this cohomological grading.

Write

$$A_X = R[X, X^{-1}, Y], \quad A_Y = R[X, Y, Y^{-1}], \quad A_{XY} = R[X, X^{-1}, Y, Y^{-1}].$$

The differential preserves each exponent pair. For a monomial $m = X^\alpha Y^\beta$, the component in A_X exists exactly when $\beta \geq 0$; the component in A_Y exists exactly when $\alpha \geq 0$. The component in A_{XY} always exists. Each existing component is the free module Rm on one generator, even if R has zero divisors.

Case $m = X^2Y^3$. This monomial occurs in all three localisations, so its component complex is

$$0 \longrightarrow Rm \oplus Rm \xrightarrow{(am, bm) \mapsto (b-a)m} Rm \longrightarrow 0.$$

The differential's kernel is the diagonal $\{(am, am) : a \in R\}$. The map is surjective, since cm is the image of $(0, cm)$. Thus

$$\check{H}_{(2,3)}^0 \cong Rm, \quad \check{H}_{(2,3)}^1 = 0.$$

There are no terms in other degrees.

Case $m = X^5Y^{-4}$. The exponent of Y is negative, so this monomial does not occur in A_X . It occurs in A_Y and A_{XY} , so its complex is

$$0 \longrightarrow 0 \oplus Rm \xrightarrow{(0, bm) \mapsto bm} Rm \longrightarrow 0.$$

The differential is an isomorphism. Therefore

$$\check{H}_{(5,-4)}^0 = 0, \quad \check{H}_{(5,-4)}^1 = 0.$$

Case $m = X^{-3}Y^{-6}$. Both exponents are negative. This monomial occurs in neither A_X nor A_Y , but does occur in A_{XY} . Its complex is

$$0 \longrightarrow 0 \longrightarrow Rm \longrightarrow 0,$$

with Rm in degree 1. Hence

$$\check{H}_{(-3,-6)}^0 = 0, \quad \check{H}_{(-3,-6)}^1 \cong Rm.$$

If $R \neq 0$, the monomial class is nonzero.

Check and pitfall. The element $X^{-3}Y^{-6}$ cannot be written as a sum of Laurent polynomials from A_X and A_Y : every monomial from A_X has nonnegative Y -exponent, and every monomial from A_Y has nonnegative X -exponent. Uniqueness of coefficients in the Laurent basis proves this without assuming that R is a field. Do not add A as a new degree-0 term: the Čech complex of this cover already starts with $A_X \oplus A_Y$.

2. The module structure on monomials with all exponents negative

Source exercise: [Exercise 27.6 in the reader](#). Exact identifier: Polynomring/HöchstelokaleKohomologie/Modulstruktur/Direkt/Aufgabe; [source revision 1083806](#). Its placement is frozen in Worksheet 27, revision 1070033.

Statement. Let K be a field and $A = K[X_1, \dots, X_d]$. On the vector space

$$H = K \langle X^\nu = X_1^{\nu_1} \cdots X_d^{\nu_d} : \nu_j \leq -1 \text{ for every } j \rangle,$$

define a natural A -module structure.

Editorial solution. We explain the case $d \geq 1$, with variables present as in the exercise. Every element of H is a finite linear combination of the displayed monomials. Ordinary Laurent multiplication by X_i does not always stay in H : if $\nu_i = -1$, the result has exponent 0. The required module action must annihilate results leaving the negative-exponent region.

For $i = 1, \dots, d$, define a linear operator $T_i : H \rightarrow H$ on the basis by

$$T_i(X^\nu) = \begin{cases} X^{\nu+e_i}, & \nu_i \leq -2, \\ 0, & \nu_i = -1, \end{cases}$$

where e_i has i th component 1 and all other components 0. These operators commute. Indeed, for $i \neq j$, increasing the i th exponent does not change whether the j th exponent is still permitted. If either of ν_i, ν_j equals -1 , both composites annihilate the monomial. If both are at most -2 , both composites yield $X^{\nu+e_i+e_j}$.

Since all T_i commute, for $p = \sum_{\alpha \in \mathbb{N}^d} c_\alpha X^\alpha \in A$ we may set

$$p \cdot h = \sum_{\alpha} c_\alpha T_1^{\alpha_1} \cdots T_d^{\alpha_d}(h).$$

This sum is finite. The map $A \rightarrow \text{End}_K(H)$ sending X_i to T_i preserves addition, multiplication, and the identity. Consequently

$$(p+q) \cdot h = p \cdot h + q \cdot h, \quad p \cdot (h+h') = p \cdot h + p \cdot h', \quad (pq) \cdot h = p \cdot (q \cdot h), \quad 1 \cdot h = h.$$

These are all the required module conditions. The formula on a single monomial is

$$p \cdot X^\nu = \sum_{\substack{\alpha \in \mathbb{N}^d \\ \nu_j + \alpha_j \leq -1 \text{ for all } j}} c_\alpha X^{\nu+\alpha}.$$

Thus all terms acquiring at least one nonnegative exponent are discarded.

To see why this action is natural, consider the Laurent module

$$L = K[X_1^{\pm 1}, \dots, X_d^{\pm 1}]$$

and the following A -submodule:

$$N = \sum_{i=1}^d K[X_1^{\pm 1}, \dots, X_{i-1}^{\pm 1}, X_i, X_{i+1}^{\pm 1}, \dots, X_d^{\pm 1}].$$

Each summand is an A -submodule: multiplication by a polynomial does not turn a nonnegative i th exponent into a negative one. The module N is spanned exactly by the Laurent monomials having at least one nonnegative exponent. Since all Laurent monomials form a K -basis of L , the classes of monomials with all exponents negative form a basis of L/N . Thus there is a vector-space isomorphism

$$H \longrightarrow L/N, \quad X^\nu \mapsto [X^\nu].$$

The action defined above is exactly the quotient A -action on L/N : a monomial leaving the negative region enters N and becomes zero. This also connects the construction with the top cohomology component computed in the lecture.

Check and pitfall. For $d = 2$,

$$X_1 \cdot (X_1^{-1} X_2^{-2}) = 0, \quad X_2 \cdot (X_1^{-1} X_2^{-2}) = X_1^{-1} X_2^{-1}.$$

Although each X_i is invertible in the Laurent ring L , its action on H is not an invertible operator. We form the quotient as an A -module, not as a module over the entire Laurent ring. If $d = 0$ is allowed, the empty-product convention gives $A = H = K$ with the usual scalar action.

3. Global sections on a plane curve

Source exercise: [Exercise 27.10 in the reader](#). Exact identifier: ProjektiveEbene/Kurve/GetwisteteStrukturgabezu d-3/GlobaleSchnitte/Aufgabe; [source revision 1097429](#). The spelling Strukturgabe in the source identifier is preserved. Its placement is frozen in Worksheet 27, revision 1070033.

Statement. Let $C = V_+(f) \subset \mathbb{P}_K^2$ be a projective plane curve of degree d over a field K . Using the long exact cohomology sequence associated to

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}_K^2}(-3) \xrightarrow{f} \mathcal{O}_{\mathbb{P}_K^2}(d-3) \longrightarrow \mathcal{O}_C(d-3) \longrightarrow 0$$

and Theorem 27.4, prove that

$$\dim_K H^0(C, \mathcal{O}_C(d-3)) = \frac{(d-1)(d-2)}{2}.$$

Editorial solution. Write $S = K[X, Y, Z]$ and $P = \mathbb{P}_K^2$. The statement that the curve is given by f means that f is a nonzero homogeneous polynomial of degree $d \geq 1$. Neither algebraic closedness of K nor smoothness of C is required for this computation.

Let $i : C \hookrightarrow P$ be the closed immersion. In the sequence of sheaves on P , the final term written $\mathcal{O}_C(d-3)$ means $i_* \mathcal{O}_C(d-3)$. This notation distinguishes the space on which the sheaf is defined without changing the exercise.

As a check on the given short sequence, multiplication by f gives an exact sequence of graded modules

$$0 \longrightarrow S(-3) \xrightarrow{f} S(d-3) \longrightarrow (S/(f))(d-3) \longrightarrow 0.$$

The first map is injective because S is an integral domain and $f \neq 0$. Localising on standard charts, taking degree-zero parts, and gluing gives the sheaf sequence in the exercise.

The beginning of the corresponding long exact cohomology sequence is

$$\begin{aligned} 0 \longrightarrow H^0(P, \mathcal{O}_P(-3)) &\longrightarrow H^0(P, \mathcal{O}_P(d-3)) \\ &\longrightarrow H^0(C, \mathcal{O}_C(d-3)) \longrightarrow H^1(P, \mathcal{O}_P(-3)). \end{aligned}$$

The identification of global sections in the pushforward term follows directly from the definition: $\Gamma(P, i_* \mathcal{O}_C(d-3)) = \Gamma(C, \mathcal{O}_C(d-3))$.

[Theorem 27.4](#), with projective-space dimension equal to 2, gives

$$H^0(P, \mathcal{O}_P(n)) \cong S_n, \quad H^1(P, \mathcal{O}_P(n)) = 0$$

for every integer n . Here $S_n = 0$ for $n < 0$. The Čech computation in that theorem computes sheaf cohomology by [Theorem 26.10](#): P is a projective scheme and $\mathcal{O}_P(n)$ is quasicoherent, with the standard affine cover.

In particular,

$$H^0(P, \mathcal{O}_P(-3)) = 0, \quad H^1(P, \mathcal{O}_P(-3)) = 0.$$

Exactness of the sequence above then gives an isomorphism

$$S_{d-3} \cong H^0(C, \mathcal{O}_C(d-3)).$$

Thus it remains only to count homogeneous monomials.

If $d \geq 3$, write $m = d - 3 \geq 0$. A basis of S_m consists of monomials $X^a Y^b Z^c$ with $a, b, c \geq 0$ and $a + b + c = m$. For each a there are $m - a + 1$ pairs (b, c) , so

$$\dim_K S_m = \sum_{a=0}^m (m - a + 1) = \frac{(m+1)(m+2)}{2} = \frac{(d-1)(d-2)}{2}.$$

If $d = 1$ or $d = 2$, then $d - 3 < 0$ and $S_{d-3} = 0$. The formula on the right is also zero for both degrees. This covers all $d \geq 1$ and proves the conclusion.

Check and pitfall. For a cubic curve ($d = 3$), the resulting space has dimension 1, represented by constant polynomials. For a quartic curve ($d = 4$), its dimension is 3, represented by X, Y, Z . Surjectivity of restriction on global sections is not automatic from sheaf surjectivity: here it follows from $H^1(P, \mathcal{O}_P(-3)) = 0$. The connection with global differential forms requires the smoothness hypothesis mentioned in the introduction to the source exercise; the dimension computation above does not add that hypothesis.

Integrative problem 1: from a parabola to an affine scheme

This problem and its solution were independently written to connect classical varieties with schemes; neither is a problem or answer from Holger Brenner's source material. The prerequisites are prime ideals, localisation, and the structure sheaf of an affine scheme. In this new explanation, Spec denotes the spectrum, also written Spek in the course.

Problem

Let k be an algebraically closed field. All rings below are commutative with identity, and all homomorphisms preserve the identity. Consider the classical parabola and the two rings

$$C = \{(a, b) \in k^2 : b = a^2\}, \quad A = k[x, y]/(y - x^2), \quad B = k[x, y]/((y - x^2)^2).$$

Write $X = \text{Spec}(A)$ and $Y = \text{Spec}(B)$.

1. Prove that the ideal of all polynomials vanishing on C is $(y - x^2)$. Construct an isomorphism $A \cong k[t]$ and its inverse.
2. Determine all points of X , the closure of each point, and the relationship between the classical points of C and the closed points of X . What does X add to the set C ?
3. Compute the global sections, the sections on $D(x)$, the stalk at $p_a = (x - a, y - a^2)$, and its residue field. Also compute the stalk and residue field at the generic point.
4. Prove that X and Y have homeomorphic underlying topological spaces but are not isomorphic as schemes. Explain why evaluating sections at all points with values in their residue fields does not detect the difference.
5. Using $A \cong k[t]$, construct the morphism $F : \mathbb{A}_k^1 = \text{Spec}(k[s]) \rightarrow X$ arising from the homomorphism $t \mapsto s^2$. Determine the image of $(s - b)$ and the image of the generic point. Explain the direction of the arrows between geometry and rings.

Complete solution

1. The coordinate ring

Division by the monic polynomial $y - x^2$, treating y as the division variable, gives

$$f(x, y) = q(x, y)(y - x^2) + r(x), \quad r(x) = f(x, x^2).$$

If f vanishes on C , then $r(a) = 0$ for every $a \in k$. An algebraically closed field is infinite: if its elements were just $a_1, \dots, a_N$, the polynomial $\prod_i (T - a_i) - 1$ would have no root in that field, a contradiction. A nonzero polynomial of degree d has at most d roots, so $r = 0$. Thus f belongs to $(y - x^2)$. The reverse inclusion follows immediately by substitution. Hence $I(C) = (y - x^2)$.

The homomorphism

$$\alpha : A \longrightarrow k[t], \quad \bar{x} \longmapsto t, \quad \bar{y} \longmapsto t^2$$

is well defined because the relation $y - x^2$ maps to zero. The homomorphism $\beta : k[t] \rightarrow A$, $t \mapsto \bar{x}$, is its inverse: $\alpha\beta(t) = t$, while $\beta\alpha$ fixes $\bar{x}$ and $\bar{y} = \bar{x}^2$. From now on we use this identification and write $x = t$ in A .

2. Closed points and the generic point

The ring $k[t]$ is a principal ideal domain. Its prime ideals are (0) and the ideals generated by irreducible polynomials. Since k is algebraically closed, those irreducible polynomials have degree one. Thus

$$|X| = \{\eta\} \cup \{p_a : a \in k\}, \quad \eta = (0), \quad p_a = (t - a).$$

The ideal p_a is maximal because $k[t]/(t - a) \cong k$. Consequently $\overline{\{p_a\}} = \{p_a\}$. In contrast, $\overline{\{\eta\}} = V(0) = X$: η is the generic point. To see this closure formula, observe that a closed set $V(J)$ contains a point $\mathfrak{p}$ exactly when $J \subseteq \mathfrak{p}$; the smallest such closed set is $V(\mathfrak{p})$.

If $J \neq 0$, choose a nonzero polynomial $h \in J$. The set $V(J)$ is contained in the finite set of roots of h . Conversely, every finite set $\{p_{a_1}, \dots, p_{a_r}\}$ equals $V(\prod_i (t - a_i))$. Thus the proper closed subsets of X are precisely the finite sets of closed points, including the empty set.

The classical point (a, a^2) corresponds to the kernel of evaluation $A \rightarrow k$, namely p_a . This gives a bijection $C \cong \{\text{closed points of } X\}$, and the classical Zariski topology agrees with the subspace topology. However, η is not a coordinate pair in k^2 . Through its closure property it represents the entire irreducible parabola, rather than an arbitrarily chosen extra classical point.

3. Sections, stalks, and residue fields

For every commutative ring R , the structure sheaf satisfies $\Gamma(D(f), \mathcal{O}) = R_f$ and $\mathcal{O}_{\text{Spec } R, \mathfrak{p}} = R_{\mathfrak{p}}$. These are [Lemma 9.12](#) and [Lemma 9.10](#) in Brenner's course; neither statement requires a Noetherian hypothesis. With $A = k[t]$, we obtain

$$\Gamma(X, \mathcal{O}_X) = k[t], \quad \Gamma(D(x), \mathcal{O}_X) = k[t, t^{-1}], \quad \mathcal{O}_{X, p_a} = k[t]_{(t-a)}.$$

Explicitly, elements of the last stalk can be written as $f(t)/g(t)$ with $g(a) \neq 0$. Its maximal ideal is $(t - a)k[t]_{(t-a)}$, and evaluation

$$\frac{f(t)}{g(t)} \longmapsto \frac{f(a)}{g(a)}$$

gives an isomorphism of residue fields $\kappa(p_a) = \mathcal{O}_{X, p_a} / (t - a)\mathcal{O}_{X, p_a} \cong k$. The stalk itself is not k : for example, $t - a$ is a nonzero element of the stalk, but its class in the residue field is zero.

At the generic point, every nonzero polynomial is allowed as a denominator:

$$\mathcal{O}_{X,\eta} = k(t), \quad \kappa(\eta) = k(t).$$

The stalk and the residue field agree here only because this local ring is already a field. The section $1/t$ is defined on $D(x)$ but not on all of X : it does not belong to $\mathcal{O}_{X,p_0}$.

4. Nilpotent structure is invisible to points alone

Set $\varepsilon = y - x^2$. The change of variables $t = x, y = t^2 + \varepsilon$ gives

$$B \cong k[t, \varepsilon]/(\varepsilon^2).$$

Every element of B has a unique expression $f(t) + \varepsilon g(t)$. In particular, $\varepsilon \neq 0$ but $\varepsilon^2 = 0$. Every prime ideal $\mathfrak{q}$ contains ε : primality and $\varepsilon^2 \in \mathfrak{q}$ imply $\varepsilon \in \mathfrak{q}$.

Taking images or inverse images under $B \twoheadrightarrow B/(\varepsilon) = A$ therefore gives a bijection of prime ideals. This bijection respects closed sets: for every ideal $J \subseteq B$, the set $V_B(J)$ corresponds to $V_A((J + (\varepsilon))/(\varepsilon))$. Thus the immersion $X \rightarrow Y$ induced by this quotient is a homeomorphism on underlying topological spaces.

However, an isomorphism of schemes would induce an isomorphism of rings of global sections. The ring $A = k[t]$ is reduced, whereas B has the nonzero nilpotent element ε . This property is preserved by ring isomorphisms. Hence X and Y are not isomorphic as schemes.

Every homomorphism from B to a field kills ε , since a field has no nonzero nilpotent elements. Thus the value of ε in every residue field is zero, although ε is not the zero section. Even at the point corresponding to $(t - a, \varepsilon)$, the stalk $B_{(t-a, \varepsilon)}$ still contains $\varepsilon \neq 0$. If a denominator $h = f + \varepsilon g$ outside the maximal ideal annihilated ε , then $h\varepsilon = f\varepsilon = 0$ would force $f = 0$; yet $f(a) \neq 0$. This is a contradiction. Thus stalk and sheaf data retain infinitesimal information lost when only residue values are examined.

5. Contravariance in an explicit example

By [Corollary 10.10](#), every homomorphism of commutative rings $\theta : R \rightarrow S$ induces a morphism of schemes $\text{Spec}(S) \rightarrow \text{Spec}(R)$, with point map $\mathfrak{q} \mapsto \theta^{-1}(\mathfrak{q})$. Take $\theta : k[t] \rightarrow k[s], t \mapsto s^2$. Then

$$\theta^{-1}((s - b)) = (t - b^2), \quad F(b) = (b^2, b^4) \in C.$$

The equality of ideals follows because composition with evaluation at $s = b$ sends t to b^2 and has kernel $(t - b^2)$. The homomorphism θ is injective: a nonzero polynomial $\sum c_i t^i$ maps to the nonzero polynomial $\sum c_i s^{2i}$. Consequently $\theta^{-1}((0)) = (0)$, and the generic point maps to the generic point. This argument also works in characteristic 2; we do not claim that the map always has two distinct preimages over each closed point.

Thus the geometric map from the line with coordinate s to the parabola pulls functions on the parabola back to functions in s . The ring arrow runs in the opposite direction to the scheme arrow.

Quick checks and pitfalls

- C corresponds to maximal ideals because k is algebraically closed; the spectrum still uses **all** prime ideals.
- Sections, stalks, and residue fields are three different objects. A zero residue value is not enough to conclude that a nilpotent section is zero.
- In part 4, a homeomorphism is not an isomorphism of schemes. The structure sheaf is part of the object that must be preserved.

Material provenance and licence

Prerequisite references: Holger Brenner, *Bündel, Garben und Kohomologie*, [Lecture 9, revision 793634](#) and [Lecture 10, revision 1003733](#). The contributor to the frozen revisions is recorded as Bocardodarapti. The new problem, synthesis, and solution above are independent editorial material, not a translation of a public source solution. Production: OpenAI Codex gpt-5.6-sol, Ultra. This new material is licensed under CC BY-SA 4.0; the credits and licences of source components remain in force. No human authorship or review is claimed, and no endorsement by the source author or institution is implied.

Integrative problem 2: affine gluing and compatibility

This synthesis problem and its solution were independently written; they are not a problem or solution from Holger Brenner's source material. The aim is to glue not only points but also structure sheaves, and to discover why pairwise conditions alone are insufficient for three charts.

Problem

Let k be any field. Take

$$U = \operatorname{Spec}(k[u, u^{-1}]), \quad V = \operatorname{Spec}(k[v, v^{-1}]),$$

and the open subsets $W_U = D(1 - u) \subset U$ and $W_V = D(1 - v) \subset V$. All morphisms below are morphisms of schemes over k .

1. Check that the homomorphism $\theta : k[v, v^{-1}, (1 - v)^{-1}] \rightarrow k[u, u^{-1}, (1 - u)^{-1}]$, $v \mapsto 1 - u$, gives an isomorphism $\varphi : W_U \rightarrow W_V$. Write down its inverse, including the images of the denominators.
2. Construct the glued space X and its structure sheaf. Prove that $X \cong \mathbb{A}_k^1$, not merely as a set of points. Locate the points $t = 0$ and $t = 1$ in the two charts.
3. Compute $\Gamma(X, \mathcal{O}_X)$ from compatible pairs of sections. Which global section is given by u on U and $1 - v$ on V ? Why is the pair formally written as $1/u$ and $1/(1 - v)$ not a pair of sections on the entirety of the two charts?
4. State the domain, inverse, and cocycle conditions for gluing three or more scheme charts. Explain the order of composition of the ring homomorphisms.
5. For three copies of $\mathbb{A}_k^1$ with coordinates t_1, t_2, t_3 , take every proposed overlap to be the whole chart, and specify the transitions by $t_2 = t_1$, $t_3 = t_2 + 1$, and $t_3 = t_1$. Pair each transition with its inverse. Prove that these data still cannot be realised as open charts with these transitions on one scheme.

Complete solution

1. The isomorphism on the overlap

For a commutative ring R , the principal open subset $D(f)$ is an affine scheme with section ring R_f ; see [Lemma 9.12](#) and [Lemma 9.13](#). Thus the rings of the two overlaps are exactly those written in the problem.

In the target ring of θ , both $1 - u$ and u are units. The substitution $v \mapsto 1 - u$ therefore extends uniquely by

$$v^{-1} \mapsto (1 - u)^{-1}, \quad (1 - v)^{-1} \mapsto u^{-1}.$$

Its inverse sends $u \mapsto 1 - v$, $u^{-1} \mapsto (1 - v)^{-1}$, and $(1 - u)^{-1} \mapsto v^{-1}$. Both composites fix every generator. By [Corollary 10.10](#), this ring isomorphism gives an isomorphism of schemes $\varphi : W_U \rightarrow W_V$. On points valued in a field extension, the transition reads $v = 1 - u$. The section homomorphism $\varphi^\#$ runs from the ring of chart V to the ring of chart U , not the other way round.

2. The glued space and sheaf

As a topological space, take the disjoint union $U \amalg V$ and identify $p \in W_U$ with $\varphi(p) \in W_V$. The quotient topology makes both charts open subsets. Indeed, for an open subset $G \subset U$, its saturation in the disjoint union is $G \amalg \varphi(G \cap W_U)$, which is open because φ is a homeomorphism between open subsets. No two distinct points within chart U are identified, and the same holds for V .

For an open subset $O \subset X$, write O_U and O_V for its inverse images in the two charts. Define

$$\mathcal{O}_X(O) = \{(a, b) \in \mathcal{O}_U(O_U) \times \mathcal{O}_V(O_V) : a|_{O_U \cap W_U} = \varphi^\#(b|_{O_V \cap W_V})\}.$$

Restriction is performed componentwise. This is indeed a sheaf: compatible local sections glue uniquely on each chart by the sheaf properties of $\mathcal{O}_U$ and $\mathcal{O}_V$; equality on the overlap can be checked locally and therefore remains valid after gluing. The restriction of this sheaf to U is isomorphic to $\mathcal{O}_U$: a section on chart U uniquely determines a section on the part of chart V identified with it through φ . The same argument applies to V . Every point therefore has an affine neighbourhood with the correct structure sheaf. By [Definition 10.1](#), X is a scheme.

Now take the affine line with coordinate t . The homomorphisms

$$k[t] \longrightarrow k[u, u^{-1}], \quad t \mapsto u, \quad k[t] \longrightarrow k[v, v^{-1}], \quad t \mapsto 1 - v$$

give isomorphisms $U \cong D(t)$ and $V \cong D(1 - t)$. On the overlap, the two formulas agree because $u = 1 - v$. The sets $D(t)$ and $D(1 - t)$ cover $\text{Spec}(k[t])$: a prime ideal cannot contain both t and $1 - t$, since their sum is 1. Their intersection is $D(t(1 - t))$, exactly the image of W_U and W_V .

Thus the map $X \rightarrow \mathbb{A}_k^1$ is a bijection whose restriction to each chart is a homeomorphism onto an open subset. It is a homeomorphism, and its sheaf homomorphism is an isomorphism on both charts. Sheaf isomorphisms can be checked on an open cover, so this map is an isomorphism of schemes.

The point $t = 0$ is not in $U = D(t)$; it appears in V as the ideal $(v - 1)$. The point $t = 1$ appears in U as $(u - 1)$ and is not in $V = D(1 - t)$. These two points are not glued to one another.

3. The global section ring as matching pairs

Use the coordinate $t = u = 1 - v$ on the overlap. The rings of the two charts can be regarded as subrings of the fraction field $k(t)$:

$$\Gamma(U, \mathcal{O}_U) = k[t, t^{-1}], \quad \Gamma(V, \mathcal{O}_V) = k[t, (1 - t)^{-1}].$$

Since all maps to the overlap ring are injective, the matching-pair condition amounts to taking the intersection of these two subrings. Suppose an element of the intersection can be written as

$$\frac{a(t)}{t^m} = \frac{b(t)}{(1 - t)^n}, \quad m, n \geq 0.$$

Then $(1 - t)^n a(t) = t^m b(t)$. The polynomials t^m and $(1 - t)^n$ are coprime in the principal ideal domain $k[t]$, so t^m divides $a(t)$. The first fraction is a polynomial. Conversely, every polynomial belongs to both rings. Therefore

$$\Gamma(X, \mathcal{O}_X) = k[t, t^{-1}] \cap k[t, (1 - t)^{-1}] = k[t].$$

The pair $(u, 1 - v)$ gives the global polynomial t . On the other hand, $1/u$ is indeed a section on U , but $1/(1 - v)$ does not belong to $k[v, v^{-1}]$. To prove this, suppose $1/(1 - v) = p(v)/v^N$. Then $v^N = (1 - v)p(v)$, which, after substituting $v = 1$, gives $1 = 0$. Thus this formula is not regular at $v = 1$, the point $t = 0$. Formal agreement on an overlap alone does not turn a rational function into a section on a whole chart.

4. Gluing and cocycle conditions

For a family of schemes U_i , choose open subsets $U_{ij} \subseteq U_i$ and scheme isomorphisms $\varphi_{ij} : U_{ij} \rightarrow U_{ji}$. The complete conditions we use are:

1. $U_{ii} = U_i$ and $\varphi_{ii} = \text{id}_{U_i}$.
2. $\varphi_{ji} = \varphi_{ij}^{-1}$.
3. For every i, j, k , $\varphi_{ij}(U_{ij} \cap U_{ik}) = U_{ji} \cap U_{jk}$, and on the domain $U_{ij} \cap U_{ik}$ we have $\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$.

The first part of condition 3 ensures that the composite has the correct domain. The second is the cocycle condition: changing charts through j or directly to k gives the same identification, including on the structure sheaves.

These conditions make the relation $p \sim \varphi_{ij}(p)$ reflexive, symmetric, and transitive. In particular, no further identifications arise that merge distinct points in the same chart. As in part 2, isomorphisms between open subsets make the map from each chart to the quotient space an open immersion. The glued sheaf is obtained from families of matching sections; the sheaf property and local affine structure are checked on each chart. This explains why the data produce a scheme, without requiring the resulting scheme to be affine or separated.

If the overlaps used are affine, write $\lambda_{ij} = \varphi_{ij}^\#$ for the ring homomorphism running from chart j to chart i . After all sections have been restricted to the same triple overlap, the cocycle condition reads

$$\lambda_{ij} \circ \lambda_{jk} = \lambda_{ik}.$$

This order is the reverse of that of the space maps $\varphi_{jk} \circ \varphi_{ij}$. When a triple overlap is not affine, the underlying equality is still an equality of sheaf morphisms there; do not replace it with an unjustified statement about a single global coordinate ring.

5. Inverse pairs are not enough

In the three-chart example, the route $1 \rightarrow 2 \rightarrow 3$ gives

$$t_3 = t_2 + 1 = t_1 + 1,$$

whereas the direct route $1 \rightarrow 3$ gives $t_3 = t_1$. These morphisms differ: they send the rational point $t_1 = 0$ to $t_3 = 1$ and $t_3 = 0$, respectively. This holds over every field because $0 \neq 1$.

Suppose there were chart immersions $j_i : U_i \rightarrow X$ realising these transitions. The transitions $1 \rightarrow 2$ and $2 \rightarrow 3$ would give $j_1(0) = j_2(0) = j_3(1)$, while the transition $1 \rightarrow 3$ would give $j_1(0) = j_3(0)$. Thus $j_3(0) = j_3(1)$, contradicting the injectivity of the open immersion j_3 . Adding all inverse transitions does not repair this failure. One can still form a quotient space by the generated equivalence relation, but the chart maps to that space are not the required immersions.

Quick checks and pitfalls

- Check denominators: $v \mapsto 1 - u$ must send every element being inverted to a unit.
- Check cocycle domains before writing a composition equation.
- Gluing schemes requires isomorphisms of structure sheaves, not merely bijections of points or agreement of rational functions on part of a region.
- The result in part 2 is affine because these charts reconstruct a cover of $\mathbb{A}_k^1$, not because every gluing of affine charts produces an affine scheme.

Material provenance and licence

Prerequisites are referenced from Holger Brenner, *Bündel, Garben und Kohomologie*, [Lecture 9, revision 793634](#) and [Lecture 10, revision 1003733](#). The contributor to the frozen revisions is recorded as Bocardodarapti. The example of the cover $D(t), D(1 - t)$, the synthesis questions, the cocycle counterexample, and the solution here are independent editorial material, not a renamed public source solution. Production: OpenAI Codex gpt-5.6-sol, Ultra. The new material is licensed under CC BY-SA 4.0, with the credits and licences of source components preserved. No human authorship or review is claimed, and no endorsement by the source author or institution is implied.

Integrative problem 3: the projective line from two affine charts

This problem brings together gluing, the Proj construction, local rings, and morphisms to affine schemes. The questions and their solutions were independently written. The example of the projective line constructed by gluing does already occur in Holger Brenner's course; that source example is not claimed as a new discovery or answer.

Problem

Let k be any field. Take two charts

$$U_0 = \operatorname{Spec}(k[t]), \quad U_\infty = \operatorname{Spec}(k[s]).$$

Glue $D(t) \subset U_0$ to $D(s) \subset U_\infty$ by the isomorphism whose section homomorphism is

$$\theta : k[s, s^{-1}] \longrightarrow k[t, t^{-1}], \quad s \longmapsto t^{-1}.$$

Call the result P .

1. Check the gluing data and prove that $P \cong \text{Proj}(k[X_0, X_1]) = \mathbb{P}_k^1$, with the standard grading $\deg X_0 = \deg X_1 = 1$.
2. Identify the points 0 and ∞ , compute the stalk and residue field at each, and explain how the generic points of the two charts become a single generic point of P .
3. Compute the global section ring as compatible pairs of polynomials. Is either rational function t or t^{-1} a global section?
4. Prove that P is not affine without using a theorem about properness or cohomology.
5. Determine all morphisms of schemes over k from P to $\mathbb{A}_k^1$. Explain concretely why two local formulas that look like coordinates do not automatically give a global morphism to the affine line.

Complete solution

1. The transition and identification with Proj

The homomorphism θ is well defined because t^{-1} is a unit in the Laurent ring. Its inverse sends $t \mapsto s^{-1}$. We therefore have an isomorphism of schemes between the two open subsets. There are only two charts; with the identity on each chart and the inverse transition, the cocycle conditions involving repeated indices hold. The structure sheaves are glued using pairs of sections that agree on the overlap, as explained in integrative problem 2.

Write $R = k[X_0, X_1]$ with its standard grading. Its irrelevant ideal is $R_+ = (X_0, X_1)$. A point of $\text{Proj}(R)$ is a homogeneous prime ideal not containing R_+ . At least one of X_0, X_1 therefore lies outside that prime ideal, so

$$\text{Proj}(R) = D_+(X_0) \cup D_+(X_1).$$

[Lemma 12.9](#) states that for a commutative graded ring and a homogeneous element f of nonzero degree, $D_+(f)$ is the affine scheme $\text{Spec}((R_f)_0)$. Here X_0 and X_1 satisfy those hypotheses. Every degree-zero monomial in R_{X_0} has the form X_1^j/X_0^j for $j \geq 0$, so

$$(R_{X_0})_0 = k[X_1/X_0] = k[t].$$

Similarly, $(R_{X_1})_0 = k[X_0/X_1] = k[s]$. On the overlap, the two coordinates are reciprocal:

$$t = X_1/X_0, \quad s = X_0/X_1, \quad st = 1, \quad (R_{X_0X_1})_0 = k[t, t^{-1}].$$

Thus the charts and transition isomorphism on $\text{Proj}(R)$ are exactly the gluing data defining P . The isomorphisms from the two charts agree on the overlap and glue to an isomorphism of locally ringed spaces. Their local inverses also agree, so the result is an isomorphism of schemes $P \cong \mathbb{P}_k^1$.

2. Two special points and the generic point

On rational points, chart U_0 sends $t = a$ to $[1 : a]$. Chart U_∞ sends $s = b$ to $[b : 1]$. Hence

$$0 = [1 : 0], \quad \infty = [0 : 1].$$

The point 0 is the ideal (t) in U_0 and does not lie in the overlap $D(t)$. The point ∞ is the ideal (s) in U_∞ and does not lie in the overlap $D(s)$. Thus these are distinct points. The notation $[a : b]$ with $a, b \in k$ describes points rational over k , not all points of the projective spectrum when k is not algebraically closed.

Taking a stalk is unchanged by restricting the space to an open neighbourhood containing the point. By [Lemma 9.10](#),

$$\mathcal{O}_{P,0} = k[t]_{(t)}, \quad \mathcal{O}_{P,\infty} = k[s]_{(s)}, \quad \kappa(0) = k = \kappa(\infty).$$

For instance, elements of $k[s]_{(s)}$ are fractions $f(s)/g(s)$ with $g(0) \neq 0$, and its residue field is obtained by evaluation at $s = 0$. This is different from replacing the whole stalk by k .

The zero ideal of $k[t]$ lies in $D(t)$, and the zero ideal of $k[s]$ lies in $D(s)$. The Laurent isomorphism identifies them. Call the resulting point η . Its closure contains all of U_0 because (0) is generic in the spectrum of the domain $k[t]$; its closure also contains all of U_∞ . Hence $\overline{\{\eta\}} = P$. Its stalk is

$$\mathcal{O}_{P,\eta} = k(t) = k(s), \quad s = t^{-1}.$$

This field is the function field of P and also the residue field at η . The identification comes from the transition, not from choosing two different generic points on the same scheme.

3. Global sections and rational functions

The sheaf property identifies global sections with pairs

$$\Gamma(P, \mathcal{O}_P) \cong \{(f(t), g(s)) \in k[t] \times k[s] : f(t) = g(t^{-1}) \text{ in } k[t, t^{-1}]\}.$$

If $f(t) = \sum_{i \geq 0} a_i t^i$ and $g(t^{-1}) = \sum_{j \geq 0} b_j t^{-j}$, uniqueness of Laurent polynomial coefficients forces $a_i = 0$ for $i > 0$, $b_j = 0$ for $j > 0$, and $a_0 = b_0$. This uniqueness follows by multiplying the equation by a sufficiently large power of t and using uniqueness of ordinary polynomial coefficients. Thus

$$\Gamma(P, \mathcal{O}_P) \cong k, \quad a \leftrightarrow (a, a).$$

The rational function t is regular on U_0 , but becomes s^{-1} on the other chart. It does not belong to $k[s]_{(s)}$: if $s^{-1} = f(s)/g(s)$ with $g(0) \neq 0$, then $g(s) = sf(s)$, which implies $g(0) = 0$, a contradiction. Thus t has a pole at ∞ and is not a global section. Interchanging the charts shows that $t^{-1} = s$ has a pole at 0 and is likewise not a global section.

It is important that this computation uses the **structure sheaf**, not all rational functions $k(t)$. The function field is not the global section ring.

4. The glued scheme is not affine

Suppose $P \cong \text{Spec}(A)$ for a commutative ring A . For an affine scheme, [Lemma 9.12](#) gives $A \cong \Gamma(P, \mathcal{O}_P) \cong k$. Since k is a field, $\text{Spec}(k)$ has exactly one point, the zero ideal. Yet P has at least two distinct points, 0 and ∞ . This contradiction shows that P is not affine.

Thus having a cover by affine schemes is much weaker than being affine. What fails if we try to reconstruct P solely from its global sections is that the chart and sheaf gluing information is lost.

5. Morphisms to the affine line

[Theorem 10.9](#) applies to a locally ringed space Z and an affine target $\text{Spec}(R)$: a homomorphism $R \rightarrow \Gamma(Z, \mathcal{O}_Z)$ determines exactly one morphism of locally ringed spaces. We apply it to the scheme $Z = P$ and $R = k[T]$.

Since the required morphisms are **over** k , the ring homomorphism must be a k -algebra homomorphism

$$k[T] \longrightarrow \Gamma(P, \mathcal{O}_P) = k.$$

Every such homomorphism is determined by one element $a \in k$, the image of T , and conversely substitution $T \mapsto a$ always gives one. The corresponding morphism is the composite

$$P \longrightarrow \text{Spec}(k) \longrightarrow \mathbb{A}_k^1,$$

where the second arrow is the rational point $(T - a)$. Thus all morphisms $P \rightarrow \mathbb{A}_k^1$ over k are these constant morphisms.

Locally, one might wish to send T to t on U_0 . To agree on the overlap, the image of T on U_∞ would have to be s^{-1} . But s^{-1} is not a section on the whole of U_∞ , as proved in part 3. If one instead chooses $T \mapsto s$, the two sections are defined on their respective charts but do not agree on the overlap: t differs from t^{-1} as an element of $k[t, t^{-1}]$. Satisfying only one of local regularity and agreement on the overlap is not enough.

Quick checks and pitfalls

- The transition relation is $st = 1$, not $s = t$. Changing it changes the gluing data.
- $\Gamma(P, \mathcal{O}_P) = k$ does not say that every stalk is k , or that P has only one point.
- Over an arbitrary field, do not identify all closed points with points valued in k .
- In the conclusion about constant morphisms, the condition “over k ” explains why the homomorphism on constants is the identity.

Material provenance and licence

References: Holger Brenner, *Bündel, Garben und Kohomologie*, [Lecture 9, revision 793634](#), [Lecture 10, revision 1003733](#), and [Lecture 12, revision 1003742](#). The contributor to the frozen revisions is recorded as Bocardodarapti. In particular, source Example 10.7 and the standard cover in Example 12.10 provide prerequisites; the integrative problem and its complete worked solution here form an independent editorial layer, not a retranslation of a public source solution. Production: OpenAI Codex gpt-5.6-sol, Ultra. This new material is licensed under CC BY-SA 4.0; all credits and licences of source components remain in force. No human authorship or review is claimed, and no endorsement by the source author or institution is implied.

Integrative problem 4: the affine line with doubled origin

This problem and solution were written as independent bridge material. Their starting point is [Example 10.6 by Holger Brenner](#). The discussion of fibres, failure of affineness, and the diagonal below is an editorial development, not an additional public solution attributed to Brenner.

Problem

Let k be a field; it need not be algebraically closed. Take two copies

$$U_0 = \operatorname{Spec} k[t_0], \quad U_1 = \operatorname{Spec} k[t_1].$$

Glue $D(t_0) \subset U_0$ to $D(t_1) \subset U_1$ by $t_0 = t_1$. Call the resulting scheme X , and call the two points arising from the ideals (t_0) and (t_1) respectively o_0 and o_1 . Write the common coordinate on the overlap of the two charts as t .

1. Describe the glued structure sheaf. Compute $\Gamma(X, \mathcal{O}_X)$, the two stalks at o_0, o_1 , and the stalk at the generic point. Does agreement of two stalks mean that the points are the same?
2. Construct the morphism $\pi : X \rightarrow \mathbb{A}_k^1$ given on each chart by the coordinate t . Compute the scheme-theoretic fibres at the origin and at the generic point. Prove that its global section homomorphism is an isomorphism, but π is not a scheme isomorphism.
3. Prove that X is not affine, although it has a cover by two affine schemes whose intersection is also affine.
4. For this problem, call a scheme over k *separated* if its diagonal $\Delta : X \rightarrow X \times_k X$ is a closed immersion. By examining the chart $U_0 \times_k U_1$, prove that X is not separated.

Use the following affine facts with their hypotheses: for a commutative ring A , [Lemma 9.10](#) gives $\mathcal{O}_{\operatorname{Spec} A, \mathfrak{p}} = A_{\mathfrak{p}}$; [Lemma 9.12](#) gives $\Gamma(D(f), \mathcal{O}) = A_f$, including $\Gamma(\operatorname{Spec} A, \mathcal{O}) = A$.

Complete solution

1. Gluing, sections, and stalks

The identification of the open subsets uses the ring isomorphism $k[t_0, t_0^{-1}] \cong k[t_1, t_1^{-1}]$. Its inverse is available; with two charts there is no additional triple-overlap condition not already determined by this isomorphism and the identities. For an open subset $W \subseteq X$, the glued sheaf is given by

$$\mathcal{O}_X(W) = \left\{ (s_0, s_1) \in \mathcal{O}_{U_0}(W \cap U_0) \times \mathcal{O}_{U_1}(W \cap U_1) \mid s_0|_{W \cap U_0 \cap U_1} = s_1|_{W \cap U_0 \cap U_1} \right\}.$$

Restriction is performed on both components. The sheaf axioms hold because sections glue uniquely on each chart, and their agreement on the overlap can then be checked locally. This sheaf restricts to the original affine structure sheaf on U_i . Thus the two charts really make X a scheme, in accordance with [Definition 10.1](#).

For $W = X$, the formula becomes

$$\Gamma(X, \mathcal{O}_X) = \{(f_0, f_1) \in k[t] \times k[t] \mid f_0 = f_1 \text{ in } k[t, t^{-1}]\}.$$

The homomorphism $k[t] \rightarrow k[t, t^{-1}]$ is injective: if a polynomial becomes zero, some power of t annihilates it in the integral domain $k[t]$, so the polynomial was already zero. The pairs above are therefore exactly the pairs (f, f) , and

$$\Gamma(X, \mathcal{O}_X) \cong k[t].$$

The neighbourhood U_i of o_i gives

$$\mathcal{O}_{X, o_0} \cong k[t]_{(t)}, \quad \mathcal{O}_{X, o_1} \cong k[t]_{(t)}, \quad \kappa(o_0) \cong \kappa(o_1) \cong k.$$

The two chart generic points, corresponding to the zero ideals, lie in $D(t)$ and glue to one point η . Its stalk is $k(t)$. The closure of $\{\eta\}$ contains both charts, so η is generic for all of X .

Nevertheless, $o_0 \neq o_1$: neither belongs to the part being glued. Indeed, the open set U_0 contains o_0 but not o_1 . Isomorphic stalks mean agreement of the type of local data, not identification of points. Every global section f has value $f(0)$ at both points, so global sections do not distinguish this pair of points.

2. The morphism to the affine line and its fibres

The maps $U_i \rightarrow \operatorname{Spec} k[t]$ arising from $t \mapsto t_i$ agree on the overlap. They therefore glue to π . On global sections,

$$\pi^\# : k[t] \longrightarrow \Gamma(X, \mathcal{O}_X), \quad f \longmapsto (f, f),$$

is the isomorphism just computed.

The scheme-theoretic fibre at the origin $0 = (t)$ is obtained by taking the fibre product with $\operatorname{Spec} k$. On each chart its ring is

$$k[t_i] \otimes_{k[t]} k \cong k[t_i]/(t_i) \cong k.$$

On the overlap, t must be both invertible and zero, so the fibre ring is the zero ring and its spectrum is empty. Hence

$$X_0 \cong \operatorname{Spec} k \amalg \operatorname{Spec} k.$$

These are two reduced points, not a single point with nilpotent elements. In contrast, over the generic point of the base, both chart fibres are $\operatorname{Spec} k(t)$ and their overlap is also the whole of $\operatorname{Spec} k(t)$. After gluing,

$$X_{\eta_{\mathbb{A}^1}} \cong \operatorname{Spec} k(t).$$

The morphism π is not an isomorphism because it sends two distinct points o_0, o_1 to the same point. Thus an isomorphism on global sections alone is insufficient to recognise an isomorphism of general schemes.

3. Why the glued scheme is not affine

Suppose X were affine. For an affine scheme, the identification $X \cong \operatorname{Spec} \Gamma(X, \mathcal{O}_X)$ follows from the definition of an affine scheme and Lemma 9.12. Under this identification, the morphism inducing the identity on the global section ring is an isomorphism. Uniqueness of that morphism is also a case of [Theorem 10.9](#): its source is a locally ringed space and its target is affine.

Since $\pi^\#$ is an isomorphism, assuming that X is affine forces π to be an isomorphism. This contradicts the two-point fibre at the origin. Therefore X is not affine.

An affine cover is a local condition in the definition of a scheme. It does not say that all charts can be replaced by a single global affine chart; the proof above exhibits precisely that failure.

4. A diagonal that is not closed

The product of the two charts has ring

$$k[t_0] \otimes_k k[t_1] \cong k[t_0, t_1],$$

because giving two k -algebra homomorphisms from one-variable polynomial rings amounts to choosing two commuting elements. Thus $U_0 \times_k U_1$ is an affine chart of $X \times_k X$.

A diagonal point in this cross-chart must come from a point of X belonging to both U_0 and U_1 , hence from $D(t)$. The diagonal ring map on this chart is

$$k[t_0, t_1] \longrightarrow k[t, t^{-1}], \quad t_0 \longmapsto t, \quad t_1 \longmapsto t^{-1}.$$

Consequently its image as a set of points is

$$\Delta(X) \cap (U_0 \times_k U_1) = V(t_0 - t_1) \cap D(t_0).$$

The closed line $V(t_0 - t_1)$ is isomorphic to $\operatorname{Spec} k[t]$. The open subset $D(t)$ is dense in it: it contains the generic point, the zero ideal of the integral domain $k[t]$. Thus the closure of this image contains the point (t_0, t_1) , namely the pair (o_0, o_1) , but that pair is not a diagonal point because $o_0 \neq o_1$.

The intersection of the diagonal image with the cross-chart is therefore not closed. If Δ were a closed immersion, this intersection would have to be closed. This contradiction proves that X is not separated. The density argument uses the generic prime point, so it remains valid when k is a finite field.

Checks and common pitfalls

- The two origins have isomorphic stalks, but open neighbourhoods distinguish them. Do not call them the same point.
- The fibre at the origin consists of two reduced points. Do not replace it by $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$.
- In checking the diagonal, the cross-chart $U_0 \times_k U_1$ is crucial: the missing point is the pair of distinct origins.

Sources and editorial status

Source references: Holger Brenner, *Bündel, Garben und Kohomologie*, [Lecture 9, revision 793634](#), especially Lemmas 9.10 and 9.12; [Lecture 10, revision 1003733](#), especially Definition 10.1, Example 10.6, and Theorem 10.9.

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Integrative problem 5: why morphisms must be local on stalks

This is an independent synthesis problem and solution, not a new public problem or solution by Holger Brenner. Its references are the [definition of a scheme morphism](#) and [Theorem 10.9 on morphisms to affine schemes](#).

Problem

All rings here are commutative, and all ring homomorphisms preserve the identity. A homomorphism of local rings $\alpha : (R, \mathfrak{m}) \rightarrow (S, \mathfrak{n})$ is called *local* if $\alpha^{-1}(\mathfrak{n}) = \mathfrak{m}$.

1. Given $\varphi : A \rightarrow B$, for $\mathfrak{q} \in \operatorname{Spec} B$ set $\mathfrak{p} = \varphi^{-1}(\mathfrak{q})$. Construct the stalk map $A_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}}$ and prove that it is local. Describe the map of residue fields.
2. Now let k be an algebraically closed field with $\operatorname{char} k \neq 2$. For $f : \operatorname{Spec} k[t] \rightarrow \operatorname{Spec} k[s]$ given by $s \mapsto t^2$, compute the stalk and residue field maps at the origin, the scheme-theoretic fibre at $s = 0$, and the fibre at $s = a \neq 0$.

3. Compute the vector space map $(s)/(s)^2 \rightarrow (t)/(t)^2$ induced at the origin. Explain why the residue field map alone does not capture this behaviour.
4. Take $Z = \operatorname{Spec} k(t)$ and $Y = \operatorname{Spec} k[t]$. Construct a morphism of *ringed spaces* $h : Z \rightarrow Y$ that sends the unique point of Z to (t) and induces the inclusion $k[t] \hookrightarrow k(t)$ on global sections. Prove that h is not a morphism of locally ringed spaces. Compare it with the scheme morphism produced by the same ring inclusion.

In part 3, the maximal ideals and their squares are taken in the respective local rings. Their quotients are the *cotangent spaces* at the points; the tangent spaces are their duals over the residue fields.

Complete solution

1. Contraction of prime ideals determines the local map

The ideal $\mathfrak{p}$ is prime because if ab maps into $\mathfrak{q}$, one of $\varphi(a), \varphi(b)$ belongs to the prime ideal $\mathfrak{q}$. Moreover, $1 \notin \mathfrak{p}$. The spectrum map

$$f : \operatorname{Spec} B \longrightarrow \operatorname{Spec} A, \quad \mathfrak{q} \longmapsto \varphi^{-1}(\mathfrak{q})$$

is continuous because $f^{-1}(D(a)) = D(\varphi(a))$.

If $s \notin \mathfrak{p}$, then $\varphi(s) \notin \mathfrak{q}$, so $\varphi(s)$ is invertible in $B_{\mathfrak{q}}$. The universal property of localisation therefore gives a homomorphism

$$f_{\mathfrak{q}}^{\#} : A_{\mathfrak{p}} \longrightarrow B_{\mathfrak{q}}, \quad \frac{a}{s} \longmapsto \frac{\varphi(a)}{\varphi(s)}.$$

This formula is well defined: the relation expressing equality of two fractions remains valid after applying the homomorphism, and denominators become units. By [Lemma 9.10](#), this is the map between the stalks of the affine structure sheaves.

In a localisation at a prime ideal, a fraction belongs to the maximal ideal if and only if its numerator belongs to the original prime ideal. Thus

$$\frac{a}{s} \in (f_{\mathfrak{q}}^{\#})^{-1}(\mathfrak{q}B_{\mathfrak{q}}) \iff \varphi(a) \in \mathfrak{q} \iff a \in \mathfrak{p}.$$

The inverse image of the maximal ideal is exactly $\mathfrak{p}A_{\mathfrak{p}}$; hence the stalk map is local. Passing to quotient rings gives

$$\kappa(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \longrightarrow B_{\mathfrak{q}}/\mathfrak{q}B_{\mathfrak{q}} = \kappa(\mathfrak{q}).$$

This map is injective: the kernel of an identity-preserving homomorphism from one field to another is a proper ideal and must therefore be zero. Its direction is opposite to that of the point map.

2. The squaring map and its scheme-theoretic fibres

The inverse image of (t) under $k[s] \rightarrow k[t]$ is (s) . The stalk map at the origin is

$$k[s]_{(s)} \longrightarrow k[t]_{(t)}, \quad \frac{a(s)}{b(s)} \longmapsto \frac{a(t^2)}{b(t^2)}, \quad b(0) \neq 0.$$

The denominator condition holds because $b(t^2)$ has value $b(0) \neq 0$ at the origin. The residue field map is the identity $k \rightarrow k$.

At the point $s = a$, the residue field of the base is k , with s acting as a . The fibre ring is therefore

$$k[t] \otimes_{k[s]} k \cong k[t]/(t^2 - a).$$

For $a = 0$, this is $k[t]/(t^2)$. It has one prime ideal, $(\bar{t})$, because every prime ideal must contain the nilpotent element $\bar{t}$. But $\bar{t} \neq 0$ and $\bar{t}^2 = 0$. Thus the fibre at the origin has only one topological point, yet its structure ring contains a nonzero nilpotent element. It is therefore nonreduced. The number of points does not record this thickening.

If $a \neq 0$, choose $b \in k$ with $b^2 = a$. Its existence uses algebraic closedness, and $b \neq 0$. Since the characteristic is not 2, the elements b and $-b$ are distinct and $2b$ is a unit. Evaluation gives

$$k[t]/(t^2 - a) \longrightarrow k \times k, \quad [g] \longmapsto (g(b), g(-b)).$$

This map is surjective: the pair (u, v) is the image of the polynomial

$$u \frac{t+b}{2b} + v \frac{b-t}{2b}.$$

Its kernel is zero. Indeed, a polynomial vanishing at b and $-b$ is divisible by the coprime factors $t - b$ and $t + b$, hence by $t^2 - a$. Thus this fibre consists of two reduced points, each with residue field k .

3. The lost infinitesimal direction

Both cotangent spaces at the origin are one-dimensional over k , with bases $[s]$ and $[t]$. The induced map satisfies

$$[s] \longmapsto [t^2] = 0 \quad \text{in } (t)/(t)^2.$$

Thus the cotangent map is zero. Its dual, the tangent space map from the source point to the target point, is also zero. In contrast, the residue field map is the identity.

There is no contradiction: the residue field remembers values at the point, whereas $\mathfrak{m}/\mathfrak{m}^2$ remembers the linear terms of functions vanishing there. Substitution $s = t^2$ preserves constants but sends a linear term on the target to a term of order two on the source.

4. A ringed-space morphism that fails to be local

The topological space Z has one point z . Define $h(z) = (t)$. This map is continuous because the inverse image of every open subset is either Z or the empty set.

For an open subset $V \subseteq Y$ containing (t) , define

$$h_V^\# : \mathcal{O}_Y(V) \longrightarrow k[t]_{(t)} \longrightarrow k(t)$$

by taking the germ at (t) and then including it in the fraction field. If $(t) \notin V$, then $h^{-1}(V) = \emptyset$; use the unique homomorphism $\mathcal{O}_Y(V) \rightarrow \mathcal{O}_Z(\emptyset) = 0$.

These maps are compatible with restriction. For two open sets containing (t) , taking the germ before or after restriction gives the same result. If the smaller set does not contain (t) , both composites map to the zero ring. We thus obtain a sheaf morphism $\mathcal{O}_Y \rightarrow h_* \mathcal{O}_Z$, so h is indeed a morphism of ringed spaces.

However, its stalk map is the inclusion

$$h_z^\# : k[t]_{(t)} \hookrightarrow k(t).$$

Since the target is a field, its maximal ideal is zero. Its inverse image is also zero, not the maximal ideal $(t)k[t]_{(t)}$ of the source. Concretely, the nonunit t in the source becomes a unit in the target. Thus this map is not local.

On global sections, $h^\#$ is still the inclusion $k[t] \hookrightarrow k(t)$. But the *scheme* morphism induced by this inclusion sends the zero ideal of $k(t)$ to the zero ideal of $k[t]$, the generic point of Y , not the origin.

This is the role of the locality condition: contraction of the maximal ideal of the stalk must agree with the target point. Theorem 10.9 asserts uniqueness in the category of locally ringed spaces; the example h does not satisfy that hypothesis and therefore does not contradict the theorem.

Checks and common pitfalls

- Check denominators before writing a stalk map: $s \notin \mathfrak{p}$ must ensure $\varphi(s) \notin \mathfrak{q}$.
- A residue field is neither a stalk nor the full ring of a scheme-theoretic fibre. At the origin in this example, the residue field is k but the fibre ring is $k[t]/(t^2)$.
- The characteristic-not-2 condition is used to distinguish b from $-b$ and divide by $2b$; do not remove it.
- Equality of global section homomorphisms does not force equality of ringed-space morphisms when the locality condition on stalks is omitted.

Sources and editorial status

References: Holger Brenner, *Bündel, Garben und Kohomologie*, [Lecture 9, revision 793634](#), Lemmas 9.10 and 9.12; [Lecture 10, revision 1003733](#), Definition 10.8, Theorem 10.9, and Corollary 10.10. The examples and solutions here are editorial additions, not quotations of source solutions.

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Integrative problem 6: Krull dimension, components, and nilpotent thickenings

This problem and solution are independent bridge material. They build on Holger Brenner's [Lemma 11.3 on prime ideals and irreducible closed subsets](#), [Definition 11.6](#), and [Lemma 11.7](#). The calculations collected here are not attributed as public source solutions.

Problem

Let k be an algebraically closed field. Consider

$$L = k[t], \quad R = k[x, y]/(xy), \quad S = k[t, \varepsilon]/(\varepsilon^2), \quad D = k[\varepsilon]/(\varepsilon^2).$$

The variable letters also denote their classes in the quotient rings. For a nonzero commutative ring A , the Krull dimension is the supremum of the lengths of chains of prime ideals

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n;$$

the length is n , the number of strict inclusions, not the number of ideals. All dimensions in this problem are finite and the suprema are attained.

1. Determine all prime ideals of each ring and compute its dimension. Give both an upper bound and a chain attaining the dimension.
2. Determine the irreducible components and reducedness of all four spectra. Explain why $\text{Spec } S$ has the same topology as the affine line but is not the same scheme.
3. At the origin, compute the local dimension and $\dim_k \mathfrak{m}/\mathfrak{m}^2$. The maximal ideals in question are respectively (t) , (x, y) , (t, ε) , and (ε) in their local rings.
4. Compute the generic stalk on the component $V(x) \subseteq \text{Spec } R$ and on $\text{Spec } S$. Is the generic stalk of an irreducible scheme always a field?

Use that $k[t]$ is a principal ideal domain, by the polynomial division algorithm. Apart from this fact, prove the required prime ideal and vector space calculations. For a commutative ring A , Lemma 11.7 identifies the dimension of the ring with the dimension of its spectrum; this affine hypothesis holds in all the examples above.

Complete solution

1. Prime ideals and chain lengths

In $L = k[t]$, the zero ideal is prime because L is an integral domain. Nonzero prime ideals are generated by irreducible polynomials. Since k is algebraically closed, these polynomials have degree one, so the ideals are exactly $(t - a)$ for $a \in k$. They are all maximal, since their quotient rings are k .

Thus no chain of prime ideals has more than one strict inclusion. The chain $(0) \subsetneq (t)$ attains length one, and $\dim L = 1$.

In R , the equation $xy = 0$ forces every prime ideal to contain x or y . If it contains x , it corresponds to a prime ideal in

$$R/(x) \cong k[y],$$

and so has the form (x) or $(x, y - b)$, $b \in k$. If it contains y , it has the form (y) or $(y, x - a)$, $a \in k$. This list is complete; the ideal (x, y) occurs on both lists and is counted only once.

The ideals (x) and (y) are incomparable minimal primes. All other ideals on the list are maximal. The longest chains therefore have length one, for example

$$(x) \subsetneq (x, y),$$

and $\dim R = 1$. Notice that (0) is **not** a prime ideal in R : x, y are nonzero but their product is zero. Thus the chain $(0) \subsetneq (x) \subsetneq (x, y)$ cannot be used to conclude that the dimension is two.

In S , every prime ideal contains ε , since $\varepsilon^2 = 0$. The ideal correspondence for the quotient ring

$$S/(\varepsilon) \cong k[t]$$

shows that all prime ideals of S are (ε) and $(\varepsilon, t - a)$ for $a \in k$. Hence

$$(\varepsilon) \subsetneq (\varepsilon, t)$$

is a maximal chain of length one, and $\dim S = 1$.

In D , every prime ideal contains ε , and $D/(\varepsilon) = k$ has only the zero prime ideal. Thus (ε) is the only prime ideal of D ; there are no strict inclusions in a prime chain, so $\dim D = 0$.

2. Components and nilpotent elements

The ring L is reduced and has one minimal prime, (0) . Its spectrum is therefore reduced and topologically irreducible. For R , the two irreducible components are $V(x)$ and $V(y)$, each an affine line. The components meet at (x, y) .

Although R has zero divisors, R is still reduced. To prove this, use the homomorphism

$$k[x, y] \longrightarrow k[x] \times k[y], \quad f \longmapsto (f(x, 0), f(0, y)).$$

Its kernel is (xy) . Indeed, if both evaluations are zero, every remaining monomial in f contains both a factor x and a factor y , so xy divides f ; the converse is immediate. Thus R embeds into a product of two integral domains. A nilpotent element of that product must be zero in both components, so R has no nonzero nilpotent elements.

In contrast, $\varepsilon \neq 0$ in both S and D , while $\varepsilon^2 = 0$. Its nonvanishing follows from the unique expressions $f(t) + \varepsilon g(t)$ in S and $a + \varepsilon b$ in D . Both rings are nonreduced. Each has one minimal prime, so its spectrum is topologically irreducible.

The quotient map $S \rightarrow S/(\varepsilon)$ gives an inclusion-preserving bijection of prime ideals. It also identifies closed sets: for an ideal $I \subseteq S$, all primes containing I already contain ε , so they correspond exactly to primes in $k[t]$ containing $(I + (\varepsilon))/(\varepsilon)$. Consequently $\text{Spec } S$ is homeomorphic to $\text{Spec } k[t]$.

They are not isomorphic as schemes. An isomorphism of schemes would give an isomorphism of global section rings, but S has a nonzero nilpotent element and $k[t]$ does not. Identifying global sections with the affine ring uses [Lemma 9.12](#). Topology and Krull dimension do not see this nilpotent thickening.

3. Local dimension and cotangent spaces

The prime ideals of the local ring $A_{\mathfrak{m}}$ correspond to the prime ideals of A contained in $\mathfrak{m}$: contraction and extension are inverse operations, since the elements outside $\mathfrak{m}$ have become units. The lists in part 1 therefore give the local dimensions at the origin: 1 for L, R, S , and 0 for D .

The residue field at each of the four origins is k . To compute $\mathfrak{m}/\mathfrak{m}^2$, it suffices to retain the linear terms. Localisation introduces no further linear relations: modulo the square of the ideal of the origin, any denominator with value $c \neq 0$ can be written $c + u$ with $u^2 = 0$, and has inverse

$$(c + u)^{-1} = c^{-1} - c^{-2}u.$$

In $L_{(t)}$, the class of t is a basis, so the cotangent dimension is one. In $R_{(x,y)}$, the maximal ideal is generated by x, y , and

$$R/(x, y)^2 \cong k[x, y]/(x^2, xy, y^2).$$

There is no linear relation between x and y in this quotient ring; every element has a unique expression $a + bx + cy$. Thus the classes of x, y form a basis of the two-dimensional cotangent space.

Similarly,

$$S/(t, \varepsilon)^2 \cong k[t, \varepsilon]/(t^2, t\varepsilon, \varepsilon^2).$$

The classes of t, ε are linearly independent and generate the cotangent space, so its dimension is two. For D , the maximal ideal (ε) already has square zero and is one-dimensional over k .

The calculations can be summarised as follows. ‘‘Cotangent’’ always refers to the specified origin.

Ring	Krull dimension	Local dimension at the origin	Cotangent dimension	Components	Reduced?
$k[t]$	1	1	1	1	Yes
$k[x, y]/(xy)$	1	1	2	2	Yes
$k[t, \varepsilon]/(\varepsilon^2)$	1	1	2	1	No
$k[\varepsilon]/(\varepsilon^2)$	0	0	1	1	No

Thus Krull dimension is not the number of variables in a presentation, nor must it equal the cotangent dimension. Even matching Krull and cotangent dimensions do not distinguish the crossing of two components from a thickening of a single component.

4. Two generic stalks with different properties

The generic point of the component $V(x)$ in $\text{Spec } R$ is the ideal (x) . In $R_{(x)}$, the element y is invertible. The equation $xy = 0$ then forces $x = 0$. Every nonzero polynomial in y also becomes a unit, so

$$R_{(x)} \cong k(y).$$

This is a field, of Krull dimension zero.

The unique generic point of $\text{Spec } S$ is $\mathfrak{p} = (\varepsilon)$. Elements outside $\mathfrak{p}$ have the form $f(t) + \varepsilon g(t)$ with $f \neq 0$. In $k(t)[\varepsilon]/(\varepsilon^2)$, such an element has inverse

$$(f + \varepsilon g)^{-1} = f^{-1} - \varepsilon g f^{-2}.$$

Since all nonzero polynomials $f(t)$ already have to be inverted, this formula proves

$$S_{(\varepsilon)} \cong k(t)[\varepsilon]/(\varepsilon^2).$$

This local ring also has dimension zero, but is not a field: ε is still nonzero and nilpotent. Thus topological irreducibility alone does not ensure that the generic stalk is a field. [Lemma 11.19](#) requires an *integral* scheme, meaning topologically irreducible and reduced; $\text{Spec } S$ fails the second condition.

Checks and common pitfalls

- To prove a dimension, one long chain gives only a lower bound. The complete lists of prime ideals above give the upper bounds.
- In R , do not include the zero ideal in a prime chain: $xy = 0$ proves that it is not prime.
- Reducedness and topological irreducibility are different properties. R is reduced but has two components; S has one component but is nonreduced.
- The cotangent space uses the square of the maximal ideal, not merely the quotient by the maximal ideal. The latter is only the residue field and removes all linear classes.

Sources and editorial status

References: Holger Brenner, *Bündel, Garben und Kohomologie*, [Lecture 9, revision 793634](#), Lemmas 9.10 and 9.12; [Lecture 11, revision 1019976](#), Lemma 11.3, Definition 11.6, Lemma 11.7, and Lemma 11.19. All additional worked examples and solutions here have independent editorial status, not the status of original public course solutions.

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Integrative problem 7: quasicoherent sheaves, localisation, and fibres

The problem and solution below are independently written editorial material, not a public problem or solution attributed to Holger Brenner. They build on [Lemma 14.4 on stalks](#), [Lemma 14.5 on sections over principal opens](#), and [Lemma 14.9 on exactness](#) in Brenner's course. All these lemmas apply to modules over commutative rings; the ring used here satisfies this hypothesis.

Problem statement

Let k be any field, $A = k[t]$, and

$$X = \text{Spec}(A), \quad T = A/(t^2), \quad M = A \oplus T, \quad \mathcal{F} = \widetilde{M}.$$

Use the cover $U = D(t)$ and $V = D(1-t)$ of X .

1. Compute the sections of $\mathcal{F}$ on $X, U, V, U \cap V$, including the restriction maps. Explain how compatible sections on U and V determine exactly one global section.
2. Determine the stalks and fibres of $\mathcal{F}$ at $p = (t)$ and at the generic point $\eta = (0)$. Is $\mathcal{F}$ quasicoherent, coherent, or locally free around p ?
3. Prove that

$$0 \longrightarrow A/(t) \xrightarrow{j} T \xrightarrow{q} A/(t) \longrightarrow 0, \quad j(\bar{a}) = \overline{ta}, \quad q(\bar{b}) = \bar{b},$$

is exact but does not split as a sequence of A -modules. Compare the effects of localisation and of taking the fibre at p .

Here the fibre of a sheaf of modules at x means $\mathcal{F}(x) = \mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x)$, not its stalk $\mathcal{F}_x$.

Complete solution

Sections and gluing

Since $t + (1-t) = 1$, no prime ideal contains both elements. Thus $D(t) \cup D(1-t) = X$, while their intersection is $D(t(1-t))$. By Lemma 14.5, sections over every principal open are obtained by localising M .

In T_t , the element t is both invertible and square-zero; consequently $1 = 0$ and $T_t = 0$. In contrast, in T we have

$$(1-t)(1+t) = 1-t^2 = 1.$$

Since $1-t$ already acts invertibly on T , localisation at $1-t$ does not change that module. With this identification we obtain

$$\begin{aligned}\Gamma(X, \mathcal{F}) &= A \oplus T, \\ \Gamma(U, \mathcal{F}) &= A_t, \\ \Gamma(V, \mathcal{F}) &= A_{1-t} \oplus T, \\ \Gamma(U \cap V, \mathcal{F}) &= A_{t(1-t)}.\end{aligned}$$

Restriction from X to U sends (a, τ) to $a/1$ and kills the torsion component. Restriction to V sends it to $(a/1, \tau)$. From U to the overlap, we localise further at $1-t$. From V to the overlap, we localise the first component at t and send the T component to zero.

Thus $a \in A_t$ and $(b, \tau) \in A_{1-t} \oplus T$ form a compatible pair exactly when $a = b$ in $A_{t(1-t)}$. All these rings lie in the fraction field $k(t)$. If

$$\frac{f}{t^r} = \frac{g}{(1-t)^s},$$

then $(1-t)^s f = t^r g$. The polynomials t^r and $(1-t)^s$ are coprime in $k[t]$, so t^r divides f . The common fraction is therefore actually a polynomial $h \in A$. This proves

$$A_t \cap A_{1-t} = A \quad \text{inside } k(t).$$

The original pair glues to $(h, \tau) \in A \oplus T$. Uniqueness follows from uniqueness of h and the fact that restriction $T \rightarrow T_{1-t}$ is an isomorphism. For example, the section $t+1$ on U and the section $(t+1, \bar{1}+t)$ on V glue to $(t+1, \bar{1}+t)$ on X .

Stalks are not fibres

Write $A_{(t)}$ for localisation at the complement of (t) . Every polynomial with nonzero constant term acts invertibly on T : modulo t^2 , it has the form $a + bt$ with $a \neq 0$, and its inverse is $a^{-1} - ba^{-2}t$. Hence

$$\mathcal{F}_p = M_{(t)} \cong A_{(t)} \oplus A/(t^2).$$

The residue field $\kappa(p)$ is k . Taking the fibre gives

$$\mathcal{F}(p) \cong M_{(t)}/tM_{(t)} \cong k \oplus k.$$

At the generic point, $\mathcal{O}_{X,\eta} = k(t)$ and t is already invertible. Thus

$$\mathcal{F}_\eta \cong k(t), \quad \kappa(\eta) = k(t), \quad \mathcal{F}(\eta) \cong k(t).$$

As the sheaf associated to an A -module, $\mathcal{F}$ is quasicohherent. The module M has two generators and $A = k[t]$ is Noetherian, so $\mathcal{F}$ is also coherent. This agrees with [Definition 14.11 of quasicohherence](#) and [Definition 14.12 of coherence](#), with the additional finiteness condition for coherence.

However, $\mathcal{F}$ is not locally free around p . The element $(0, \bar{1}) \in M_{(t)}$ is nonzero and annihilated by $t^2 \neq 0$. A free module over the integral domain $A_{(t)}$ has no such torsion: multiplication by a nonzero element is injective in each coordinate. If $\mathcal{F}$ were locally free on a neighbourhood of p , its stalk at p would be free, contradicting this observation. On $D(t)$, by contrast, this sheaf is free of rank one.

Exactness, localisation, and loss of injectivity on fibres

The map j is well defined because replacing a by $a + tc$ changes ta only by t^2c . If $j(\bar{a}) = 0$, then $ta \in (t^2)$, so $a \in (t)$ and $\bar{a} = 0$. Thus j is injective. The map q is surjective, with kernel $(t)/(t^2)$, exactly the image of j . The sequence is exact.

Suppose there were an A -linear map $s : A/(t) \rightarrow T$ with $q \circ s$ the identity. Then $s(\bar{1})$ would have to be $\overline{1 + ct}$ for some $c \in k$. Linearity with respect to t requires

$$ts(\bar{1}) = s(t\bar{1}) = s(0) = 0.$$

But $t\overline{1 + ct} = \bar{t} \neq 0$ in T . This contradiction proves that the sequence does not split.

Localisation preserves exactness. After localising at t , all three modules become zero. After localising at $A \setminus (t)$, the sequence remains the same nonsplit exact sequence on stalks at p . By Lemma 14.9, the associated sheaves also form a short exact sequence.

Taking fibres differs from localising. Tensoring with $\kappa(p) = A/(t)$ produces

$$k \xrightarrow{0} k \xrightarrow{\text{id}} k \longrightarrow 0.$$

The first map is zero because its generator was sent to $\bar{t}$, which becomes zero after quotienting T by tT . The second map is the identity because the class $\bar{1}$ still maps to $\bar{1}$. The tensor sequence remains exact at the last two terms, but its first map is no longer injective. There is no contradiction: tensoring is generally only right exact, whereas localisation is exact.

Checks and pitfalls

The two fibre dimensions are 2 at p and 1 at η ; both are consistent with torsion disappearing after t is inverted. The torsion component must not be discarded on $D(1 - t)$, since that open set contains p . The statement that exactness of a sheaf sequence can be checked on **stalks** cannot be replaced by a statement that all maps on **fibres** must remain injective.

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Integrative problem 8: a Čech calculation with signs and quotient classes

This problem and solution were independently written to synthesise localisation methods and Čech cohomology. The referenced theory is Holger Brenner's: [Definition 26.3](#), [Example 26.7](#), and [Lemma 26.8](#). This is not an addition to the public solutions of the source worksheet.

Problem statement

Let k be any field and $R = k[x, y, z]$. Take

$$X = \text{Spec}(R) \setminus V(x, y), \quad U_0 = D(x), \quad U_1 = D(y).$$

Thus the entire line defined by $x = y = 0$ is removed, not just a single point. We work with the structure sheaf $\mathcal{O}_X$ and the index order $0 < 1$.

1. Write down the Čech complex, including the signs of its differential. Compute $\check{H}^0$ and $\check{H}^1$ as modules over $k[z]$.

2. Determine the normal representative of the class given by

$$c = \frac{z}{xy} + \frac{1}{x^2y} + \frac{x^2}{y^3} + \frac{y}{x^4}.$$

Show explicitly which terms are coboundaries.

3. Explain why the quotient for $\check{H}^1$ is a quotient module, not a quotient ring. Determine whether the class $z/(xy)$ is zero; then determine its image in the calculation obtained by setting $z = 0$.
4. Check the hypotheses identifying $\check{H}^1$ with $H^1(X, \mathcal{O}_X)$, and conclude that X is not affine even though $\Gamma(X, \mathcal{O}_X) = R$.

Complete solution

The complex and its signs

The intersection is $U_0 \cap U_1 = D(xy)$. By the description of sections through localisation, the complex is

$$0 \longrightarrow R_x \oplus R_y \xrightarrow{\delta^0} R_{xy} \longrightarrow 0, \quad \delta^0(a, b) = b - a.$$

Both terms on the right of the formula are taken after restriction to $D(xy)$. The sign comes from $(\delta^0 s)_{01} = s_1|_{01} - s_0|_{01}$, following the order $0 < 1$. Since the cover has only two members, $\check{C}^2 = 0$; every element of R_{xy} is therefore a degree-one cocycle.

To recall the convention, for three ordered cover members $0 < 1 < 2$, the next differential would be

$$(\delta^1 c)_{012} = c_{12}|_{012} - c_{02}|_{012} + c_{01}|_{012}.$$

Substituting $c_{ij} = s_j - s_i$ gives $(s_2 - s_1) - (s_2 - s_0) + (s_1 - s_0) = 0$. This explains the sign cancellation; for our cover there is no degree-two term to compute.

The kernel and quotient module

The three localisation rings can be viewed as subrings of the Laurent polynomial ring

$$R_{xy} = k[z][x, x^{-1}, y, y^{-1}].$$

As a $k[z]$ -module, this last ring has a monomial basis $x^i y^j$ with $(i, j) \in \mathbb{Z}^2$. Every element is a **finite** linear combination of these monomials. In R_x , the exponent of y must be nonnegative; in R_y , the exponent of x must be nonnegative. Uniqueness of Laurent expressions gives

$$R_x \cap R_y = k[z][x, y] = R.$$

Thus the kernel of δ^0 consists of pairs (f, f) , $f \in R$, and

$$\check{H}^0(\{U_0, U_1\}, \mathcal{O}_X) \cong R.$$

The image of δ^0 is $R_x + R_y$: the minus sign does not change the generated submodule because R_x is closed under negation. This submodule consists exactly of combinations of monomials with at least one of i, j nonnegative. Hence

$$\begin{aligned} \check{H}^1(\{U_0, U_1\}, \mathcal{O}_X) &= R_{xy} / (R_x + R_y) \\ &\cong \bigoplus_{a \geq 1, b \geq 1} k[z][x^{-a} y^{-b}]. \end{aligned}$$

A normal representative exists by discarding all monomials whose x or y exponent is nonnegative. Its uniqueness follows from uniqueness of Laurent coefficients: a combination of monomials with both exponents negative cannot equal a combination of monomials in the complementary set. Thus this is not merely a generating list but a genuine basis of a free module over $k[z]$.

Reducing the given cocycle

Set

$$a = -\frac{y}{x^4} \in R_x, \quad b = \frac{x^2}{y^3} \in R_y.$$

Our chosen sign convention gives

$$\delta^0(a, b) = \frac{x^2}{y^3} + \frac{y}{x^4}.$$

Therefore

$$[c] = \left[\frac{z}{xy} + \frac{1}{x^2y} \right] = z[x^{-1}y^{-1}] + [x^{-2}y^{-1}].$$

This is the required normal representative. Its class is nonzero: the coefficient of the basis element $[x^{-2}y^{-1}]$ is 1, nonzero over any field. This remains true in characteristic two; the sign convention is still valid, although negation coincides with the identity.

Why this is not a quotient ring

The submodule $R_x + R_y$ is stable under multiplication by R , but is not an ideal of R_{xy} . It contains 1 but does not contain $1/(xy)$. An ideal containing 1 must be the whole ring. Thus the quotient notation above must be read as a quotient of R -modules, or in particular of $k[z]$ -modules, not as a quotient ring of R_{xy} .

An example of using the correct module structure is

$$xy[x^{-1}y^{-1}] = [1] = 0$$

even though $[x^{-1}y^{-1}] \neq 0$. This does not mean that arbitrary classes can be multiplied using multiplication in R_{xy} ; that multiplication does not descend to this quotient.

The class $z[x^{-1}y^{-1}]$ is nonzero because its coefficient z is nonzero in $k[z]$ and the module above is free over $k[z]$. Substitution $z = 0$ defines a map from our complex to the complex

$$0 \longrightarrow k[x, y]_x \oplus k[x, y]_y \longrightarrow k[x, y]_{xy} \longrightarrow 0.$$

The map is compatible with $b - a$, so it induces a cohomology map replacing every coefficient $f(z)$ by $f(0)$. The image of $z[x^{-1}y^{-1}]$ is zero, whereas the image of $[c]$ is $[x^{-2}y^{-1}]$, still nonzero. This conclusion follows directly from the complexes; no unproved cohomology base-change theorem is needed.

From Čech to sheaf cohomology

The spaces U_0 , U_1 , and $U_0 \cap U_1$ are affine, with rings R_x , R_y , and R_{xy} . All three are integral domains, being localisations of the integral domain $k[x, y, z]$. The restriction of $\mathcal{O}_X$ to each open is its structure sheaf. [Lemma 25.7](#) therefore gives

$$H^1(U_0, \mathcal{O}_X) = H^1(U_1, \mathcal{O}_X) = H^1(U_0 \cap U_1, \mathcal{O}_X) = 0.$$

The hypotheses of [Lemma 26.8](#) hold, so the $\check{H}^1$ computed above equals $H^1(X, \mathcal{O}_X)$. In particular, this group is nonzero.

As a nonempty open subset of the integral scheme $\text{Spec}(R)$, X is integral. If X were affine, its coordinate ring would be an integral domain and Lemma 25.7 would force $H^1(X, \mathcal{O}_X) = 0$, contradicting the class $[x^{-1}y^{-1}]$ we found. Thus X is not affine. The identity $\Gamma(X, \mathcal{O}_X) = R$ remains true, since global sections form the kernel of the Čech complex by the sheaf gluing property. The global section ring alone does not recover this nonaffine scheme.

Checks and pitfalls

A denominator containing both x and y does not automatically yield a nonzero class: after cancellation, both exponents must genuinely be negative. The order $0 < 1$ fixes the sign $b - a$. Using $a - b$ consistently gives an isomorphic complex, but mixing the conventions spoils the representative calculation. The statement about sheaf H^1 is used only after the comparison hypotheses have been checked.

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Integrative problem 9: computable projective cohomology

This is an independent editorial problem and solution, not an additional public solution attributed to Holger Brenner. The calculation connects [Theorem 26.10](#), on comparison with Čech cohomology for the standard affine cover, and [Theorem 27.4](#), on cohomology of twisted structure sheaves, with evaluation of sections at a point.

Problem statement

Let k be any field,

$$S = k[X_0, X_1, X_2], \quad P = \mathbb{P}_k^2 = \text{Proj}(S), \quad \mathcal{E} = \mathcal{O}_P(-4) \oplus \mathcal{O}_P(2).$$

For $n \in \mathbb{Z}$, use the description $\Gamma(D_+(X_i), \mathcal{O}_P(n)) = (S_{X_i})_n$; the subscript n denotes the homogeneous part of degree n .

1. Write down the Čech complex for the cover $U_i = D_+(X_i)$, ordered by $0 < 1 < 2$. Use Laurent monomials to explain why $H^1(P, \mathcal{O}_P(n)) = 0$ and how bases for H^0 and H^2 are obtained.
2. Compute all $H^q(P, \mathcal{E})$ and give concrete bases for the nonzero groups, not merely their dimensions.
3. Take the rational point $p = [1 : 1 : 0]$. In the trivialisation on U_0 , use the basis X_0^{-4} for $\mathcal{O}_P(-4)$ and X_0^2 for $\mathcal{O}_P(2)$. Compute the image and kernel of evaluation

$$\text{ev}_p : H^0(P, \mathcal{E}) \longrightarrow \mathcal{E}(p).$$

Is $\mathcal{E}$ generated by its global sections?

Complete solution

The Čech complex and decomposition by monomials

The standard affine cover and all its intersections give the complex

$$\begin{aligned} 0 \longrightarrow \bigoplus_{i=0}^2 (S_{X_i})_n &\xrightarrow{\delta^0} (S_{X_0X_1})_n \oplus (S_{X_0X_2})_n \oplus (S_{X_1X_2})_n \\ &\xrightarrow{\delta^1} (S_{X_0X_1X_2})_n \longrightarrow 0. \end{aligned}$$

All terms are viewed inside the common Laurent ring; the restriction maps are inclusions. With components ordered 01, 02, 12,

$$\delta^0(s_0, s_1, s_2) = (s_1 - s_0, s_2 - s_0, s_2 - s_1), \quad \delta^1(c_{01}, c_{02}, c_{12}) = c_{12} - c_{02} + c_{01}.$$

Their composite is zero by direct cancellation. The scheme P is projective over the commutative ring k , and $\mathcal{O}_P(n)$ is invertible, hence quasicoherent. Thus the hypotheses of Theorem 26.10 hold: the cohomology of this complex is the required sheaf cohomology.

Fix a monomial

$$m = X_0^{\alpha_0} X_1^{\alpha_1} X_2^{\alpha_2}, \quad \alpha_0 + \alpha_1 + \alpha_2 = n,$$

and write $N = \{i : \alpha_i < 0\}$. This monomial occurs in $S_{\prod_{i \in J} X_i}$ exactly when $N \subseteq J$: negative exponents are allowed only for variables that have been inverted. The differentials do not mix distinct monomials, so the complex decomposes as a direct sum of coefficient complexes for individual monomials. Every element is still a finite sum; we are not using infinite Laurent series.

There are four possibilities.

1. If $N = \emptyset$, the coefficient complex is $k^3 \rightarrow k^3 \rightarrow k$ with the maps above. The kernel of the first map consists of (a, a, a) and has dimension one. Its image has dimension two. The second map is surjective with two-dimensional kernel; since $\delta^1 \delta^0 = 0$, this kernel is exactly the image of the first map. Thus only H^0 contributes one copy of k .
2. If $\#N = 1$, only one degree-zero term and two degree-one terms occur. The complex has the form $k \rightarrow k^2 \rightarrow k$. The first map is injective, with each coefficient equal to 1 or -1 ; the second map is surjective. Its kernel is one-dimensional and contains the image of the first map, so they agree. All cohomology in this case is zero.
3. If $\#N = 2$, only one degree-one term and one degree-two term occur. The map between them is 1 or -1 , hence an isomorphism. All cohomology is zero.
4. If $\#N = 3$, only the degree-two term occurs. This monomial contributes one copy of k to H^2 .

The dimension arguments hold in every characteristic: the coefficients 1 and -1 remain nonzero even when they coincide. We obtain

$$\begin{aligned} H^0(P, \mathcal{O}_P(n)) &= S_n, \\ H^1(P, \mathcal{O}_P(n)) &= 0, \\ H^2(P, \mathcal{O}_P(n)) &= \bigoplus_{\substack{\alpha_0, \alpha_1, \alpha_2 < 0 \\ \alpha_0 + \alpha_1 + \alpha_2 = n}} k [X_0^{\alpha_0} X_1^{\alpha_1} X_2^{\alpha_2}]. \end{aligned}$$

There is no cohomology in degrees $q \geq 3$, since the Čech complex ends in degree two. This is also the case $R = k, d = 2$ of Theorem 27.4; the monomial calculation exhibits the bases and exactness directly.

Cohomology of the two twists

There are no polynomial monomials of degree -4 , so $H^0(P, \mathcal{O}_P(-4)) = 0$. For H^2 , write $a_i = -\alpha_i$. The conditions become $a_i \geq 1$ and $a_0 + a_1 + a_2 = 4$. Their solutions are exactly $(2, 1, 1), (1, 2, 1), (1, 1, 2)$. Thus a basis is

$$\left[\frac{1}{X_0^2 X_1 X_2} \right], \quad \left[\frac{1}{X_0 X_1^2 X_2} \right], \quad \left[\frac{1}{X_0 X_1 X_2^2} \right].$$

Square brackets denote classes in the quotient of the degree-two term by the image of the degree-one term; these three fractions are not global sections of $\mathcal{O}_P(-4)$.

For $n = 2$, a basis of S_2 is

$$X_0^2, \quad X_0 X_1, \quad X_0 X_2, \quad X_1^2, \quad X_1 X_2, \quad X_2^2.$$

Three negative integers cannot sum to 2, so $H^2(P, \mathcal{O}_P(2)) = 0$. Both twists have $H^1 = 0$. The complex for the direct sum $\mathcal{E}$ is the direct sum of the two complexes, so its kernels, images, and cohomology also decompose. The complete conclusion is

$$H^q(P, \mathcal{E}) \cong \begin{cases} k^6, & q = 0, \\ 0, & q = 1, \\ k^3, & q = 2, \\ 0, & q \geq 3. \end{cases}$$

A basis for H^0 consists of the six pairs $(0, m)$ with m in the list of quadratics above. A basis for H^2 consists of the three pairs $(\xi, 0)$ with ξ in the list of fraction classes above.

Evaluation at a point and global generation

On U_0 , use coordinates $u = X_1/X_0$ and $v = X_2/X_0$. The point p is given by $(u, v) = (1, 0)$ and has residue field k . The specified trivialisation identifies $\mathcal{E}(p)$ with $k \oplus k$.

A general global section has the form $(0, f)$ with

$$f = a_{00}X_0^2 + a_{01}X_0X_1 + a_{02}X_0X_2 + a_{11}X_1^2 + a_{12}X_1X_2 + a_{22}X_2^2.$$

Its local coefficient in the positive summand is $f/X_0^2 = a_{00} + a_{01}u + a_{02}v + a_{11}u^2 + a_{12}uv + a_{22}v^2$. Hence

$$\text{ev}_p(0, f) = (0, a_{00} + a_{01} + a_{11}).$$

Its image is the line $\{0\} \oplus k$, since the section $(0, X_0^2)$ maps to $(0, 1)$. Its kernel is defined by the single equation $a_{00} + a_{01} + a_{11} = 0$ and has basis

$$\begin{aligned} (0, X_0X_1 - X_0^2), & \quad (0, X_1^2 - X_0^2), \\ (0, X_0X_2), & \quad (0, X_1X_2), \quad (0, X_2^2). \end{aligned}$$

These five vectors are linearly independent because each of the monomials X_0X_1 , X_1^2 , X_0X_2 , X_1X_2 , and X_2^2 occurs in only one vector; all satisfy the kernel equation.

Since the evaluation image is one-dimensional while the fibre is two-dimensional, global sections do not generate $\mathcal{E}$ at p . Thus $\mathcal{E}$ is not globally generated. The summand $\mathcal{O}_P(2)$ itself is globally generated: at any point, at least one X_i is nonzero, and the section X_i^2 is a generator in the trivialisation on $D_+(X_i)$. The failure for $\mathcal{E}$ comes from the summand $\mathcal{O}_P(-4)$, which has a one-dimensional fibre at every point but no nonzero global sections.

Checks and pitfalls

The three generators of $H^2(\mathcal{O}_P(-4))$ all have degree -4 and **three** negative exponents. A fraction with any nonnegative exponent is a coboundary in degree two. Do not confuse sheaf rank, the dimension of the global section space, and cohomology dimensions: here $\mathcal{E}$ has rank 2, but $h^0(\mathcal{E}) = 6$ and $h^2(\mathcal{E}) = 3$. Even six global sections need not generate a two-dimensional fibre.

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Integrative problem 10: Euler characteristic and a thickened point

The following problem and solution were independently written; they are not a source problem or solution by Holger Brenner. The foundations used are [Theorem 26.10](#), [Theorem 27.4](#), [Definition 27.8](#), and [Lemma 27.9](#) in the BGK translation. The source statement of [additivity](#) applies to short exact sequences of coherent sheaves on a projective scheme over a field.

Problem

Let K be any field, $X = \mathbb{P}_K^1$ with coordinates $[X_0 : X_1]$, $m \geq 1$, and $n \in \mathbb{Z}$. Write $P = [1 : 0]$ and

$$Z = \text{Proj}(K[X_0, X_1]/(X_1^m)), \quad i : Z \hookrightarrow X.$$

The notation $\mathcal{O}_Z(n)$ means $i^*\mathcal{O}_X(n)$.

1. Prove that multiplication by X_1^m gives a short exact sequence $0 \rightarrow \mathcal{O}_X(n-m) \rightarrow \mathcal{O}_X(n) \rightarrow i_*\mathcal{O}_Z(n) \rightarrow 0$. Determine the ring of Z on the chart containing it.
2. Compute h^0 and h^1 of all three sheaves, and their Euler characteristics. Explain why all higher cohomology vanishes.
3. For the cover $U_0 = D_+(X_0)$ and $U_1 = D_+(X_1)$, determine the connecting homomorphism $\delta : H^0(Z, \mathcal{O}_Z(n)) \rightarrow H^1(X, \mathcal{O}_X(n-m))$. Determine when restriction of global sections to Z is surjective.
4. Work out the case $m = 3, n = 0$ explicitly. Explain why the length of Z , rather than its number of topological points, occurs in the Euler formula.

Solution

1. Exactness on two charts

On U_0 use $t = X_1/X_0$; on U_1 use $s = X_0/X_1 = t^{-1}$. Write the local frames of $\mathcal{O}_X(r)$ as $e_0^{(r)} = X_0^r$ and $e_1^{(r)} = X_1^r$ for every $r \in \mathbb{Z}$. For $r < 0$, this notation denotes a frame of an invertible sheaf, not a global polynomial. On the overlap, $e_1^{(r)} = t^r e_0^{(r)}$.

In these frames, multiplication by X_1^m is given by

$$K[t] \xrightarrow{t^m} K[t] \quad \text{on } U_0, \quad K[s] \xrightarrow{1} K[s] \quad \text{on } U_1.$$

The first map is injective because $K[t]$ is an integral domain; the second is an isomorphism. Their quotients are $K[t]/(t^m)$ and 0, respectively. They are compatible on the overlap, since t is a unit there. Thus the required sheaf sequence is exact. All its sheaves are coherent: the first two are locally free of rank one, while the quotient has a finite presentation on these Noetherian charts.

The scheme Z lies entirely in U_0 , and

$$Z = \text{Spec}(K[t]/(t^m)).$$

Its only prime ideal is (t) , so its topological space has only the point P . However, $1, t, \dots, t^{m-1}$ form a basis of its ring as a vector space over K .

2. Cohomology and Euler characteristic

Theorem 26.10 applies because X is projective over K and the sheaves above are quasicoherent. In the frame $e_0^{(r)}$, the Čech complex for $\mathcal{O}_X(r)$ has differential

$$K[t] \oplus K[t^{-1}] \rightarrow K[t, t^{-1}], \quad (a, b) \mapsto t^r b - a.$$

This uses the sign convention $s_1 - s_0$. There are no terms of degree two or higher. The kernel identifies with $K[t] \cap t^r K[t^{-1}]$, while the cokernel is

$$K[t, t^{-1}] / (K[t] + t^r K[t^{-1}]).$$

For the kernel, the available monomials are exactly t^j with $0 \leq j \leq r$. For the cokernel, the surviving monomials are exactly t^j with $r < j < 0$. Linear independence of Laurent monomials gives

$$h^0(\mathcal{O}_X(r)) = \max(r+1, 0), \quad h^1(\mathcal{O}_X(r)) = \max(-r-1, 0), \quad H^q(\mathcal{O}_X(r)) = 0 \quad (q \geq 2).$$

This calculation is also the case $d = 1$ of Theorem 27.4. Hence $\chi(\mathcal{O}_X(r)) = r+1$ for every integer r , including $r < 0$.

The frame $e_0^{(n)}$ trivialises $\mathcal{O}_Z(n)$. The Čech complex for $i_*\mathcal{O}_Z(n)$ has only the term $K[t]/(t^m)$ in degree zero: its values on U_1 and on the overlap are zero. Therefore

$$h^0(i_*\mathcal{O}_Z(n)) = m, \quad H^q(X, i_*\mathcal{O}_Z(n)) = 0 \quad (q > 0), \quad \chi(i_*\mathcal{O}_Z(n)) = m.$$

On Z itself the result is the same: its topological space has one point, so the global section functor equals the stalk functor at that point and is exact. Its positive cohomology therefore vanishes. The additivity formula now becomes the identity

$$\underbrace{n+1}_{\chi(\mathcal{O}_X(n))} = \underbrace{n-m+1}_{\chi(\mathcal{O}_X(n-m))} + \underbrace{m}_{\chi(i_*\mathcal{O}_Z(n))}.$$

Why does additivity follow from exactness? The cohomology sequence ends as

$$0 \rightarrow H^0(\mathcal{O}_X(n-m)) \rightarrow H^0(\mathcal{O}_X(n)) \rightarrow H^0(\mathcal{O}_Z(n)) \xrightarrow{\delta} H^1(\mathcal{O}_X(n-m)) \rightarrow H^1(\mathcal{O}_X(n)) \rightarrow 0.$$

In a finite exact sequence of finite-dimensional vector spaces, the dimension of each term is the sum of the dimensions of the incoming and outgoing images. The alternating sum cancels every image dimension twice with opposite signs. The result is exactly the Euler formula above, not additivity of h^0 alone.

3. The connecting homomorphism

Represent a section on Z uniquely by $q(t) = \sum_{r=0}^{m-1} c_r t^r$. Lift it to $q(t)e_0^{(n)}$ on U_0 and to 0 on U_1 . The Čech difference of the two lifts is $-q(t)e_0^{(n)}$. Dividing by the multiplier t^m gives

$$\delta([q]) = [-t^{-m}q(t)] \quad \text{in} \quad \frac{K[t, t^{-1}]}{K[t] + t^{n-m}K[t^{-1}]}.$$

Changing the lifts changes this representative by a coboundary. In particular, replacing q by $q + t^m a$ adds $-a \in K[t]$, so its class does not change.

The class $\delta([t^r])$ is nonzero exactly when $n-m < r-m < 0$, that is, $r > n$ for $0 \leq r < m$. These nonzero classes are linearly independent. Consequently

$$\dim \ker \delta = \min(m, \max(n+1, 0)), \quad \dim \operatorname{im} \delta = m - \min(m, \max(n+1, 0)).$$

This is also visible directly: for $n \geq 0$, global sections of $\mathcal{O}_X(n)$ are polynomials in t of degree at most n , and restriction to Z takes their classes modulo t^m . For $n < 0$ there are no nonzero global sections. Restriction is surjective exactly when $n \geq m-1$.

4. One point of length three

For $m=3$ and $n=0$, the global sequence is

$$0 \longrightarrow K \xrightarrow{a \mapsto (a, 0, 0)} K^3 \xrightarrow{(a, b, c) \mapsto (-b, -c)} K^2 \longrightarrow 0,$$

with basis $1, t, t^2$ of $K[t]/(t^3)$ and basis $[t^{-2}], [t^{-1}]$ of $H^1(\mathcal{O}_X(-3))$. Thus t and t^2 are sections on Z that cannot extend to global functions on X . The Euler formula gives

$$1 = (-2) + 3.$$

The filtration $K[t]/(t^3) \supset (t)/(t^3) \supset (t^2)/(t^3) \supset 0$ has three successive quotients isomorphic to K ; its length is three. In general, the filtration by powers of t has m such quotients. Euler characteristic records the dimension of sections and therefore counts this length m , not merely the single point of the topological space.

Checks and material provenance

Surjectivity of sheaves is checked locally; surjectivity on global sections is measured by δ and must not be inferred automatically. No assumption that K is algebraically closed is needed, since P is an explicitly specified rational point.

Holger Brenner's BGK Lectures 26 and 27 use the frozen parent revisions [793619](#) and [1070036](#). The revision of the cohomology formula entity used is [1102393](#); other transclusion identities remain those in the edition's frozen manifest. This independent problem, bridge exposition, and solution: CC BY-SA 4.0. Model provenance: OpenAI Codex gpt-5.6-sol, Ultra. The credits and licences of source components remain in force; no human authorship or review, or endorsement by the source author, is claimed.

Integrative problem 11: a classical conic and its scheme

This problem and solution are independent material, not a source problem by Holger Brenner. The classical comparison starts from [Lemma 28.2 on affine charts](#) and [Definition 28.3 of regular functions](#) in Lecture 28 of *Algebraische Kurven*. Their translation IDs are br-ak-2012-l28-lem-01 and br-ak-2012-l28-def-02, respectively. On the scheme side, use [BGK Lemma 12.9](#) and [BGK Lemma 12.17](#): Proj charts are spectra of degree-zero components of localisations, computed by dehomogenisation.

Problem

Let K be an algebraically closed field of arbitrary characteristic. Distinguish

$$C_{\text{kl}} = \{[x : y : z] \in \mathbb{P}^2(K) \mid xz - y^2 = 0\}$$

with its Zariski topology and classical regular functions from the scheme

$$C = \text{Proj}(A), \quad A = K[X, Y, Z]/(XZ - Y^2).$$

1. Compute the charts $U_X = D_+(X)$ and $U_Z = D_+(Z)$ and the gluing on their overlap. Prove that they cover the entire scheme, not just its K -points.
2. Prove that $\nu : \mathbb{P}_K^1 \rightarrow C$, given on points by $[u : v] \mapsto [u^2 : uv : v^2]$, is a scheme isomorphism.
3. List the closed points and generic point of C . Compute the stalks of the structure sheaf and their residue fields, and compare them with the classical local rings. Determine $\Gamma(C, \mathcal{O}_C)$.
4. Determine the scheme-theoretic intersection with the line $Z = 0$ and explain what information is lost if only the set of intersection points is retained.
5. Explain why A is not the global function ring of C and why $\nu^*\mathcal{O}_C(1) \cong \mathcal{O}_{\mathbb{P}_K^1}(2)$.

Solution

1. Two charts and one inversion

Lemma 12.17 applies because A is standard graded with a homogeneous relation of degree two. On U_X , put $t = Y/X$ and $z = Z/X$. Then

$$(A_X)_0 = K[t, z]/(z - t^2) \cong K[t].$$

On U_Z , put $s = Y/Z$ and $x = X/Z$. Then

$$(A_Z)_0 = K[s, x]/(x - s^2) \cong K[s].$$

If a homogeneous prime ideal $\mathfrak{p}$ contains X and Z , the relation $Y^2 = XZ$ forces it to contain Y . It contains the irrelevant ideal (X, Y, Z) and is therefore not a Proj point. Thus U_X and U_Z cover every scheme point.

On U_X , the condition $Z \neq 0$ means invertibility of t^2 , which is equivalent to invertibility of t . Since

$$s = \frac{Y}{Z} = \frac{Y/X}{Z/X} = \frac{t}{t^2} = t^{-1},$$

the overlap is $\text{Spec}(K[t, t^{-1}])$, and the gluing homomorphism is determined by $s \mapsto t^{-1}$.

2. An isomorphism, not merely a bijection on points

On the chart $u \neq 0$ of $\mathbb{P}_K^1$, the coordinate is v/u . The formula for ν gives $t = Y/X = v/u$ and $Z/X = (v/u)^2$. Thus the chart map is the isomorphism $K[t] \rightarrow K[v/u]$, $t \mapsto v/u$. On the chart $v \neq 0$, the corresponding map is $K[s] \rightarrow K[u/v]$, $s \mapsto u/v$.

These maps are compatible on the overlap because both gluings use inversion. The local maps and their inverses therefore glue to mutually inverse scheme morphisms. This proves that ν is a scheme isomorphism without deducing it merely from a bijection on closed points.

The inverse formula on classical points is $[1 : t]$ on U_X and $[s : 1]$ on U_Z . Both are regular in their charts, so they also give an isomorphism of classical varieties. No step divides by 2; the result remains valid in characteristic two.

3. Points, stalks, and functions

Since K is algebraically closed, the prime ideals of $K[t]$ are (0) and $(t - a)$ for $a \in K$. All closed points of C are therefore

$$P_a = [1 : a : a^2] \quad (a \in K), \quad P_\infty = [0 : 0 : 1].$$

The homogeneous prime ideals representing them in A are $(Y - aX, Z - a^2X)$ and (X, Y) . The generic points of the charts, namely the zero ideals of $K[t]$ and $K[s]$, are identified on the overlap. The result is one generic point η of C , whose closure is all of C .

To see that the zero ideal of A is indeed prime, map X, Y, Z to u^2, uv, v^2 in $K[u, v]$. Modulo $XZ - Y^2$, every polynomial has a representative $B(X, Z) + YC(X, Z)$. Monomials in the image of the first term have both exponents even; those in the image of the second have both exponents odd. Distinct monomials in either group remain distinct, so this map has zero kernel. Thus A is an integral domain, and η is represented by (0) in Proj . The chart argument also shows that there are no other points.

The local rings and residue fields are

$$\begin{aligned} \mathcal{O}_{C, P_a} &= K[t]_{(t-a)}, & \kappa(P_a) &= K, \\ \mathcal{O}_{C, P_\infty} &= K[s]_{(s)}, & \kappa(P_\infty) &= K, \\ \mathcal{O}_{C, \eta} &= K(t), & \kappa(\eta) &= K(t). \end{aligned}$$

For example, the stalk at P_a contains fractions $f(t)/g(t)$ with $g(a) \neq 0$; the map to the residue field evaluates them at a . A stalk is not its residue field: the element $t - a$ is nonzero in $K[t]_{(t-a)}$ but has zero image in K .

The classical definition of a regular function gives the same local ring at each P_a and P_∞ . What the scheme adds is the generic point as an actual point, not a replacement of the local rings at classical points. The topology on the closed-point set, as a subspace of C , agrees with the classical topology: on both charts, closed sets are defined by the same polynomial equations. The classical space itself has no generic point.

Global sections of the structure sheaf are pairs $f(t) \in K[t]$, $g(s) \in K[s]$ with $f(t) = g(t^{-1})$ on the overlap. In the Laurent ring, the first polynomial has only nonnegative exponents and the second only nonpositive exponents. Hence

$$\Gamma(C, \mathcal{O}_C) = K[t] \cap K[t^{-1}] = K.$$

The same calculation applies to classical global regular functions.

4. Intersection with the tangent line

For $Z = 0$, the conic relation gives $Y^2 = 0$. Every prime on the intersection contains Y and cannot also contain X ; hence the whole intersection lies in U_X . The line equation there is $z = 0$, while on the conic $z = t^2$. Thus the scheme-theoretic intersection is

$$C \cap V_+(Z) = \text{Spec}(K[t]/(t^2)).$$

The classical intersection point set is just $\{P_0\}$. The intersection scheme has length two, since the classes of $1, t$ are linearly independent and $t^2 = 0$ with $t \neq 0$. This nilpotent structure is invisible in the one-point set.

This does not mean that the conic is singular. On chart U_X , the equation $z - t^2$ has partial derivative 1 with respect to z ; the chart itself is isomorphic to the affine line. The line equation $z = 0$ vanishes to order two along the parameter t , thus recording tangency. The same argument remains valid in characteristic two.

5. Homogeneous degree does not give global functions

The ring A is graded and contains homogeneous coordinates of positive degree. Multiplying all coordinates of a point by λ multiplies the values of X, Y, Z by λ ; these coordinates therefore do not define K -valued functions on projective points. They are sections of $\mathcal{O}_C(1)$. Regular functions, in contrast, locally use homogeneous fractions of degree zero. Since $\Gamma(C, \mathcal{O}_C) = K$, the two rings are plainly different.

The frames of $\mathcal{O}_C(1)$ on U_X and U_Z are X and Z , with $Z = t^2 X$ on the overlap. Their pullbacks to $\mathbb{P}_K^1$ have frames u^2 and v^2 and transition $v^2 = (v/u)^2 u^2$. These are exactly the gluing data for $\mathcal{O}_{\mathbb{P}_K^1}(2)$.

As a numerical check, its global sections are represented by polynomials in t of degree at most two, with basis $1, t, t^2$. The Čech complex has cokernel

$$K[t, t^{-1}] / (K[t] + t^2 K[t^{-1}]) = 0,$$

since the two subspaces together contain every Laurent monomial. Thus $h^0(\mathcal{O}_C(1)) = 3$, $h^1(\mathcal{O}_C(1)) = 0$, and $\chi(\mathcal{O}_C(1)) = 3$, in accordance with [BGK Theorem 27.4](#). The degree two of the conic is visible in the pulled-back twist, although the conic as a scheme is isomorphic to the projective line.

Checks and material provenance

Algebraic closedness is used to identify all closed points with K -valued points. The chart calculations, scheme isomorphism, and length-two intersection hold over any field. Three objects that must not be confused are the homogeneous coordinate ring, the global section ring, and the function field.

The frozen parent revision of classical Lecture 28 is [1052516](#); that of BGK Lecture 12 is [1003742](#). The BGK entities on [Proj charts](#) and [dehomogenisation](#) have their own frozen identities. This independent problem, bridge exposition, and solution: CC BY-SA 4.0. Model provenance: OpenAI Codex gpt-5.6-sol, Ultra. The credits to Holger Brenner and revision contributors, and the licences of source components, remain in force; no human authorship or review, or endorsement by the source author, is claimed.

Integrative problem 12: source navigation and reconstruction of hypotheses

This is an independently written mathematical navigation problem, not an exercise from Holger Brenner's source material. Its aim is to turn an overly broad argument into a correct statement by tracing results and hypotheses in the frozen source. All necessary information is supplied below; no new web search is needed.

Reading material and source identities

Use the following four locations in the BGK reader. The theorem numbers in the text and the suffixes of digital IDs do not always agree.

Result	Translation location	Frozen semantic-entity revision
Theorem 26.10: Čech comparison	br-bgk-2019-l26-thm-01	1088414
Theorem 27.5: finiteness on projective space	br-bgk-2019-l27-thm-03	1088405
Theorem 27.7: finiteness on projective schemes	br-bgk-2019-l27-thm-05	1088399

Result	Translation location	Frozen semantic-entity revision
Definition 27.8: Euler characteristic	br-bgk-2019-l27-def-01	1091410

The parent lecture-page revisions are [793619](#) for [Lecture 26](#) and [1070036](#) for [Lecture 27](#). The parent pages include other entities by transclusion. The complete identity of the frozen edition therefore also includes those entity revisions, the manifest, and the expanded text.

Problem

Consider the following proposed argument, which is deliberately incorrect:

Let R be a commutative ring, $X = \text{Proj}(R[X_0, \dots, X_N]/\mathfrak{a})$ with $\mathfrak{a}$ homogeneous, and let $\mathcal{F}$ be quasi-coherent. The standard affine cover computes cohomology, so all $H^q(X, \mathcal{F})$ are finite-dimensional over R . Consequently $\sum_q (-1)^q \dim_R H^q(X, \mathcal{F})$ is always an integer-valued Euler characteristic.

1. Separate the claims about computing cohomology, finite generation of modules, and vector space dimension. Find the correct hypotheses and exact source for each claim.
2. Test the finiteness claim on $X = \mathbb{P}_K^1$ and $\mathcal{F} = \bigoplus_{j \geq 0} \mathcal{O}_X$. Test the use of the word “dimension” when $R = \mathbb{Z}$ and $\mathcal{F} = \mathcal{O}_{\mathbb{P}_\mathbb{Z}^1}$.
3. In the induction step of the proof of Theorem 27.5, suppose we are given $\mathcal{E} = \bigoplus_{j=1}^r \mathcal{O}(\ell_j)$ with r finite, a surjection $\mathcal{E} \rightarrow \mathcal{F}$, and coherent kernel $\mathcal{G}$. For an integer $q \geq 1$, suppose R is Noetherian and both $H^{q-1}(\mathcal{E})$ and $H^q(\mathcal{G})$ are finitely generated. Reconstruct the step proving finite generation of $H^{q-1}(\mathcal{F})$, checking the image and kernel of each map.
4. Explain why the parent-page revision alone does not fix the entire lecture content. What can and cannot be proved by a source SHA-256 hash?

Solution

1. Three claims with three scopes

First, Theorem 26.10 applies to a projective scheme over a commutative ring R and a quasicoherent sheaf $\mathcal{F}$. Its conclusion is agreement of sheaf cohomology with Čech cohomology for the standard affine cover, not finite generation of modules. In this cover, every nonempty intersection is $D_+(\prod X_i)$ and is affine. Quasicoherence allows the vanishing of positive cohomology on those affine intersections to be used.

For an embedding in $\mathbb{P}_R^N$, this Čech complex has no terms of degree greater than N . Thus it gives $H^q(X, \mathcal{F}) = 0$ for $q > N$. But having only finitely many terms in a complex does not mean that each term or its cohomology is finitely generated.

Second, Theorem 27.7 requires R to be Noetherian, X projective over R , and $\mathcal{F}$ coherent. Its conclusion is that $H^q(X, \mathcal{F})$ is a finitely generated R -module. Theorem 27.5 gives the same conclusion when $X = \mathbb{P}_R^N$. The word “quasicoherent” in the proposed argument must be strengthened to “coherent” to apply this result.

Third, to use $\dim_K$ as vector space dimension, take $R = K$ to be a field. A field is automatically Noetherian. For X projective over K and $\mathcal{F}$ coherent, finite generation of modules becomes finite dimensionality of vector spaces. Definition 27.8 then gives

$$\chi(X, \mathcal{F}) = \sum_{q=0}^{\dim X} (-1)^q \dim_K H^q(X, \mathcal{F}).$$

The source definition uses the vanishing of cohomology above the dimension. To ensure that the infinite sum is actually finite, the bound $q > N$ from the Čech complex also suffices in this setting with a fixed embedding. No algebraic-closedness hypothesis on K is needed.

The corrected statement, briefly, is: *on a projective scheme over a field, a coherent sheaf has finite-dimensional cohomology and only finitely many nonzero cohomology groups; the alternating sum of their dimensions is its Euler characteristic.* Computing through Čech cohomology is a proof tool, not a substitute for the finiteness hypotheses.

2. Two tests separating the hypotheses

On $U_0 = \text{Spec}(K[t])$ and $U_1 = \text{Spec}(K[t^{-1}])$, the sheaf $\mathcal{F} = \bigoplus_{j \geq 0} \mathcal{O}_X$ is associated to a free module of infinite rank. It is quasicoherent but not coherent: its stalk at the generic point, $\bigoplus_{j \geq 0} K(t)$, is not finitely generated.

Sections on each chart are tuples of finite support. A matching pair on the overlap uses only finitely many indices in total, since there are two charts. For each index, the gluing equation is $a_j(t) = b_j(t^{-1})$, so both polynomials must be constant. Consequently

$$H^0(X, \mathcal{F}) = \bigoplus_{j \geq 0} K.$$

This has infinite dimension. Theorem 26.10 still applies, but Theorem 27.7 cannot be applied to this sheaf. This is a concrete example showing that quasicoherence is insufficient for the finiteness conclusion on a projective scheme.

Now use $\mathbb{P}_{\mathbb{Z}}^1$. Matching polynomial pairs again give

$$H^0(\mathbb{P}_{\mathbb{Z}}^1, \mathcal{O}) = \mathbb{Z}[t] \cap \mathbb{Z}[t^{-1}] = \mathbb{Z}.$$

This module is generated by 1, in accordance with the finiteness theorem, since $\mathbb{Z}$ is Noetherian and $\mathcal{O}$ is coherent. However, $\mathbb{Z}$ is not a field, so “vector space dimension over $\mathbb{Z}$ ” is undefined. Module rank can be discussed separately, but replacing dimension with rank is not a literal application of Definition 27.8 and must not be introduced without defining a new invariant.

3. Recovering the image–kernel step

The given short exact sequence is

$$0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0.$$

The relevant part of its long exact cohomology sequence is

$$H^{q-1}(\mathcal{E}) \xrightarrow{\epsilon} H^{q-1}(\mathcal{F}) \xrightarrow{\delta} H^q(\mathcal{G}).$$

Exactness means $\ker \delta = \text{im } \epsilon$. The first isomorphism theorem gives a short exact sequence

$$0 \longrightarrow \text{im } \epsilon \longrightarrow H^{q-1}(\mathcal{F}) \longrightarrow \text{im } \delta \longrightarrow 0.$$

The last term is the **image of δ** , not $\ker \delta$. The translated source [proof of Theorem 27.5](#) retains two inconsistencies already flagged by edition notes: the ambient index $\mathbb{P}_R^d$ in one place although the space under discussion is $\mathbb{P}_R^N$ after renaming the indices; and a repeated assertion about finite generation of the kernel where finite generation of the image of δ is needed. The reconstruction below is an independent explanation, not a silent alteration of the frozen text.

The module $\text{im } \epsilon$ is a quotient of $H^{q-1}(\mathcal{E})$ and is therefore finitely generated: the image of a finite generating set still generates. The module $\text{im } \delta$ is a submodule of $H^q(\mathcal{G})$. Since R is Noetherian, a submodule of a finitely generated module is also finitely generated. To recall why, first prove the claim for a submodule $M \subseteq R^d$ by induction on d : projection to the last coordinate has image a finitely generated ideal, while its kernel is a submodule of R^{d-1} . Generators of the kernel together with lifts of generators of the image generate M . For an arbitrary finitely generated module, take a surjection from R^d and apply this result to the inverse image of the submodule in question. Thus $\text{im } \delta$ is finitely generated. This is where the Noetherian hypothesis is used.

Take generators $a_1, \dots, a_s$ of $\text{im } \epsilon$ and $b_1, \dots, b_t$ of $\text{im } \delta$. Choose lifts $\tilde{b}_i$ in $H^{q-1}(\mathcal{F})$. For every v in the middle module, write $\delta(v) = \sum_i r_i b_i$. Then $v - \sum_i r_i \tilde{b}_i \in \ker \delta = \text{im } \epsilon$, so it is a combination of $a_1, \dots, a_s$. The finite set

$$a_1, \dots, a_s, \tilde{b}_1, \dots, \tilde{b}_t$$

therefore generates $H^{q-1}(\mathcal{F})$. All sheaf maps in this step are on the same ambient space $\mathbb{P}_R^N$. For this subproblem, the existence of $\mathcal{E}$, coherence of $\mathcal{G}$, and the two cohomological finiteness statements are given data; the argument does not assume that they follow merely from exactness.

4. Source identity is not a theorem certificate

Parent revision 1070036 of Lecture 27 dates from 6 February 2026. In its frozen material, the Theorem 27.5 entity has revision 1088405, dated 30 May 2026, and [its proof entity](#) has revision 1101592, dated 17 June 2026. This does not mean that the parent page changed on those latter two dates: the entities it includes have their own revision histories. Opening the parent revision alone does not reliably reconstruct the entire transclusion closure used by the edition.

To identify the reading material, record the course title, result number, reader ID, parent-page revision, statement or proof entity revision, and the identities of the manifest and expanded text. In the freeze used by this problem, the SHA-256 hash of the expanded Lecture 27 text is

```
ee29b146c2ad0c14c25ea7c97662c294a2
ec0424791999580bec98448531316.
```

Join the two lines without spaces to obtain one 64-digit hexadecimal hash. A matching hash ties the check to the same bytes as the recorded identity; it is not proof that every mathematical argument in those bytes is correct. The image–kernel inconsistency in the preceding part must still be resolved through exactness and module algebra. Conversely, a correct mathematical repair must not be reported as though it were an exact quotation from the source revision.

Self-check and material provenance

A complete answer distinguishes four things: the cover that computes cohomology, the finiteness conditions, the scalars that allow vector space dimension, and the identity of the referenced text. A quick test: if “coherent” is removed, the infinite direct-sum example defeats finiteness; if “field” is removed, vector space dimension notation is not automatically meaningful.

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One capstone: an affine neighbourhood containing zero and infinity

The aim of this exercise is to find the correct statement, recover hypotheses recorded on a preceding page, and fill one gap in a proof with a calculation of your own. There is still just one object: the projective line glued from two affine lines. This is one integrated exercise, not a tag inventory or extra mastery-bank problems.

The following problem and solution were independently written. The motivating example comes from **The Stacks Project Authors**; the prerequisites on sheaves, localisation, and morphisms come from **Holger Brenner**’s course. Stacks is used as a downstream reference, not as a translated source lecture.

Verified reading map

The following official pages were checked on **31 August 2026**. Use tags as reference identities; chapter and lemma numbers displayed on the pages help navigation but may change. A permanent tag is not a frozen copy either: record the access date when reconstructing a source.

Official tag	Identity displayed when checked	Role in the exercise
01HR	Section 26.5, <i>Affine schemes</i>	Context for the spectrum construction with its sheaf.
01HW	Definition 26.5.5	Meaning of an affine scheme as a locally ringed space.

Official tag	Identity displayed when checked	Role in the exercise
01HV	Lemma 26.5.4	Sections on principal opens and stalks.
01JA	Section 26.14, <i>Glueing schemes</i>	Complete hypotheses for glueing data.
01JB	Lemma 26.14.1	Existence of the glued locally ringed space and its mapping properties.
01JC	Lemma 26.14.2	Why glueing scheme pieces produces a scheme.
01JE	Example 26.14.4, <i>Projective line</i>	The example leaving an affine-neighbourhood proof to be supplied.

All six candidate tags are relevant, but they are not interchangeable: in particular, 01HW is not a glueing lemma. For the explicit assertion about $\Gamma(D(f), \mathcal{O})$ and stalks, the section reference 01HR is sharpened to 01HV, linked directly from that section. No additional tags outside this route are needed.

Integrated problem

Take any field k , without assuming algebraic closedness or characteristic zero. Put

$$X_0 = \operatorname{Spec}(k[x]), \quad X_\infty = \operatorname{Spec}(k[y]).$$

Identify $D(x) \subseteq X_0$ with $D(y) \subseteq X_\infty$ through the ring isomorphism

$$k[y, y^{-1}] \longrightarrow k[x, x^{-1}], \quad y \longmapsto x^{-1}.$$

Write P for the glueing. The point 0 is represented by the ideal (x) ; the point ∞ by (y) . The point 1 is represented by $(x - 1)$ or $(y - 1)$. The final objective is to prove, as schemes, that

$$U = P \setminus \{1\} \cong \operatorname{Spec}(k[s]), \quad s = \frac{1}{x-1} = \frac{y}{1-y},$$

and to explain why P itself is not affine. Work through the following seven stages as one argument.

1. From the actual titles and statements, distinguish a definition, hypotheses, an existence lemma, a local lemma, and an example. What remains to be read when 01JB merely says that glueing data are given?
2. Write the directions of the space map and ring map, all domains, inverses, and the three-index conditions. Explain why P is a scheme.
3. Compute $\Gamma(P, \mathcal{O}_P)$ and prove that P is not affine without assuming that k is infinite.
4. Show that U is open and contains 0 and ∞ . Find the section rings on $A = U \cap X_0$ and $B = U \cap X_\infty$, then prove that the two formulas for s agree.
5. Construct isomorphisms $A \cong D(s)$ and $B \cong D(s+1)$ in $\operatorname{Spec}(k[s])$, including inverse ring maps. Check their agreement on the overlap and finish the proof that U is affine.
6. Determine the s coordinates, local rings, and residue fields of 0, ∞ , and the generic point. Why must these three kinds of data not be interchanged?
7. Correct two false shortcuts: “its global section ring is a polynomial ring, so the space is affine” and “the glueing lemma ensures that every glueing of affine pieces is still affine”. Identify the part of the proof that actually closes each gap.

Complete solution

1. Recovering a statement before using it

Tag 01HW gives a definition, not a theorem deducing affineness from the global section ring. Tag 01JB uses the data described in the introduction to 01JA. Reading the lemma page alone therefore does not recover all its

hypotheses. We need locally ringed spaces X_i , open subsets $U_{ij} \subseteq X_i$, and isomorphisms $\varphi_{ij} : U_{ij} \rightarrow U_{ji}$, with $U_{ii} = X_i$. For every i, j, k we require

$$\varphi_{ij}^{-1}(U_{ji} \cap U_{jk}) = U_{ij} \cap U_{ik}, \quad \varphi_{jk} \circ \varphi_{ij} = \varphi_{ik} \quad \text{on } U_{ij} \cap U_{ik}.$$

The first equality gives the composite in the second equality the correct domain. The conclusion of 01JB is still a locally ringed space. The additional hypothesis that each piece is a scheme allows 01JC to be applied. Tag 01JE supplies a particular example, not a substitute for the general hypotheses. This is the required chain of source use, following the [introduction 01JA](#), [Lemma 01JB](#), and [Lemma 01JC](#).

2. Two pieces and gluing as a scheme

The space map $\varphi_{0\infty} : D(x) \rightarrow D(y)$ runs in the opposite direction to $\varphi_{0\infty}^\# : k[y, y^{-1}] \rightarrow k[x, x^{-1}]$. The inverse ring map sends $x \mapsto y^{-1}$; both composites are identities on generators and therefore on the entire rings. Set $U_{00} = X_0$, $U_{\infty\infty} = X_\infty$ with their identities, and $U_{0\infty} = D(x)$, $U_{\infty 0} = D(y)$.

With only two indices, every three-index condition reduces to an identity or the inverse relations above. The image of $D(x)$ is exactly $D(y)$, so the composition domains also agree. A ring isomorphism induces a scheme isomorphism, not just a point map; this also follows from [Brenner, Corollary 10.10](#). Tag 01JB then provides the gluing and its open cover. Every point lies in one of the affine pieces; this is the local reason in 01JC that P is a scheme. This construction is the projective line in [Example 01JE](#).

3. Global does not mean affine

The sheaf condition gives

$$\Gamma(P, \mathcal{O}_P) = \{(f(x), g(y)) \in k[x] \times k[y] : f(x) = g(x^{-1}) \text{ in } k[x, x^{-1}]\}.$$

In the Laurent ring, the monomials x^n , $n \in \mathbb{Z}$, are linearly independent over k . The left-hand side of the equation has only nonnegative powers; the right-hand side has only nonpositive powers. Equality forces all coefficients except that of power zero to vanish. The two constants must agree, so $\Gamma(P, \mathcal{O}_P) \cong k$.

If P were affine, write $P \cong \text{Spec}(R)$. The global section formula in [01HV](#) gives $R \cong k$, so P would have only one point. Yet 0 and ∞ are distinct: neither lies in the region identified during gluing. This is a contradiction. The proof also works over finite fields; we are not counting only k -rational points and do not need an argument that P is infinite.

4. A section replacing the coordinate

The point 1 is the same on both charts because the inverse of $1 \in k$ is still 1. Its complement on each chart is a principal open, so U is open in the glued space, and

$$A = D(x - 1) \subseteq X_0, \quad B = D(y - 1) \subseteq X_\infty,$$

$$\Gamma(A, \mathcal{O}) = k[x, (x - 1)^{-1}], \quad \Gamma(B, \mathcal{O}) = k[y, (1 - y)^{-1}].$$

Using $1 - y$ or $y - 1$ does not change the localisation, since -1 is a unit. The points 0 and ∞ are not removed because $0 \neq 1$ in a field. On $A \cap B$, both x and $x - 1$ are invertible and $y = x^{-1}$. Hence

$$\frac{y}{1 - y} = \frac{x^{-1}}{1 - x^{-1}} = \frac{1}{x - 1}.$$

The two local sections are compatible, so the sheaf axiom gives a unique section $s \in \Gamma(U, \mathcal{O}_U)$. This axiom is [Brenner, Definition 4.1](#); the localisation formula is [Lemma 9.12](#).

5. Proving affineness with two local inverses

Write $T = \text{Spec}(k[s])$, with s on this side an indeterminate. The section just constructed determines a morphism $h : U \rightarrow T$ through $k[s] \rightarrow \Gamma(U, \mathcal{O}_U)$. Its existence and uniqueness use [Brenner, Theorem 10.9](#): the source is a locally ringed space and the target is an affine scheme.

On A , the section $s = (x - 1)^{-1}$ is a unit, so $h|_A$ factors through $D(s) \subseteq T$. Its ring map and inverse are

$$\begin{aligned} k[s, s^{-1}] &\longrightarrow k[x, (x - 1)^{-1}], & s &\longmapsto (x - 1)^{-1}, & s^{-1} &\longmapsto x - 1, \\ k[x, (x - 1)^{-1}] &\longrightarrow k[s, s^{-1}], & x &\longmapsto 1 + s^{-1}, & (x - 1)^{-1} &\longmapsto s. \end{aligned}$$

Both composites are identities on generators; in particular, $(1 + s^{-1}) - 1 = s^{-1}$. Thus $A \cong D(s)$. On B we have $s + 1 = (1 - y)^{-1}$, so the following inverse pair gives $B \cong D(s + 1)$:

$$\begin{aligned} k[s, (s + 1)^{-1}] &\longrightarrow k[y, (1 - y)^{-1}], & s &\longmapsto \frac{y}{1 - y}, & (s + 1)^{-1} &\longmapsto 1 - y, \\ k[y, (1 - y)^{-1}] &\longrightarrow k[s, (s + 1)^{-1}], & y &\longmapsto \frac{s}{s + 1}, & (1 - y)^{-1} &\longmapsto s + 1. \end{aligned}$$

Here $1 - s/(s + 1) = 1/(s + 1)$; substitution on y and s also returns the original generators. No division is by an element not already inverted in the ring concerned.

The two opens of T cover it: the ideal $(s, s + 1)$ is the unit ideal because $(s + 1) - s = 1$. Now check the overlap, not just two separate formulas. On $D(s(s + 1))$, the inverse of the first chart gives

$$x = 1 + s^{-1} = \frac{s + 1}{s},$$

and the inverse of the second gives $y = s/(s + 1)$. Both are units and $xy = 1$, exactly the transition relation forming P . Under $A \cong D(s)$, the overlap $A \cap B = D(x(x - 1))$ corresponds to $D(s(s + 1))$, because $s + 1 = x/(x - 1)$. The same statement on B follows from $s = y/(1 - y)$.

Thus the two local inverses agree as morphisms to U on the overlap. They glue to $g : T \rightarrow U$. On the cover $D(s), D(s + 1)$, the composite $h \circ g$ is the identity; on the cover A, B , $g \circ h$ is the identity. Equality of morphisms can be checked on an open cover: the point maps and section pullbacks agree locally, hence globally by uniqueness of gluing. Therefore $g = h^{-1}$ and

$$U \cong \text{Spec}(k[s]), \quad \Gamma(U, \mathcal{O}_U) \cong k[s].$$

The order matters: here affineness is proved by a scheme isomorphism, and the global section ring is obtained afterwards. This supplies the details omitted in [Example 01JE](#).

6. Coordinates, germs, and values

At 0, we have $x = 0$, so $s = -1$; at ∞ , we have $y = 0$, so $s = 0$. The charts have the same generic point after gluing, since the function-field isomorphism identifies $y = x^{-1}$. The isomorphism in stage 5 gives the following table.

Point of U	Ideal of $k[s]$	Local ring	Residue field
0	$(s + 1)$	$k[s]_{(s+1)}$	k
∞	(s)	$k[s]_{(s)}$	k
Generic η	(0)	$k(s)$	$k(s)$

The stalk formula used applies to any prime point, not just closed points; see [01HV](#). At 0, the element $s + 1$ is a nonzero germ but has value zero after quotienting by the maximal ideal. At η , the value of s is a transcendental element of $k(s)$, not a chosen member of k . Thus a coordinate as a section, a germ as an element of a local ring, and a value as an element of a residue field are different types of objects.

7. Checking the two shortcuts

A global section ring does not determine an arbitrary scheme. Already in stage 3, P and $\mathrm{Spec}(k)$ have isomorphic global section rings but are not isomorphic schemes. To conclude that U is affine, we genuinely need stage 5: correct domains, inverse ring maps, agreement on the overlap, and a cover of all of T .

Likewise, the conclusion of 01JC is *scheme*, not *affine scheme*. The object P itself refutes that erroneous strengthening. Definition 01HW requires a global isomorphism to a spectrum; the existence of affine charts alone only satisfies the definition of a scheme. No finiteness, algebraic-closedness, or characteristic hypothesis on the field is hidden in this calculation.

Oral-proof rubric for self-study

Close the solution and explain one connected proof in about ten minutes, writing down the necessary ring maps. Score each aspect 0 if you cannot yet explain it, 1 if the idea is correct but one gap remains, and 2 if all its requirements are met.

Aspect	Requirements for a score of 2
Source identity	Distinguish 01HW, 01JB, and 01JC, and find the hypotheses of 01JB in 01JA.
Domains and directions	Write $D(x) \rightarrow D(y)$ with the opposite-direction ring map and its inverse.
Global argument	Compute $\Gamma(P, \mathcal{O}_P) = k$ using Laurent polynomials and disprove affineness with two points.
Affine-neighbourhood proof	Give both pairs of inverse maps, show $D(s) \cup D(s+1) = T$, and check $xy = 1$ on the overlap.
Object types and hypotheses	Distinguish sections, stalks, and residue fields, and explain why an arbitrary field suffices.

A score of 8–10 indicates that you can reconstruct the main argument; choose an aspect scored below 2 to revisit. A score of 4–7 suggests repeating stages 4–5, writing out every localisation. A score of 0–3 suggests returning to the prerequisite bridge on sheaves and morphisms, then redoing stages 1–2. This score is only a self-study diagnostic: not a certification, not a claim of human review, and not a condition for constructing, validating, or publishing the material.

Credits and boundaries of the original layer

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